Tunatrike: A Booking Application for Tricycle in General Santos City

A Thesis
Presented to the Faculty of
Information and Communications Technology Program
STI College General Santos, Inc.

In Partial Fulfilment
of the Requirements for the Degree
Bachelor of Science in Computer Science

Celso Onnele T. Arancon
Ariel M. Guerra
Adrian D. Macaculop
Raffy G. Simon
Alvin A. Tampus Jr.

October 2024

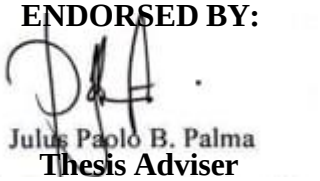
ENDORSEMENT FORM FOR ORAL DEFENSE

TITLE OF RESEARCH: **Tunatrike: A booking application for tricycle in General Santos City**

NAME OF PROPONENTS: Celso Onelle T. Arancon
Ariel M. Guerra
Adrian D. Macaculop
Raffy G. Simon
Alvin A. Tampus Jr.

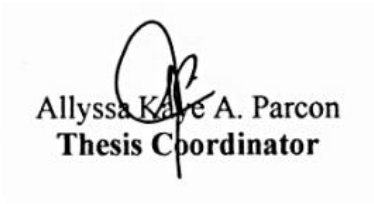
In Partial Fulfilment of the Requirements
for the degree Bachelor of Science in Computer Science
has been examined and is recommended for Oral Defense.

ENDORSED BY:



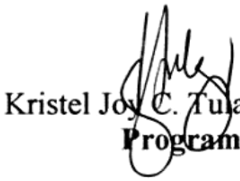
Julius Paolo B. Palma
Thesis Adviser

APPROVED FOR ORAL DEFENSE:



Allyssa Kaye A. Parcon
Thesis Coordinator

NOTED BY:



Kristel Joy C. Tulagan, LPT, MIT
Program Head

October 2024

APPROVAL SHEET

This thesis titled **TunaTrike: A booking application for tricycle in General Santos City**, prepared and submitted by **Celso Onnele T. Arancon; Ariel M. Guerra; Adrian D. Macaculop; Raffy G. Simon;** and **Alvin A. Tampus Jr.**, in partial fulfillment of the requirements for the degree of Bachelor of Science in Computer Science, has been examined and is recommended for acceptance and approval.

Julius Paolo B. Palma
Thesis Adviser

Accepted and approved by the Thesis Review Panel
in partial fulfillment of the requirements for the degree of
Bachelor of Science in Computer Science

Anthony Mark F. Silong, MIT
Panel Member

Mark Joel M. Umadhay, MIT
Panel Member

Kristel Joy C. Tulagan, MIT, LPT
Lead Panelist

APPROVED:

Allyssa Kaye A. Parcon
Thesis Coordinator

Kristel Joy C. Tulagan, MIT, LPT
Program Head

November 2024

ACKNOWLEDGEMENTS

The proponents would like to extend our heartfelt thanks to everyone who contributed their time and effort to this project. Without you, it would not have been possible. Above all, we are eternally grateful to the Almighty God for the countless blessings we received throughout the development of our project.

I would like to thank our Thesis Coordinator, Ma'am Allyssa Kaye Parcon, and Sir Julus Paolo B. Palma, for their invaluable guidance, unwavering support, and encouragement throughout this research. Their expertise and insights have been instrumental in shaping the direction of this thesis.

Our deepest gratitude to our Thesis Adviser Sir Julus Paolo B. Palma who assisted the researchers in coming up with and selecting intriguing, realistic, and relevant study topics. His unceasing efforts, helpful critiques, and timely guidance have substantially enhanced our work. His perseverance, patience, and attention were invaluable in helping us finish our thesis task successfully.

The Proponents would also like to express gratitude to their respected teachers, fellow students, and advisors. Your guidance, suggestions, and support have been extremely helpful. Your combined knowledge, skills, and different viewpoints have played a significant role in molding our project into something we can all be proud of.

The individuals and Local Government Office who participated in the survey for the researchers' thesis also deserve thanks. Your involvement has been crucial in assisting them in collecting the necessary data to finalize their study.

Lastly, express deepest gratitude to our parents, friends, and guardians: Mr. And Mrs. Alvin Tampus Sr. and Esterberta Tampus; Mr. And Mrs. Ariel Guerra Sr and Richel Guerra; Mr. And Mrs. Arthur Macaculop and Rosenda Macaculop; Mr. And Mrs, Celso Arancon and Onnele Arancon; Mr. And Mrs. Rufino Simon and Marissa Gabriel.

ABSTRACT

Title of research: **TunaTrike: A booking application for tricycle in General Santos City**

Researchers: **Celso Onelle T. Arancon**
Ariel M. Guerra
Adrian D. Macaculop
Raffy G. Simon
Alvin A. Tampus Jr.

Degree: **Bachelor of Science in Computer Science**

Date of Completion: **December 2024**

Keywords: **Urban Transportation, Tricycle, Fare calculation Geolocation, Mobile Application,**

This paper proposes the design and implementation of a specialized fare meter for tricycles, a widely used mode of transportation in urban and rural areas in Philippines. Tricycles, also known as "trike taxis," are vital for providing affordable and accessible transport to millions daily. However, the lack of standardized fare calculation systems often results in disputes and inconsistencies between passengers and drivers. To address this issue, the proposed fare meter offers a transparent and reliable method for determining fares based on distance traveled and time spent on the ride. Utilizing advanced technology, the meter accurately measures distance, trip duration, traffic conditions, and waiting time. Key features include a user-friendly interface for both passengers and drivers, real-time fare calculation, and adherence to local fare regulations. By enhancing transparency and reducing fare-related disputes, the fare meter aims to improve the efficiency and fairness of tricycle services. Additionally, it is expected to bolster the professionalism of tricycle drivers and elevate the passenger experience. This fare meter represents a significant advancement in modernizing and standardizing fare calculation in the tricycle transportation industry, benefiting both passengers and drivers.

TABLE OF CONTENTS

	Page
Title Page	i
Endorsement Form for Proposal Defense	ii
Approval Sheet	iii
Acknowledgments	iv
Abstract	v
Table of Contents	vi
List of Tables	vii
List of Figures	viii
List of Appendices	ix
Introduction	1
Definition of terms	2
Background of the problem	4
Overview of the current state of technology	5
Objectives of the study	8
Scope and limitations of the study	10
Review of related literature/studies/systems	13
Methodology	21
Results and Discussions	33
Conclusion	37
References	39
Appendices	35

LIST OF TABLES

Table		Page
1	Demographic Profile	57
2	User App (Passengers) Survey Results	58
3	Driver App (Drivers) Survey Results	59
4	Hardware Requirements Admin, Commuters and Tricycle Drivers	37
5	Software Requirements	40
6	Actual Thesis Expenses	41

LIST OF FIGURES

Figure		Page
1	Agile: Scrum framework	51
2	System architecture design	52
3	Entity relationship diagram	53
4	Context flow diagram	54
5	Diagram lvl 0	55

LIST OF APPENDICES

Appendix		Page
A	Hardware/Software Requirements	37
B	Gantt Chart	42
C	Actual Thesis Expenses	44
D	Data Interpretation of the survey	46
E	List of Figures	50
F	List of Tables	56
G	Formal Letters	60
H	Revision's List	63
I	Grammarian's Certification	70
J	Statistician's Certificate	74
K	System Validation Certificate	76
L	Fare Metrics of General Santos City	79
M	User's Manual	84
N	Curriculum Vitae of Researchers	106
O	30% Code from the System	112

INTRODUCTION

The intricate network of transit systems that kept people moving keeps the pulse of urban life alive in congested cities. But outdated and inaccurate methods of determining fare prices have frequently clouded the urban commute, leading to fare disparities and, occasionally, unfair pricing for commuters. According to Spieler, C. (2020), this problem is most noticeable in urban areas with underfunded and ineffective transit systems, where many people are compelled to use more costly and inefficient forms of transportation like Tricycle. It affects household budgets and serves as a lifeline for people from all walks of life in this fast-urbanizing age when access to jobs, education, and basic services is crucial.

The researchers therefore put forward an innovative proposal in response to this urgent demand for a more accessible, equitable, and efficient urban transportation solution: a modern mobile application designed to solve the problems encountered by urban commuters, whether they are in the world's most developed cities or its emerging ones. With the help of cutting-edge geolocation technologies, this innovative application will transform real-time distance measurement and fee computation. The application is also ready to work seamlessly with the current modes of transportation, adhering to laws such as "RA 11314" or the "Student Fare Discount Act," which guarantees that students, and for Persons with Disabilities (PWD) in the Philippines is mandated by Republic Act No. 10754, also known as the Act Expanding the Benefits and Privileges of Persons with Disability receive their 20% fare discount, making it easier for them to plan and pay for their entire travels. The researchers' application provides transparent fare estimates, real- time travel updates, and a very intuitive user interface that can be used by people with varying levels of technological proficiency, (The Philippine Star, 2023)

The Researchers firmly believe that this cutting-edge technology has the power to bring in a new era of low-cost, readily available, and ecologically responsible urban transportation. With their goals firmly established, the researcher will learn about the innovative work that promises to revolutionize urban transportation and meet the constantly changing demands of the city, making efficient and equitable commuting accessible to all. Real-time distance monitoring will be revolutionized by the researchers' application. Imagine living in a

society where commutes are predictable and affordable, where hidden fees are a thing of the past. This not only empowers individuals financially, but also reduces stress and uncertainty associated with travel, allowing everyone to focus on the things that matter most.

Definition of Terms

Tricycle – Is a type of transportation, a three–wheeled motorcycle, used in some places where other modes of transport may not be readily available or affordable (IATSS Research, 2023).

Geolocation Technologies - Advanced methods that use geographical data to determine precise locations, aiding in accurate fare calculations and real-time tracking of commutes (Utilities One, 2023).

Student Fare Discount Act (Republic Act 11314) - Refers to a specific law or regulation ensuring fare discounts (20% in this case) for students, aimed at making transportation more accessible for these groups (Republic Act No. 11314, 2019).

PWD Fare Discount (Republic Act 10754, Section 6.7) - This will cover the real fare for travel by land transportation, including but not restricted to taxis, shuttle services, public railways, including Light Rail Transit, Metro Rail Transit, and Philippine National Railways, public utility buses or jeepneys, taxis, Asian Utility Vehicles (AUVs), and the like.

Together with ensuring that restaurants and recreation centers under their control adhere to the law, the concerned Local Government Units will work with the Leagues of Cities and Municipalities to include a 20% discount and VAT exemption for individuals with disabilities in the terms and conditions attached to business permits for the establishments.

Transparent Fare Estimates - Providing clear and easily understandable calculations of transportation costs to users, eliminating hidden fees and ensuring clarity in pricing (Faster capital, 2023).

Fare Matrix – A guide provided by the local government unit (LGU) that suggests the appropriate fare prices based on different distance ranges and service types.

Feedback – A system within the app that allows users to provide feedback about their ride experience, offering valuable data for ongoing improvements and optimization of the app.

Background of the problem

The functioning of cities is impacted by how people move within them, and tricycles serve as a common mode of transportation across many urban areas. These three-wheeled vehicles offer locals a flexible and convenient way to travel short distances. However, the way fares are determined for tricycle services often lacks a systematic approach, leading to pricing structures that are not optimal.

As cities grow, ensuring fair and efficient transportation for everyone becomes increasingly crucial. With more people migrating to urban areas, the need for accessible transport becomes paramount. However, a significant issue arises from the unfair or inaccurate determination of fares, impacting commuters across various demographics. Visitors to a city, particularly tourists, face unique challenges when navigating the transportation system. Unfamiliarity with fare systems, routes, and available modes of transportation can lead to confusion and potential over-payment for their journeys. Additionally, tourists might be subject to higher rates due to their lack of local knowledge or inability to access discounted fares available to residents.

Many urban transportation systems continue to use antiquated fare calculating techniques, depending on set rates or conjecture that ignore important variables including the kind of conveyance, the distance traveled, and required discounts for individuals with impairments, pensioners, and students. Because of this, travelers frequently spend more than is necessary, which puts a strain on their finances and restricts their access to jobs, vital services, and metropolitan opportunities—all of which threaten equity and inclusivity.

According to Edwin Fernandez (August 2022), In General Santos City, the issue of tricycle fare determination has long been a point of contention, with complaints of overcharging and a lack of standardization. Recent efforts by the local government aimed at regulating this sector have highlighted the complexity and need for a more structured system. For example, in 2022, city officials implemented a new fare matrix to address excessive fare charges, setting a base fare of PHP18 for the first four kilometers and additional charges for longer distances In General Santos City, the issue of tricycle fare determination has long been a point of contention, with complaints of overcharging and a lack of

standardization. Recent efforts by the local government aimed at regulating this sector have highlighted the complexity and need for a more structured system. For example, in 2022, city officials implemented a new fare matrix to address excessive fare charges, setting a base fare of PHP18 for the first four kilometers and additional charges for longer distances.

However, despite these efforts, challenges remain, particularly regarding compliance. Some drivers have been reported charging passengers above the established rates, especially during adverse conditions like heavy rain, while others have even altered fare matrices to justify overcharging. These issues are exacerbated by the city's growing population and the overwhelming number of tricycles in the area—over 42,000 tricycles, compared to an ideal number of only 9,000 units, further complicates fare regulation.

Overview of the current state of the technology

The research emphasizes that traditional ways of deciding how much people pay for city transportation here in General Santos City often led to unfairness. This especially impacts individuals and families living in areas with limited financial resources, as these fixed pricing structures can strain household budgets. Additionally, these outdated systems can hinder access to crucial services for those with financial constraints.

General Santos City's technical environment is experiencing a radical change aimed at resolving the disparities from using conventional techniques to calculate transportation expenses. It has been determined that the current models are insufficient since they frequently place an unfair financial burden on the populace, especially those who live in economically challenged areas. In addition to making household finances more difficult, these antiquated pricing practices also contribute to the larger problem of those with limited means having limited access to necessary services.

Given the shortcomings of these traditional methods, there is an increasing need to introduce more advanced and inclusive technology in urban transportation. The study highlights the need for creative solutions that consider the wide range of economic origins that the people in the city come from. Adopting contemporary technology offers the chance to build a more adaptable and just system, such as digital platforms and dynamic pricing

schemes. One of the primary difficulties raised is the impact of fixed pricing systems on households with limited financial resources. These conventional methods, which disregard the varying economic capacities of the population, unfairly spread the costs of city mobility. Families in low-income areas find it more difficult to pay for regular transport services because of the burden this puts on household resources.

Moreover, tourists face challenges when navigating unfamiliar transportation systems in cities. The lack of knowledge about fare structures and routes often results in confusion and, in some cases, overcharging due to the absence of transparent pricing. To rectify these issues, the study proposes the development of a modern mobile application. This app would revolutionize the transportation experience by leveraging technologies like geolocation. Through these technologies, the app would accurately determine a user's location, facilitate ride bookings, and precisely calculate fares based on the actual distance traveled. This innovative approach aims to replace the existing rigid and potentially unfair fare systems prevalent in cities.

The primary objective of introducing this new app is to enhance urban transportation for all individuals, regardless of whether they are residents or visiting tourists. By addressing fare-related challenges, the app strives to make travel more equitable and convenient. For tourists, the goal is to ensure they are not overcharged and have a hassle-free experience while navigating city transportation. Overall, the focus is on creating a fairer and more transparent system that benefits both locals and visitors alike.

This mobile app is developed using Android Studio and the Flutter framework to create a versatile and efficient system for fare management. Android Studio serves as the primary development environment, providing powerful tools for coding, testing, and debugging, ensuring the app’s compatibility with Android, the dominant platform in the Philippines. Flutter's widget-based design also supports the creation of intuitive interfaces, making the app user-friendly and visually appealing. By integrating geolocation services, the app determines users' precise locations and calculates fares based on distance traveled, a key feature to ensure fair and accurate fare assessment. Additionally, the app utilizes a real- time database to store route, fare, and discount eligibility data, enabling the dynamic

application of fare reductions for students, seniors, and persons with disabilities, creating a more equitable pricing model. Combining geolocation data and user profiles, the app introduces a dynamic pricing system that adjusts fares according to travel distance and user-specific factors. This modernized approach not only addresses issues related to fixed fare structures, especially for low-income households, but also provides tourists with a transparent and hassle-free way to navigate city transportation. Overall, the app's goal is to promote fairness, transparency, and convenience in city travel, delivering an equitable and positive experience for all users.

Objectives of the study

To develop, maintain, and implement “Tunatrike: A booking application for tricycles in General Santos City” by Geolocation. Initially, the researchers carefully designed the system, detailing how it will be constructed. Then put it together using both radiational and modern techniques, aiming to highlight the benefits of this combined approach in making a functional and innovative “Tunatrike: A Booking application for tricycle in General Santos City”

The following are the specific objectives:

1. To create a fare calculation system that uses geolocation technologies to precisely calculate the distance traveled.

   The system will encompass the development of an advanced infrastructure aimed at swiftly and accurately computing fair and transparent tricycle fares. This will be achieved through the intelligent integration of geolocation technology. Geolocation technology will ensure accurate location tracking, forming the bedrock for the system's core functionalities: fare calculation, enhanced by the seamless geolocation capabilities.

2. To develop a mobile application that integrates a fare calculation system compliant with transportation regulations, including RA 11314 (Student Fare Discount Act) and RA 10754 (Benefits for Persons with Disabilities), while providing a user- friendly interface for fare information and ride management.

   The system will encompass the creation of a comprehensive mobile application, strategically designed to seamlessly integrate the fare calculation system while diligently adhering to a spectrum of transportation regulations, including the stringent guidelines outlined in "RA 11314" and "RA 10754" This application will play a pivotal role in ensuring full compliance with these regulations while serving as a user-friendly interface. It will offer convenient access to fare-related information and efficient ride management functionalities, thereby emphasizing the app's significant role in delivering both the fare calculation system and meeting the specified regulatory requirements.

3. To develop a system that integrates a user feedback feature into the mobile application so that users may share their experiences and contribute useful information for ongoing system improvement.

   The system will prioritize the integration of a robust user feedback mechanism within the mobile application. This mechanism will serve as a vital conduit for users to effortlessly report their experiences, offer suggestions for improvements, and contribute valuable insights. By enabling this interactive feedback loop, the system aims to continuously refine and enhance its functionalities based on user input. This emphasis on the app's feedback loop highlights its pivotal role in facilitating ongoing system optimization and improvement, ensuring that user experiences and preferences are integral to the system's evolution and refinement.

Scope of the study

The system being developed for tricycle transportation management focuses on providing an efficient, transparent, and secure platform for both drivers and passengers in General Santos City. It includes an **Admin (LTO) Module** for managing driver registrations, verifying licenses, monitoring driver performance, and ensuring only qualified drivers can access the system. The **Driver Module** allows tricycle drivers to create and maintain their profiles, manage ride requests, handle navigation, and track earnings. A **Passenger User Module** facilitates ride booking, fare estimation, payment processing, and access to emergency assistance, ensuring a safe and convenient travel experience. Passengers and drivers can communicate seamlessly through the platform, improving coordination and service quality.

The scope of the study are as follows:

Admin (LTO) Module Driver Management

A registration module will be part of the system, making it easier to manage and register tricycle drivers and keep an eye on their actions within the system. Drivers can enter their information and access the platform via this module. Administrators may ensure that only qualified and confirmed drivers are a part of the system by reviewing, approving, and verifying each driver's credentials through the Verification and approval module before providing them access. Administrators will also be able to keep an eye on consumer evaluations of the drivers' performance thanks to the rating performance Module, which will facilitate ongoing evaluation and enhancement of the offered service.

Driver (Tricycle) Module: Profile Management

Drivers will be able to create and manage their profiles by entering vital information about themselves, their vehicles, and their licenses, according to the system's Registration and Setup Module. This module makes sure drivers can keep their profiles current. Furthermore, licenses and car registrations can be uploaded and verified by drivers using

the Document Submission Module, which will speed up the validation process and guarantee that all required paperwork is safely kept and easily available within the system.

Ride Handling

A Navigation and Trip Completion Module will be included in the system, giving drivers the opportunity to mark a ride as complete once a trip has been successfully completed and navigation aids to locate pick-up and drop-off locations. This guarantees a clear and seamless experience for both drivers and passengers. Furthermore, drivers will be able to efficiently manage ongoing rides, accept passenger bookings, and receive incoming ride requests thanks to the Receiving and Accepting Ride Requests Module. This module makes sure that drivers can manage their schedules effectively and react to requests quickly.

Fare Information and Payment

Drivers will be able to access fare estimations depending on variables including the type of service rendered and the distance traveled thanks to the system's Fare Calculations Module. This module guarantees fare management transparency and aids drivers in understanding the cost structure of each ride. Additionally, the Access to revenues Module will give drivers real-time access to financial summaries and facilitate efficient income tracking by enabling them to examine and manage their revenues from completed rides. This lesson helps drivers arrange their finances more effectively and with greater clarity.

Passenger User Module Ride Booking and Payment

A Ride Booking Module built into the system will make it simple for users to request trips by giving pick-up and drop-off locations and indicating the kind of tricycle service they need. By streamlining the booking procedure, this module makes sure that travelers have a positive experience. The Fare Estimation and Payment Module will also enable safe payments, give passengers access to trip details, and estimate fares accurately, all of which will add to the ride's transparency and convenience. In the event of an emergency, passengers would be able to quickly access emergency assistance by using the Emergency Hotline Module to get in touch with local law enforcement PNP-Gensan. To improve safety and accountability, this module also provides a means for passengers to report any drivers who are breaking the law.

Communication and Feedback

Drivers and passengers can arrange rides over the phone thanks to the Driver-Passenger Communication Module, enabling smooth communication between the two parties. During the voyage, this module will allow drivers and passengers to talk about specifics like locations, pick-up times, and any other issues. It will also provide a platform for problem reporting, allowing passengers to express concerns immediately, and for feedback input, ensuring that both parties can share experiences and improve service quality and safety.

Limitations of the study

The system is limited to registered tricycle drivers, relies on a fare matrix suggested by the LGU-Gensan, and is not available for iOS users. Additionally, drivers cannot accept multiple bookings simultaneously, and maintaining consistent service quality remains a challenge.

The system is limited on the following areas:

1. The system is specifically designed to cater exclusively to registered tricycle drivers only and exclude from services the unregistered tricycle, drivers that have no driver's license and no Official Receipt and Certificate of Registration.
2. The system has a fare matrix suggested by the LGU-Gensan.
3. The app does not support IOS.
4. The system cannot control the capacity of the commuter, it will depend on the driver if they want to add more passengers.
5. Ensuring consistent quality service from drivers, including their behavior, adherence to traffic rules, and professionalism, is an ongoing challenge for these platforms.
6. The driver cannot accept multiple bookings.

LITERATURE REVIEW

Urban Transportation and Fare Systems

According to Dr Daniel Oviedo (2021) mobility plays a significant role in shaping the urban form of our cities and the way the researchers experience them. Transport infrastructures and public transport services determine such a role and how these infrastructures and services fit with our cities' socioeconomic, functional, and spatial structure. The distribution of different means for urban mobility determines individuals' ability in various social and economic positions to access opportunities and exercise their right to the city. Latin America is a unique setting for examining the links between urban mobility and social equity. This volume compiles contributions from scholars researching a diverse set of cases across the Americas concerning the social impacts of urban mobility and discussing concepts and methods that can help us examine the distributional effects of transport policies in the region. By reviewing how spatial and social mobilities contribute to the reproduction of both spatial and social inequities in Latin American cities, the book provides an exciting collection of research that expands on different fronts on the current literature in and about the region.

According to Damilola Oladimeji (2023). As technology continues to evolve, our society is becoming enriched with more intelligent devices that help us perform our daily activities more efficiently and effectively. One of the most significant technological advancements of our time is the Internet of Things (IoT), which interconnects various smart devices (such as smart mobiles, intelligent refrigerators, smartwatches, smart fire alarms, smart door locks, and many more) allowing them to communicate with each other and exchange data seamlessly.

An Act Expanding the Benefits and Privileges of Persons with Disability (PWD), Amending Section 32 of Republic Act No. 7277.

On July 27, 2015, Republic Act No. 10754, known as “An Act Expanding the Benefits and Privileges of Persons with Disability (PWD), Amending Section 32 of Republic Act No.

7277.” To implement said law, corresponding Rules and Regulations (IRR) have been issued and portion of said IRR read:

Section 6.7 Land Transportation Travel- this shall apply to actual fare for land transportation travel such as, but not limited to, public utility buses or jeepneys, taxis, Asian Utility Vehicles (AUVs), shuttle service, public railways including Light Transit, Metro Rail Transit and Philippine National Railways and Transportation Network Vehicle Services (TNVS).

The Department of Transportation including those agencies attached and under its administrative supervision and control shall issue the corresponding guidelines, circulars or directives related to domestic air, sea and land transportation travel and disseminate such information to all concerned.

Student Fare Discount Act (RA 11314)

According to former President Rodrigo Duterte who approve this act on April 17, 2019. Declaration of Policy. -Section 1, Article XIV, of the 1987 Constitution states: "The State shall protect and promote the right of all citizens to quality education at all levels and shall take appropriate steps to make such education accessible to all. In addition, Paragraph 3, Section 2 of the same Article directs the State to establish and maintain a system of scholarship grants, student loan programs, subsidies, and other incentives, which shall be available to deserving students in both public and private schools, especially to the underprivileged.

Data Privacy and Security

According to Yaqi Gong, Ashley Schroeder (2022). Smart tourism has gained popularity in recent years. While we praise the benefits it brings, we should not neglect the associated challenges of data privacy and security. This study intends to investigate the scope of data privacy and security research on smart tourism by analyzing 38 relevant contributions in multi-disciplinary fields systematically. Historical trends, disciplines, regions, themes, relevance, methods, and theories were analyzed to map out the existing research on data privacy and security in smart tourism. Findings reveal that although relevant studies have

grown in frequency in recent years, there is still a lack of attention to tourists' information privacy and security in smart tourism research. Specifically, privacy- based conceptualized studies, theory-driven empirical research, and legal analysis in data protection deserve immediate attention. Tourism scholars should also consider proper methods to raise tourists' data privacy and security awareness and build an ethical relationship between tourists and smart tourism applications.

According to Muhammad Ridwan Farizqi, Anna Dina Kalifia, Rr. Hajar Puji Sejati (March 3, 2024) A motorbike is a two-wheeled vehicle powered by an engine. Motorbike users are required to service their cars because the components on the motorbike will experience a decrease in performance along with the length of time they use the motorbike. However, this can be overcome by servicing your motorbike regularly. However, sometimes motorbike users have to face queues at workshops that are long and quite time-consuming. Then, an Android-based online motorbike service booking system was created, which did not require customers to come to the seminar directly to make a booking. With this application, customers must book via Android by selecting the desired schedule. The application was created using Android Studio with the Kotlin programming language. The database section uses the Firebase Real-time Database to update the data in real-time. The service provider will also easily confirm orders via the website connected to Firebase. In short, it is hoped that this application can help motorbike users book services online, so they don't have to spend time queuing.

Review of Related Studies and/or systems

According to Neil P. Balba, Nova Marie F. Rosas PJ Zeder Drilon, Sarah Jane F. Fallaria, Julie F. Fallaria (August 2019), Romblon is one of the provinces in the Philippines located in region 4B under MIMAROPA (Mindoro, Marinduque, Romblon and Palawan) region. It is an archipelagic province located at the middle of the Philippines composed of three major islands namely the Romblon, the capital of the province, Tablas, and Sibuyan along with seventeen other smaller islands. Among the islands, Tablas is the biggest airport in which it is located. To reach the place from Manila, the visitors should ride on a bus for about 2 hours to Batangas Port. From the port, the visitor should take a Ferry to Odiongan,

the main town of Tablas Island, for about 8 hours.Transport is straightforward on Romblon Island. Cars are not that many and to get around, most people use motorbikes for private use and “trikes” for public use. “Trikes” or Tri-bike, a motorcycle attached to a side car, is used by public commuters in transporting from one place to another.

The purpose of the study is to design an online system that can be used in booking of tri-bikes applying the Global Positioning System (GPS) technology for navigation and location finding. Since there are several problems being encountered in the booking of tri- bikes such as long queue, difficult chance of riding during peak time, limited trikes during unholy hours and the stress in the waiting of booking trips, an automated booking system applying current technologies was designed. The system runs in a client server technology that will run in any means of available communication devices particularly using the Internet. The internet connections in different areas on the island are already improving and most of the people have their own mobile phones that could facilitate the designed technology even in the remote area. This is on-time to the current upgrading of Internet facilities in the remote areas in the Philippines. The system iSAKAY is similar in the principles of Grab and Uber usually being used in the big cities all over the Philippines. The big cities have lots of cars and different means of transportation, the applications have become successful. The researcher’s designed system is dedicated to a tri-bike or tricycles available in the remote areas in the Philippines, especially in Tablas, Romblon. Using the mobile phone, the app iSAKAY can be used to book directly with the trikes operators via networks. Because of GPS technology, the driver can easily locate the position of the customers and can directly go to the given directions via phone of the trike driver.

According to Rathee, D.S., Rathee, K., Singh, R.S., Gemeda, K.A. (2024), Addressing the pressing issue of transportation for individuals with disabilities, this article aims to develop a customized three-wheeled vehicle equipped with an electric motor and pedals, to cater to the mobility requirements of people with disabilities. The proposal focuses on designing, manufacturing, and deploying a solar-powered E-Tricycle using readily available components from the Adama market, enhancing both sustainability and accessibility. To bolster security and track tricycles, a GPS-GSM system has been also integrated in this design. This innovative approach offers an affordable prototype featuring a solar panel on

the roof, providing shade and effortless charging. The project encompasses the creation of a sustainable, safe, and robust structure, assembly of mechanical components on the frame, and integration of electrical circuits, including indicators, on/off switches, lights, motors, batteries, battery management systems (BMS), solar panels, and their controllers. Furthermore, a mathematical model has been developed to calculate the power required for E-Tricycle movement, accounting for the rider and their possessions.

According to Nurul Farah Izzah Zailani (2020). Transportation or transport is the movement of people, livestock and objects from one location to another. In other words, the transportation activity is characterized as a particular move of an entity or object from point A to point B. Transportation modes include air, land (rail and road), water, cable, pipeline and space. You can break the field into facilities, equipment and operations. Transport is important because it enables people-to-people exchange which is central to civilization growth. Transport infrastructure consists of fixed facilities including highways, bridges, airways, waterways, canals and pipelines, and terminals such as airports, railway stations, bus stations, warehouses, truck terminals, refueling warehouses (including docks and fuel stations) and seaports. Terminals can be used for passenger and freight interchanges as well as for maintenance. Vehicles that operate on these networks may include automobiles, motorcycles, buses, trains, boats, helicopters, watercraft, spacecraft, and aircraft. Operations deal with the manner in which vehicles are handled and the procedures for this purpose, including financing, legalities and policies. In the transport sector, service activities and management may be either public or private, depending on the country and mode.

According to Zahib, Maulidatul Aulia, Veronikha Effendy, and Eko Darwiyanto (2022), The online transportation ordering mobile application is an application that helps users in ordering transportation facilities that can be accessed by anyone using mobile phones and internet users are no exception for people with visual impairments. Users with visual impairments with limited accessibility will have difficulty accessing the application. Based on the results of the initial survey, it was found that the online grab transportation ordering application is currently still less accessible and an increase in accessibility is needed. This study aims to design the right user experience to improve the accessibility of the application

so that users with visual impairments can access the application properly. In this research, the method used is User Centered Design (UCD) which focuses on user needs and characteristics who are visually-impaired users. The study's findings are recommendations for the interaction design of the Grab application. Based on the needs and characteristics of the users, the resulting accessibility improvements are in the form of a simpler design than the previous design and features designed to overcome the pain points of users with visual impairments. Furthermore, the design that has been made is tested using the Cognitive Walkthrough method, which contains user scenarios to achieve certain goals.

According to Neil P. Balba (2019). Romblon is one of the provinces in the Philippines located in Region 4Bunder MIMAROPA (Mindoro, Marinduque, Romblon and Palawan) region. It is an archipelagic province located at the middle of the Philippines composed of three major islands namely the Romblon, the capital of the province, Tablas, and Sibuyan along with seventeen other smaller islands. Among the islands, Tablas is the biggest in which airport is located. To reach the place from Manila, the visitors should ride on a bus for about 2hours to Batangas Port. From the port, the visitor should take a Ferry to Odiongan, the main town of Tablas Island, for about 8 hours. Transport is straightforward on Romblon Island. Cars are not that many and to get around, most people use motorbike for private use and “trikes” for public use. “Trikes” or Tri-bike, a motorcycle attached to a side car, is used by public commuters in transporting from one place to another.

Synthesis

The synthesis of city transport fare with a combination of various technological approaches introduces a novel paradigm for fare systems. Unlike traditional models characterized by distance- or flat-based architectures, this new approach integrates the strengths of diverse technologies, encompassing not just deep learning but also other methodologies like probabilistic modeling, rule-based systems, or ensemble methods.

This approach leverages the adaptability and flexibility of deep learning while incorporating the strengths of other techniques. It assesses extensive and diverse datasets encompassing passenger behavior, economic indicators, traffic patterns, and other multi- modal data sources. This fusion allows for a more dynamic and responsive method, enabling real-time adjustments to urban dynamics and facilitating fair and effective fare pricing. Comparing this method with techniques like reinforcement learning and neural networks helps to delineate its unique advantages and limitations. It highlights how this amalgamation of approaches enhances the accuracy and adaptability of fare synthesis within municipal transit systems.

Real-world case studies demonstrating the application of these technological combinations in fare systems reveal both successes and challenges. These implementations provide valuable insights for future enhancements, highlighting the need for transparency, accountability, and addressing any biases in algorithmic decision-making to maintain public trust. While deep learning alone presents its own advancements, this integrated approach in municipal transport fare systems introduces a disruptive trajectory. It not only fills gaps in traditional models but also presents new opportunities for research and innovative expression, laying the groundwork for a more responsive, equitable, and technologically sophisticated urban transportation fare ecology.

The review of related literature (RRL) and related studies (RRS) on urban transportation fare systems share a common goal of exploring how technological advancements can address the limitations of traditional fare calculation methods. Both highlight that distance- based or flat-rate systems often fail to meet the complex and dynamic needs of modern cities, advocating instead for innovative, data-driven approaches. The literature emphasizes

the potential of technologies like deep learning and other advanced algorithms to provide fair and adaptive fare structures, drawing on a wide range of data, including passenger behavior, economic indicators, and multi-modal transport patterns.

However, the RRL and RRS differ in their focus and depth. The RRL delves into theoretical frameworks, comparing diverse technological approaches such as deep learning, probabilistic modeling, and rule-based systems and contrasting them with traditional models to establish a more adaptive fare paradigm. It evaluates each method's strengths and weaknesses in meeting the demands of urban mobility. In contrast, the RRS centers on real-world applications, showcasing case studies that reveal both successes and challenges in implementing these technologies. This practical focus highlights crucial concerns, including transparency, accountability, and bias, all essential for building public trust and achieving equitable fare systems.

Together, the RRL and RRS offer a comprehensive view: the RRL provides theoretical insights for developing fairer, more adaptable systems, while the RRS underscores the practical considerations and obstacles in real-world implementation. This combined perspective emphasizes not only the importance of adopting diverse technological approaches but also the need to address practical issues responsibly to ensure sustainable urban transportation solutions.

METHODOLOGY

This chapter based on figure 1 in Appendix E, lays out the SDLC methodology that will power the development of researchers' revolutionary mobile application, set to change how urban transportation works. By the strengths of Agile methodologies, the researchers will navigate a flexible, efficient, and risk-managed development journey.

The researcher's approach harnessed the strengths of Agile methodologies, employing an approach infused with the adaptability and efficiency inherent in Scrum methodology. Using the Scrum methodology for the development of the TunaTrike application aligned well with the objectives and scope of the study due to its iterative and flexible approach. Scrum, a popular agile framework used in software development, emphasizes adaptability, collaboration, and continuous improvement.

By leveraging Scrum methodology, the study ensured that the development of the TunaTrike application remained agile, adaptable, and responsive to user needs and regulatory requirements. It enabled the team to deliver a more refined, user-friendly, and functionally robust transportation solution by incorporating feedback and making iterative improvements throughout the development process.

During the **Analysis Phase**, the researchers conducted a comprehensive survey to gather insights directly from tricycle users, commuter users, and the Local Government Office (LGU). This survey was designed to collect information on user preferences, expectations, and any challenges they faced with the current fare calculation system. The goal was to ensure that the developed system not only met technical requirements but also aligned with the needs and preferences of the actual users.

In the **Design Phase**, the researchers turned the feedback from the Analysis Phase into a plan for the mobile app. They focused on creating an easy-to-use app that would calculate fares accurately and follow all the rules. The team designed the app's interface to be simple and accessible for everyone, including people who might not be very familiar with technology. They also made sure it would support the required fare discounts for students and persons with disabilities.

During the **Development Phase**, the researchers focused on implementing the plans from the design phase. The main objectives were to build a system for precise fare calculation and incorporate a user feedback mechanism. This phase involved creating the mobile application, integrating geolocation technology, and other essential components, with the aim of developing a system that could calculate tricycle fares accurately while remaining user-friendly and compliant with regulations.

The Iteration Phase involved making updates, enhancements, and refinements to the system based on user feedback. This iterative process was crucial for staying responsive to evolving user needs and preferences. The feedback loop established during the **Deployment Phase** ensured that the system continued to adapt and evolve to meet the changing demands of its users. Insights gathered during this phase informed subsequent testing phases, allowing the team to focus on areas that were critical or problematic in real-world usage.

In the **Testing Phase**, the team addressed the goal of creating a strong and reliable system. Rigorous testing methods were employed to ensure the application worked effectively and satisfied users. The focus was on ensuring the fare calculation system was accurate and adhered to the rules outlined in "RA 11314." This phase also included a user feedback mechanism to continuously improve the system based on user insights.

During the **Deployment Phase**, the team maintained regular communication with users, continually collecting feedback to refine and enhance the system over time. This ongoing dialogue with users was part of a continuous improvement loop, ensuring that user experiences and preferences shaped the system's evolution. Connecting back to the testing phase, this user engagement loop further informed the testing methodologies, ensuring the system remained responsive to user needs and preferences.

Based on the Figure 2 in Appendix E, the diagram outlines a robust and scalable architecture for a ride-sharing platform. Riders initiate ride requests through a mobile app, which communicates with the system's backend. The system employs a sophisticated matching algorithm to connect riders with nearby drivers, taking into account factors such as location, availability, and preferences.

Drivers have the flexibility to accept or decline ride requests based on their availability and desired routes. Their real-time location data is continuously tracked, enabling the system to provide accurate (ETA) estimated time arrival, estimates, and optimize the matching process. To ensure a seamless experience, the system incorporates trip management services that monitor the entire trip lifecycle. These services send timely updates to both riders and drivers, keeping them informed about the ride's progress, estimated arrival time, and any potential changes.

Notifications are delivered through a dedicated notification service, ensuring that users receive timely alerts and updates. The system utilizes Firebase as its primary database, providing a scalable and reliable solution for storing and retrieving user data, ride information, and historical records to handle the increasing load and maintain optimal performance, the architecture employs load balancers that distribute incoming requests across multiple servers. This ensures efficient resource utilization and prevents system bottlenecks. Web servers are responsible for processing requests related to user profiles, trip history, and other essential information.

Accurate location data is crucial for the system's functionality. Location services are integrated to provide precise location coordinates, which are periodically batched and written to a database table for further analysis and processing.

By incorporating these elements, the ride-sharing system ensures scalability, reliability, security, and a user-friendly experience. It is designed to handle a growing number of users and rides while maintaining high performance and availability. The system's architecture also prioritizes data security to protect user information and prevent unauthorized access.

Based on the Figure 3 in Appendix E, The Entity-Relationship Diagram (ERD) consists of various entities represented by rectangles, connected by relationships shown as diamonds. The lines between these entities are labeled to indicate the type of relationship between them.

The process begins with the user logging into the app. After logging in, the commuter can access their dashboard, which displays their past rides and profile information. If the commuter wants to book a ride, they click the "Ride Booking" button, which takes them to a screen where they can enter their pick-up and drop-off locations. Once this information is entered, the commuter clicks the "Find Driver" button. The system then searches for an available driver to take the commuter to their destination. Upon finding a driver, the commuter is shown the driver's name, photo, and tricycle plate number. The commuter can then confirm the booking by clicking the "Book Ride" button.

For the Driver Entity, the diagram includes the process for a driver to sign up on the platform. This involves submitting personal information and uploading necessary documents, such as their driver's license and vehicle registration.

For the Admin Entity, the diagram covers actions like adding or removing vehicles, uploading vehicle documents, and managing vehicle insurance information.

Once the ride is complete, the commuter has the option to rate the driver and leave a comment about the ride. Other features of the app are also included in the diagram, such as the ability for users to send messages to drivers and view their past rides.

This ERD provides a clear and structured overview of the app's user flow, driver interactions, and admin functionalities.

The Entity Relationship Diagram (ERD) also includes elements such as:

1. Notifications
2. Approvals
3. Feedback mechanisms.
4. Reports

Based on the Figure 4 in Appendix E, In the Tuna Trike ride-sharing system, collaboration occurs among three key participant groups: drivers, commuters, and administrators. These entities exchange essential information, including names, email addresses, and payment details, facilitating the functionality of the system. Users can perform various tasks through the system, such as booking rides, where commuters initiate requests by specifying pickup and drop-off locations. Upon receiving these requests, drivers have the option to either accept or reject them.

A crucial aspect of the system is the calculation of the fare, determined based on factors like distance traveled. This information is pertinent for both commuters and drivers to ensure transparency and fairness in the payment process. Additionally, the system estimates the arrival time of the driver, providing commuters with a projected timeframe for their journey.

Feedback plays a significant role in the continuous improvement of the system. Both drivers and commuters can share their experiences and insights, contributing to enhancements in service quality. The system retains account details for drivers and commuters, as well as comprehensive information about each ride, encompassing details such as date, time, and route. The collection of data will be stored in on web- based Database, clarifying this aspect would provide a more comprehensive understanding of how the TunaTrike system manages and safeguards user-related information and ride details. Overall, the system aims to streamline the ride-sharing experience for both drivers and commuters while maintaining transparency, efficiency, and user satisfaction.

Based on the Figure 5 in Appendix E, the diagram outlines the intricate processes involved in a ride-sharing platform. It begins with a commuter initiating a ride request by inputting their personal information and the desired ride details. The system then calculates the fare based on factors such as distance, time, and demand. Once the fare is determined, the commuter is notified, and the system proceeds to assign a suitable driver. The driver receives a notification about the incoming ride request. They have the option to accept or reject the request based on various factors, including their availability, location, and the estimated fare. If the driver accepts, the system assigns them to the commuter, and the ride is scheduled.

The driver then proceeds to provide the ride service. Upon completion, the commuter is prompted to provide feedback on the ride experience, including ratings for the driver's performance and overall satisfaction. This feedback is valuable for the system as it helps to maintain quality standards and improve the overall user experience.

The system processes the commuter's feedback and updates the driver's rating accordingly. This rating system incentivizes drivers to provide excellent service and helps commuters make informed choices when selecting drivers. Additionally, the system collects and analyzes ride data to optimize operations, improve efficiency, and enhance the platform's overall performance.

In summary, the diagram depicts a comprehensive overview of the ride-sharing process, from the initial request to the final feedback. It highlights the key interactions between commuters, drivers, and the system, showcasing the interconnectedness of the various components involved in providing a seamless ride-sharing experience.

The smartphone app has two sections: one for drivers and one for passengers. Drivers receive trip requests, accept rides, and navigate to pick-up locations, while passengers request rides, set destinations, and track drivers. Both apps use encrypted connections to communicate with the server for real-time location updates, matching, and booking. GPS ensures accurate pick-ups and drop-offs, while mobile data or the internet is essential for continuous operation. The server matches passengers with available drivers, and a database stores user details, car information, trip history, and fare data to support app functionality.

overall satisfaction. This feedback is valuable for the system as it helps to maintain quality standards and improve the overall user experience.

The system processes the commuter's feedback and updates the driver's rating accordingly. This rating system incentivizes drivers to provide excellent service and helps commuters make informed choices when selecting drivers. Additionally, the system collects and analyzes ride data to optimize operations, improve efficiency, and enhance the platform's overall performance.

In summary, the diagram depicts a comprehensive overview of the ride-sharing process, from the initial request to the final feedback. It highlights the key interactions between commuters, drivers, and the system, showcasing the interconnectedness of the various components involved in providing a seamless ride-sharing experience.

There are two sections to the smartphone app, one for drivers and one for passengers. Drivers use the app to get trip requests, accept rides, and find their way to pick- up locations; passengers use it to request rides, establish destinations, and track drivers. For real-time location updates, driver- passenger matching, and booking, both apps communicate with the server over encrypted connections. For uninterrupted contact to ensure that everything goes as planned, mobile data or the internet are essential. To locate drivers and passengers on a map and guarantee precise pick- ups and drop-offs, GPS is necessary. The server locates a driver who is available when passengers schedule a ride. The database helps the server handle requests and deliver information to the apps by storing all user information, car information, trip history, and fare data.

There is an admin interface to manage drivers, vehicles, fares, and system settings, monitoring real-time system performance. GPS data is analyzed to pinpoint passenger and driver locations on the map for efficient ride matching. A geolocation API enhances location accuracy. The fare calculation service figures out ride costs based on distance, time, and dynamic pricing. The back- end server is like the brain, handling all core tasks: processing bookings, matching drivers, communicating with apps, calculating fares, storing data, and managing payments.

Technically, it follows a client-server architecture, reacting to events like bookings and location updates, potentially using a microservices architecture for scalability. Scalability is crucial to handle high traffic volumes with minimal downtime. Security measures, like secure APIs and data encryption, protect user information and transactions. The system offers convenient ride booking, real-time tracking, dynamic pricing, and collects data for insights to improve efficiency and provide personalized service.

There is an admin interface to manage drivers, vehicles, fares, and system settings, monitoring real-time system performance. GPS data is analyzed to pinpoint passenger and driver locations on the map for efficient ride matching. A geolocation API enhances location accuracy. The fare calculation service figures out ride costs based on distance, time, and dynamic pricing. The back- end server is like the brain, handling all core tasks: processing bookings, matching drivers, communicating with apps, calculating fares, storing data, and managing payments.

Technically, it follows a client-server architecture, reacting to events like bookings and location updates, potentially using a microservices architecture for scalability. Scalability is crucial to handle high traffic volumes with minimal downtime. Security measures, like secure APIs and data encryption, protect user information and transactions. The system offers convenient ride booking, real-time tracking, dynamic pricing, and collects data for insights to improve efficiency and provide personalized services

RESULTS AND DISCUSSIONS

Demographic Profile

Based on the Table 1 in Appendix F, The Developers conducted a survey and obtained the demographic profile of the users who partake in the beta testing of the under develop Tunatrike: A booking application for tricycle in General Santos City.

According to this table, a somewhat higher proportion of students took the beta test, with male students accounting for the largest group (34%), followed by female students (25%). The non-student category was smaller, consisting of 17% female non-students and 24% male non-students. This demographic variety offers important information about how various user groups used and interacted with the Tunatrike app during its test stage.

A survey of General Santos City drivers and passengers was done to assess the "Tunatrike" booking application's effectiveness and user satisfaction. Important topics included in the poll were accuracy of the fare calculation, user interface, and feedback systems. The results, which are based on survey data, are shown below.

Based on the Table 2 & 3 in Appendix F, survey conducted with both passengers (users of the "TunaTrike" app) and drivers (those providing the service through the app). The survey aimed to gather valuable feedback on various aspects of the app, including fare calculation, ease of use, customer support, communication, and overall user experience.

For passengers, the survey results showed a strong positive reception, with 70% of respondents finding the fare calculation based on geolocation to be "Very Accurate," indicating a high level of trust in the app's pricing mechanism. Additionally, 90% of users reported that booking rides and viewing fare information was "Very Easy" or "Easy," reflecting a smooth and user-friendly experience. The emergency hotline feature also received positive feedback, with 55% of passengers rating it as "Very Satisfied." Importantly, 90% of users suggested that allowing drivers to accept multiple bookings could improve service efficiency, highlighting an area for potential improvement.

Similarly, the survey results from drivers indicated a positive experience with the app. 90% of drivers found the registration process straightforward, and 90% were satisfied with the communication between drivers and passengers. Earnings transparency was also a strong point, with 90% of drivers feeling that the app clearly displayed their earnings. The survey also showed that 100% of drivers found the app effective in handling ride requests, ensuring efficient connections between drivers and passengers.

Only a small minority of respondents pointed out areas for improvement, indicating that both drivers and passengers generally have a positive opinion of the "TunaTrike" app. In order to better understand user needs and direct future updates to improve the functionality and speed of the app, this input is extremely helpful.

CONCLUSION

This study set out to create Tunatrike, a mobile booking application designed specifically for tricycles in General Santos City. By integrating geolocation technology for fare calculations and ensuring compliance with local transportation regulations, the app aims to enhance the riding experience for both passengers and drivers. According to the survey results, the fare calculation system has proven to be largely accurate, with most passengers expressing satisfaction with its transparency and reliability. Users also found the app easy to navigate, and while many appreciated the effectiveness of the feedback mechanism, some pointed out that it could use a bit of improvement. Overall, both passengers and drivers seem to enjoy using the app, particularly for its ability to efficiently connect them. However, many passengers suggested that allowing drivers to accept multiple bookings would improve efficiency and make rides more accessible.

From the conclusions drawn in this study, It's clear that the fare calculation system based on geolocation technology is effective, as most passengers found it fair and transparent. This demonstrates that leveraging advanced technology for fare computation has resonated well with both passengers and drivers. Furthermore, the app's usability received high praise, indicating that it provides a smooth experience for its users. Still, there's room for improvement in the feedback mechanism, which should be refined to facilitate better communication and faster issue resolution. Many users expressed interest in a ride-sharing feature, specifically the ability for drivers to accept multiple bookings, which could enhance efficiency and availability.

With these findings in mind, several recommendations emerge. First, the team behind Tunatrike should continue to refine the geolocation-based fare system to ensure it consistently delivers accurate fare calculations across various routes and conditions. Second, improving the feedback mechanism will be essential to allow users to report issues and share suggestions more effectively.

In the case of a problem or emergency during a ride, the Tunatrike app's emergency mode is intended to guarantee passenger safety by facilitating direct communication with local

authorities. Based on the study, the emergency module has been integrated with PNP-General Santos (PNP-Gensan) to streamline response times and enhance overall safety for both passengers and drivers.

Lastly, introducing a multiple booking feature for drivers could significantly improve efficiency by enabling them to serve more passengers on shared routes, benefiting both drivers and riders. Lastly, expanding the app's compatibility to include iOS devices would increase accessibility, allowing more users in the city to take advantage of this innovative service.

REFERENCES

D. Marra, H. Becker, K. W. Axhausen and F. Corman., (2019) Developing a passive GPS tracking system to study long-term travel behavior, Transportation Research Part C: Emerging Technologies, pp. 348-368, IEEE.

D. Davis., (1999) Encouraging Commuter Student Connectivity," Business Communication Quarterly, vol. 62, no. 2, pp. 74-78, IEEE.

Ferro, P. S., Behrens, R., & Wilkinson, P. (2013). Hybrid urban transport systems in developing countries: Portents and prospects. Research in Transportation Economics, 39(1), 121-132

G. H. Costa and F. Baldo., (2015) Generation of road maps from trajectories collected with smartphone – A method based on Genetic Algorithm, Applied Soft Computing, vol. 37,pp. 799-808, IEEE.

G. Zhang, F. Wei, N. Jia, S. Ma and Y. Wu., (2019) Information adoption in commuters' route choice in the context of social interactions, Transportation Research Part A: Policy and Practice, vol. 130, pp. 300-316, IEEE.

Ghose, A., Kwon, H. E., Lee, D., & Oh, W. (2019). Seizing the commuting moment: Contextual targeting based on mobile transportation apps. Information Systems Research, 30(1), 154-174.

Goldberg, D. (2021). The New Geography of Fare-Free Public Transport: Examining the Suspension of Fares in the United States During COVID-19. Brussels: Masters in urban studies (ULB/VUB).

https://fastercapital.com/content/Transparent-Price-Band--Promoting-Trust-and-Fairness- in-Pricing.html

J. J. Newbold., (2015) Lifestyle Challenges for Commuter Students, New Directions for Student Services, vol. 2015, no. 150, pp. 79-86, IEEE. https://lawphil.net/statutes/repacts/ra2019/ra_11314_2019.html

https://utilitiesone.com/bridging-the-gap-how-geolocation-connects-people-to-public-transport

https://www.ucl.ac.uk/bartlett/development/publications/2021/jan/urban-mobility-and-social-equity-latin-america-evidence-concepts-methods

J. P. Pullen., (2015) These Are the 10 Best Apps for People Who Commute, Time.com, p. 1, IEEE.

J. Vijayan., (2016) Google Enhances Maps App for Android, iOS and Desktop Users, eWeek, pp. 5-5, IEEE.

Kokaz, K., & Rogers, P. (2002). Urban transportation planning for air quality management: Case study in Delhi, India, of role of social and economic costs in welfare maximization of mobility choice. Transportation research record, 1817(1), 42-49.

L. Wang and D. Y. Ju., (2015) Concurrent Use of an In-Vehicle Navigation System and a Smartphone Navigation Application, Social Behavior & Personality: an international journal, vol. 43, no. 10, pp. 1629-1639, IEEE.

Liu, X., & Yu, L. (2009). Development of geolocation data compression for transportation target identification. Transportation research record, 2105(1), 71-82.

M. Ezzat, M. Sakr, R. Elgohary and M. E. Khalifa., (2018) Building road segments and detecting turns from GPS tracks, Journal of Computational Science, vol. 29, pp. 81-93,

Spieler, C. (2020). Racism has shaped public transit, and it's riddled with inequities. UrbanEdge,KIfU Research, Editor.

The case of "Tricycles" and "Habal-habal in Davao City Philippines https://www.sciencedirect.com/science/article/pii/

The Philippine Star April 25, 2023, https://kpmg.com/ph/en/home/insights/2023/04/para-po.html

Wilson, C. (2003). Computer attack and cyber terrorism: vulnerabilities and policy issues for congress. Focus on Terrorism, 9(1), 1-42..

Yuval Cohen1, Stephann Makri, Steffen Reymann, Ioannis Kaparias June 2017, https://eprints.soton.ac.uk/416826/1/4_49.pdf

Balba, N., Zeder, N. M. F. R. P., Drilon, S. J. F. F., & Fallaria, J. F. (2019). ISAKAY: Android Based Booking System for Tri-Bike Operators and Drivers Association with Cloud-Based Data Analytics. *Volume-8 Issue-10*, 1265-1269.

de Jesus, M. J. M., Gutierrez, A., & Muñoz, M. (2018). An Analysis of the Factors Influencing Commuter Decisions in Using the Angkas: Motorcycle Ride Booking Transportation Service. *ISSN No. 2362-7832*, 46.

Gooi Sai Weng, Suhaiza Zailani, Mohammad Iranmanesh, Sunghyup Sean Hyun. () Mobile taxi booking application service's continuance usage intention by users

Chavez, E., Hernandez, A., Niguidula, J., & Caballero, J. (2017). Ride express: An online seat reservation and vehicle location tracker with estimated time of arrival. In *Proceedings of the 2017 Cebu International Conference on Computers, Electrical and Electronics Engineering* (pp. 148-152).

Detera, E. I. Z., & Hanbal, I. F. (2024). Development of a Utility Vehicle Express System Framework Using Design Thinking Process for the Province of Albay.

Muhammad Ridwan Farizqi, Anna Dina Kalifia, Rr. Hajar Puji Sejati (March 3, 2024) Design of an Android-Based Motorcycle Service Booking Application.

Rathee, D. S., Rathee, K., Singh, R. S., & Gemeda, K. A. (2024, January). Smart Solar E-Tricycle Design and Fabrication for Especially Abled Persons. In *International Conference on Advances in Distributed Computing and Machine Learning* (pp. 305- 316). Singapore: Springer Nature Singapore.

RA 10754: https://ncda.gov.ph/disability-laws/implementing-rules-and-regulations-irr/irr- of-ra-10754-an-act-expanding-the-benefits-and-privileges-of-persons-with-disability- pwd/

Zailani, Nurul Farah Izzah, et al. "Factors influencing consumer perception on ride-sharing application services: a case study of grab car." Psychology and Education 57.9 (2020): 2490-2495.

Zahib, Maulidatul Aulia, Veronikha Effendy, and Eko Darwiyanto. "Designing User Experience for Improving Mobile Application Accessibility Online Transport Booking for Visually-Impaired User With User-Centered Design: A Case Study of Grab." Journal of Information System Research (JOSH) 4.1 (2022): 42-52.

Balba, N., et al. "ISAKAY: Android Based Booking System for Tri-Bike Operators and Drivers Association with Cloud-Based Data Analytics." Volume-8 Issue-10 (2019): 1265- 1269. Edwin Fernandez "Gensan trike drivers urged to obey new fare matrix". (August, 2022). https://www.pna.gov.ph/index.php/articles/1182073

Mark Y. TACDERASa , Anne Clarice L. NGb , Niki Jon Y. TOLENTINOc , Noriel Christopher TIGLAOd , and Chrysologus E. HERRERAe "Examining the Implementation of the Public Utility Vehicle Modernization Program (PUVMP) in General Santos City, Philippines:An IndustryPerspective". (December, 2021). https://www.ncts.upd.edu.ph/tssp/wp-content/uploads/2022/02/PTJ-V4-N1- TSSP-2021- Vol4-No1-04-Tacderas-et-al.pdf

APPENDICES

APPENDIX A. HARDWARE/SOFTWARE REQUIREMENTS ADMIN

Table 4

Hardware Requirements Admin, Commuters and Tricycle Drivers

Hardware Requirements	Minimum Requirements
Computer	*Cpu: AMD Ryzen 3 3200g* *Ram: 8 GB DDR4* *Storage: SSD 256 GB* *Mouse: A4tech* *Keyboard: A4tech* *Monitor: 60hz*
Mobile Phone	At least 4gb of ram *At least 16GB of free internal storage* *Android 11*

Note. Specifications may vary based on specific use cases.

Table 5

Software Requirements

Android Studio	Android Studio is the integrated developmen environment for Google's Android platform. Versions o Android Studio are compatible with some Apple Windows and Linux operating systems. and Android Studio provides a unified environment where you can build apps for Android phones, tablets, Android Wear Android TV, and Android Auto. Structured code modules allow you to divide your project into units o functionality that you can independently build, test, and debug.
Flutter	Flutter is an open-source UI software development ki (SDK) created by Google, designed for building natively compiled applications for mobile (iOS, Android), web and desktop (Windows, macOS, Linux) from a single codebase. It features a "hot reload" function that allows developers to see changes in real-time without restarting the application, thereby enhancing the developmen process. Flutter is built around a comprehensive set of widgets, which are highly customizable and can be composed to create complex user interfaces. Its cross platform nature enables developers to write code once and deploy it across multiple platforms,

	ensuring consistency while reducing development time and costs. Performance- wise, Flutter apps are compiled directly to native ARM code, providing performance levels close to native applications. Additionally, Flutter boasts a strong and growing community, supported by a plethora o packages and plugins that extend its functionality.
Dart	Dart, the programming language used to write Flutter applications, is also developed by Google. It is optimized for client development, including web and mobile apps and offers a declarative and reactive programming model Dart supports both Ahead-Of-Time (AOT) and Just-In Time (JIT) compilation; AOT ensures fast startup times and predictable performance for Flutter apps, while JIT facilitates development with features like hot reload. Dart's syntax is clean and easy to understand, making it accessible to developers familiar with languages such as Java JavaScript, and C#. As a statically typed language, Dar helps catch errors early in the development cycle Furthermore, it includes a comprehensive standard library that supports asynchronous programming, collections math, and more.

Firebase	Firebase is a comprehensive app development platform provided by Google that offers a wide array of tools and services designed to help developers build, improve, and grow their applications. Originally developed by Firebase Inc. and acquired by Google in 2014, it has since become an integral part of Google's cloud services. Firebase is particularly popular among mobile and web developers fo its ease of use and rich feature set.

Note. This software was used during development.

APPENDIX B. GANTT CHART

GanttChart of Activities

	FEBRUARY 2024				MARCH 2024				APRIL 2024				MAY 2024				JUNE 2024				JULY 2024				AUGUST 2024				SEPTEMBER 2024				OCTOBER 2024				NOVEMBER 2024				DECEMBER 2024			
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
Formation of Group																																												
Formation of ResearchTitle																																												
Selection of adviser																																											
Title Defense																																											
Documentation (Chapters I, II, III)																																											
Writing of Formal Letter for LGU Transportation Office																																											
Creating ERD Diagram																																											
System Wireframe																																											
Mock Defense with Adviser and Panelist																																											
Revisionof Documents after Mock Defense																																											
Adjusting the endorsement form to be sent to the Thesis Coordinator																																											
Database Management																																											
Chapter3: Methodology																																											
Methodsof Research&Thesis 1 Final Defense																																											
Thesis 0 Final Defense																																											
System Development																																											
Database Diagrams																																											
ChapterIV: Methodology, Results and Documents																																											
Chapter V: Conclusions and Recommendation																																											
Thesis II Final Defense																																											

Legend

Blue - Completed Activities

Yellow - On Going

APPENDIX C. ACTUAL THESIS EXPENSES

THESIS EXPENSES

Quantity	Specifics	Approximate Cost
3 reams	Bond Paper	690
2 months	Commuting Fare and Gasoline	2,485
2 months	Internet Connection	2,798
1,800	Google cloud console	1,000
2pcs	Brother DT60 Ink	800
	Grammarian	**1,700**
	SMS Verification	**600**
	Domain	**75**
	Hostinger	**295**
Total		**10,443**

Table 6. Thesis Expenses

Prepared by:

Celso Onnele T. Arancon **Ariel M. Guerra**

Adrian D. Macaculop **Raffy G. Simon**

Alvin A. Tampus Jr

Noted by:

Celso Arancon **Ariel Guerra Sr.**

Arthur Macaculop **Rufino Simon**

Ma. Shena A. Tampus.

Noted by:

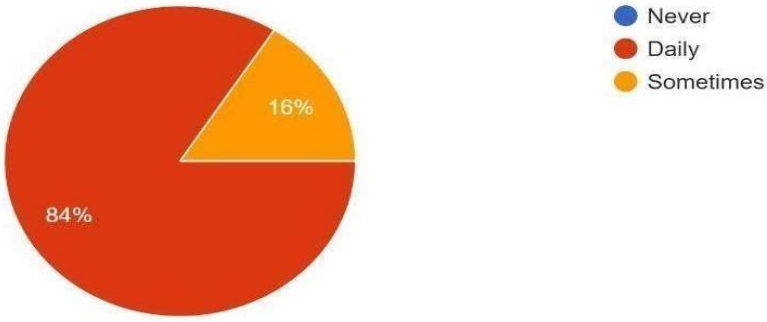
Julius Paolo B. Palma
Thesis Adviser

Allyssa Kaye A. Parcon
Thesis Coordinator

APPENDIX D. DATA INTERPRETATION OF THE SURVEY

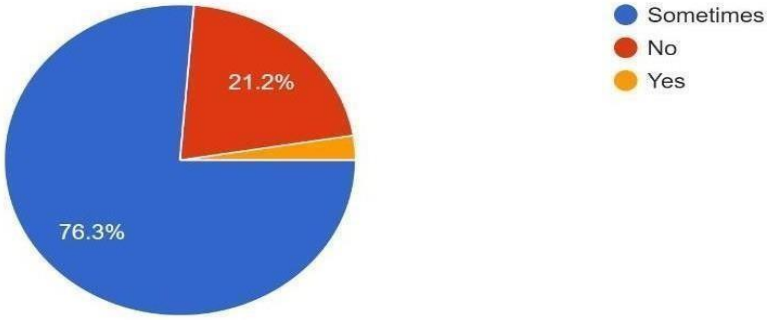
In a typical day how often you ride a tricycle?

119 responses



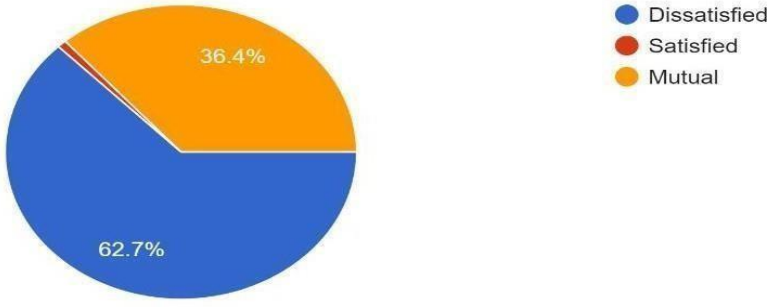
Do you feel safe when riding a tricycle during day/night?

118 responses



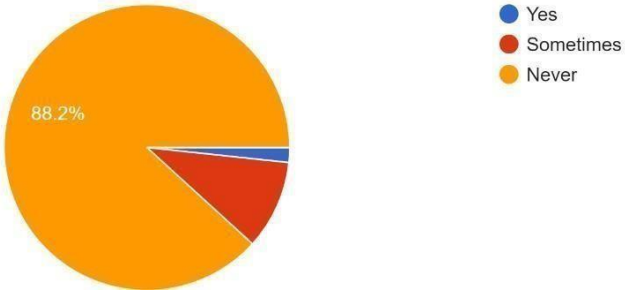
What do you feel about price of the tricycle fare in GenSan?

118 responses



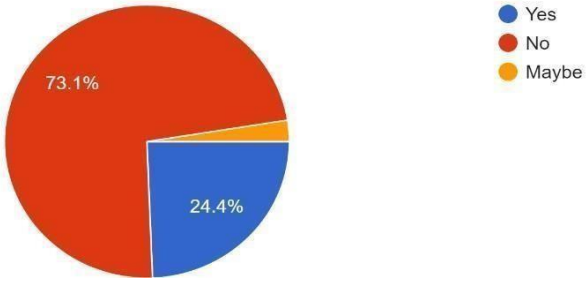
Are there discounts for seniors, people with disabilities, and students who ride tricycles?

119 responses



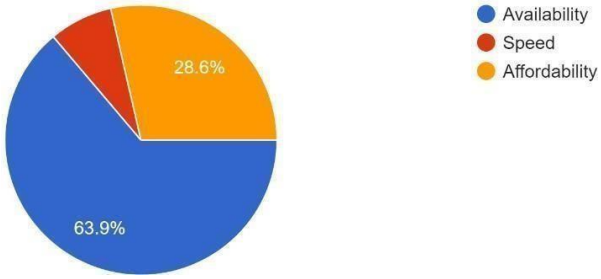
What would you think if there was a tricycle booking app that offered senior, PWD, and student discounts?

119 responses



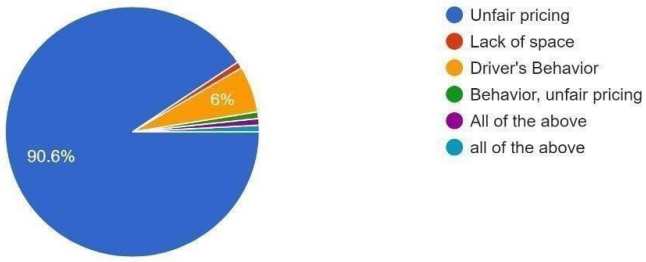
What is your main reason for choosing a tricycle for commuting?

119 responses



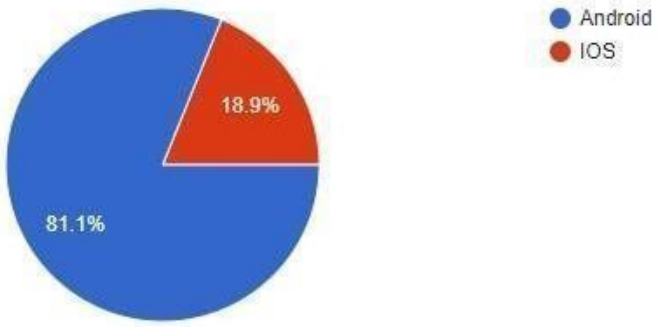
What are the common problems when riding a tricycle?

117 responses



What mobile device do you use?

90 responses



APPENDIX E. LIST OF FIGURES

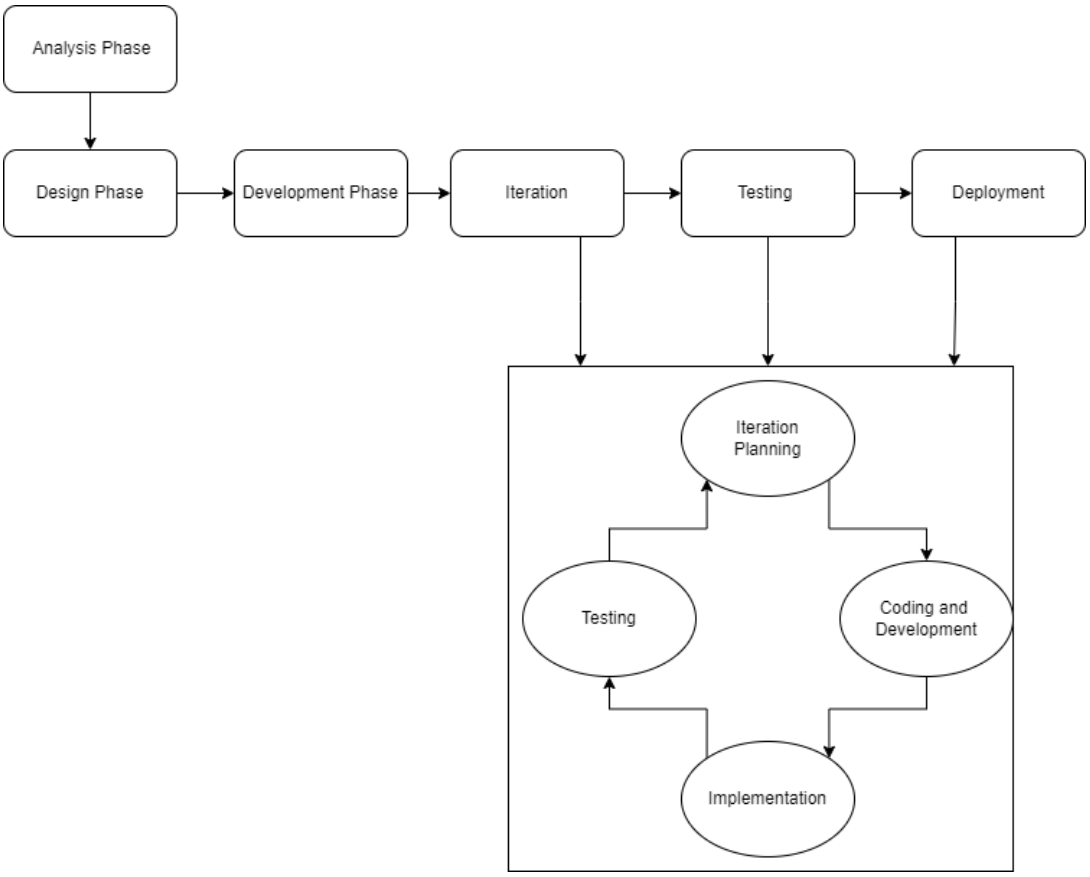


Figure 1. Agile: Scrum framework

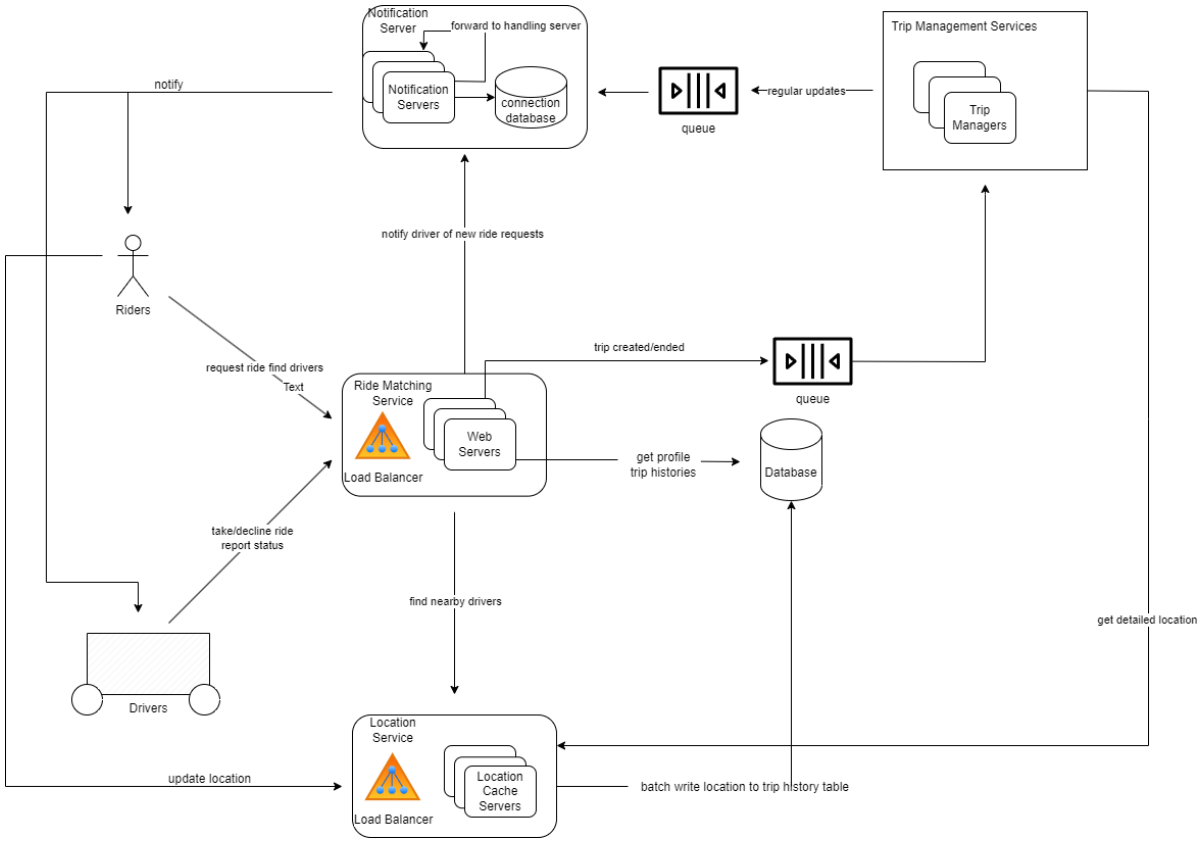


Figure 2. System architecture design

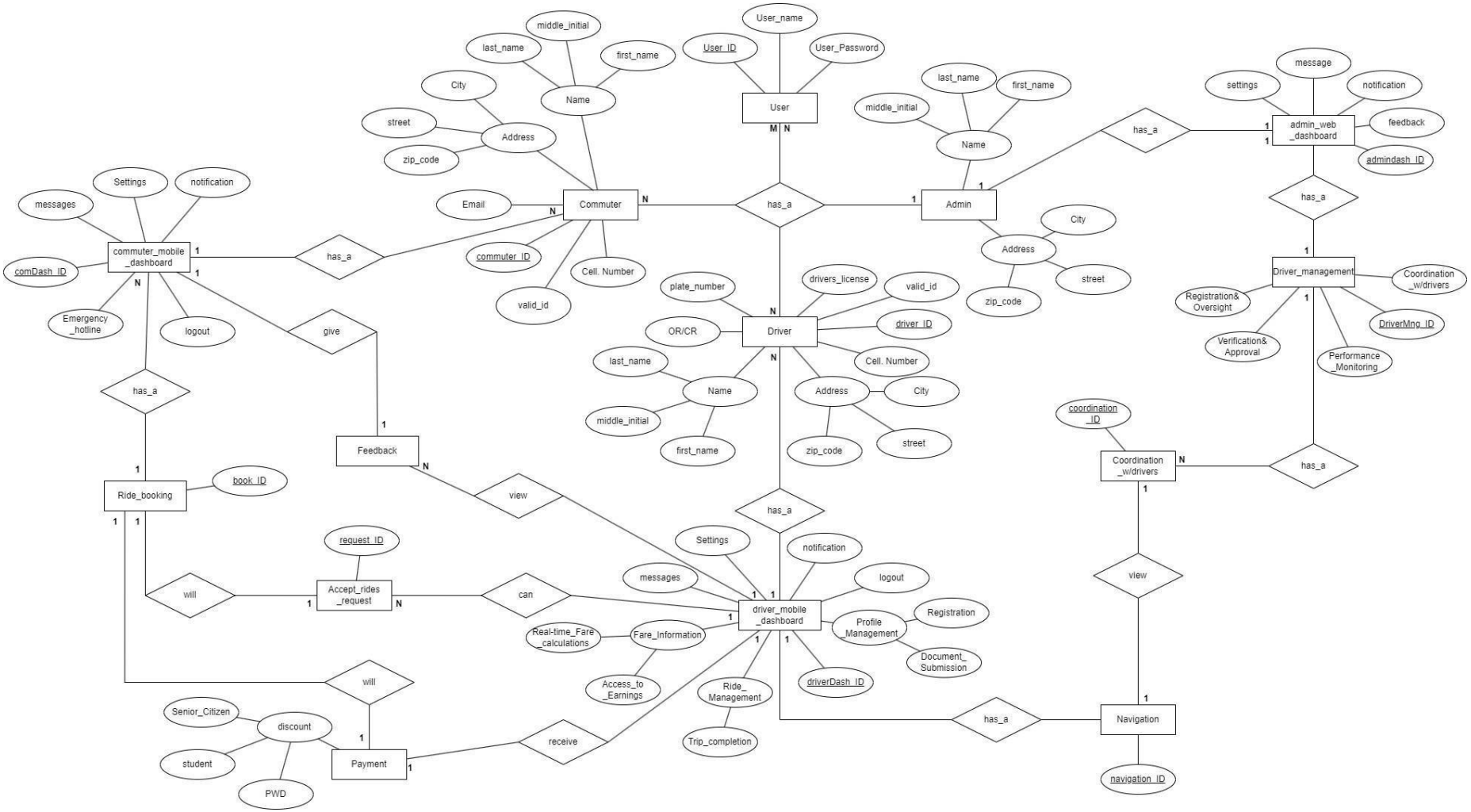


Figure 3. Entity-Relationship Diagram

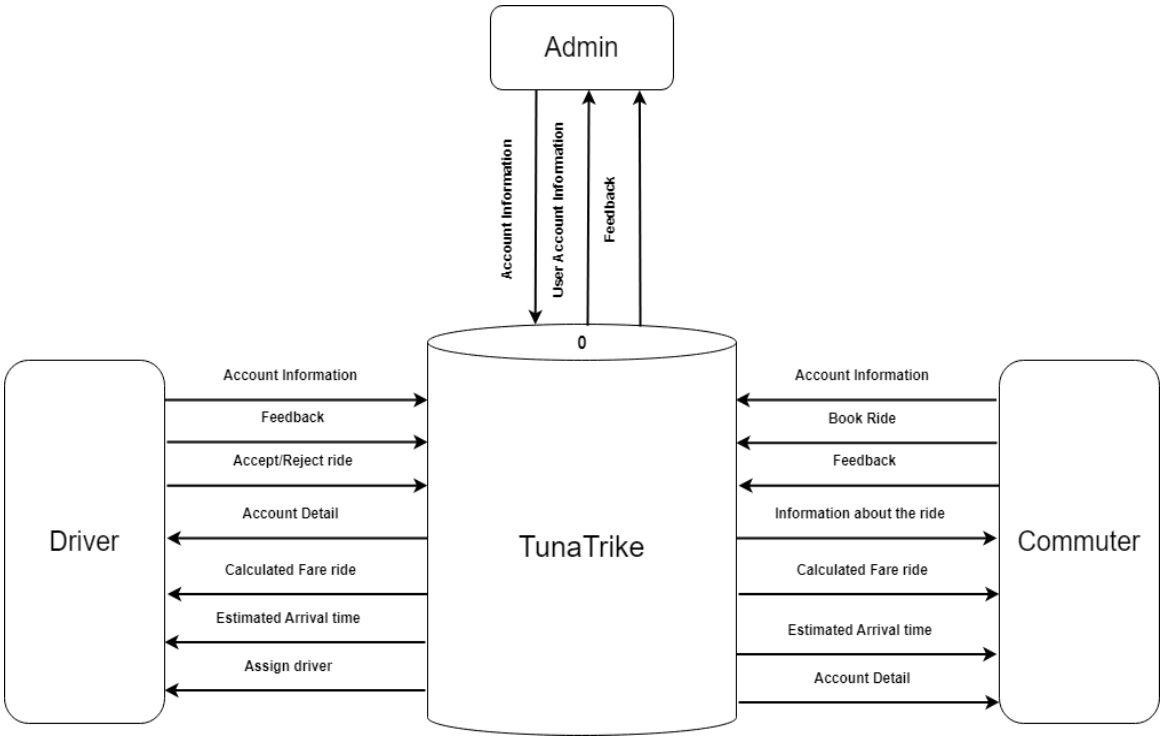


Figure 4. Context Flow Diagram

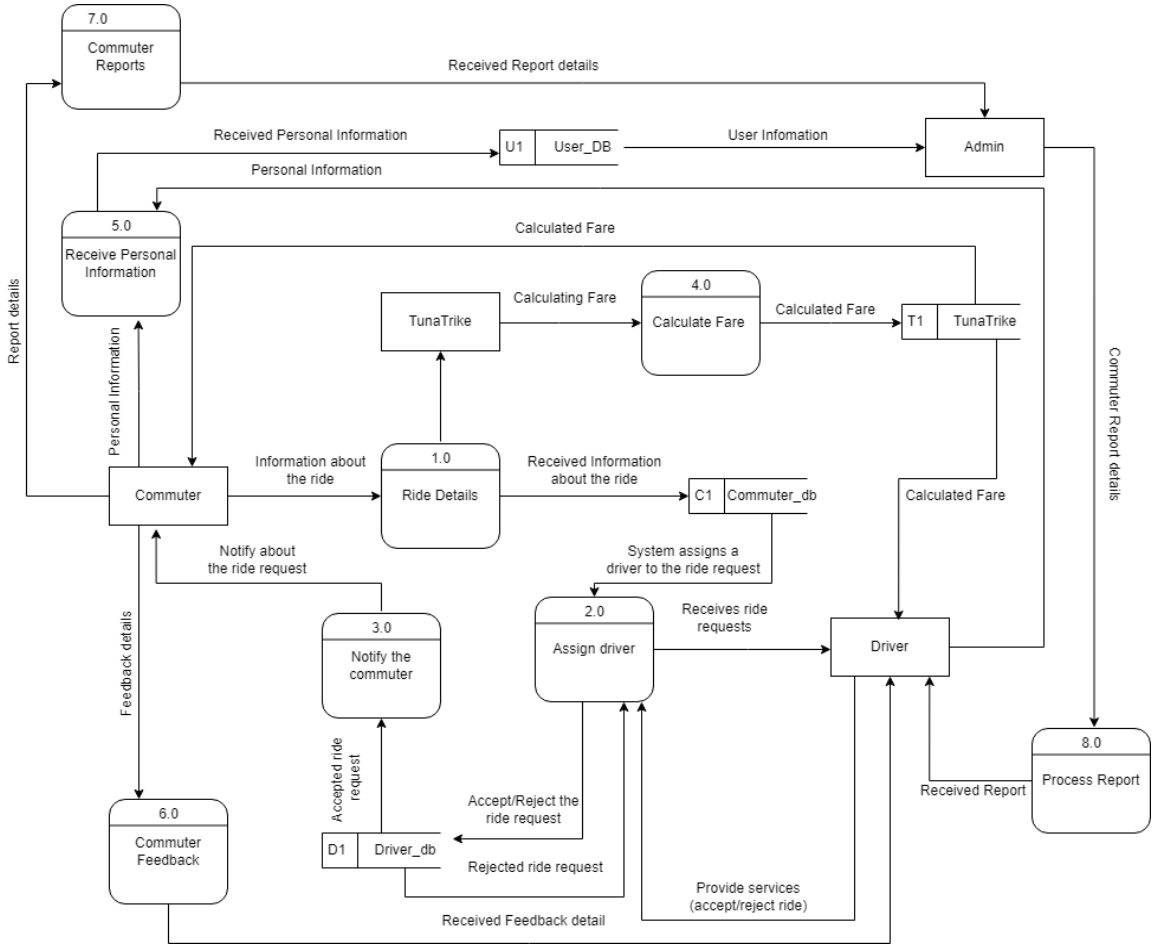


Figure 5. Diagram level 0

APPENDIX F. LIST OF TABLES

Table 1

Demographic profile

Indicators Gender	Respondents	Percentage
Male (student)	40	34%
Female (student)	30	25%
Male (non-student)	28	24%
Female (non-student)	20	17%
Student	70	59%
Non-Student	48	41%
Total	118	100%

Note. This table shows the demographic profile of respondents by gender and student status, with percentages based on a total of 118 respondents.

Table 2

User App (Passengers) Survey Results

Survey Questions	Response Options	Percentage
1. How accurate do you find the fare calculation based on geolocation technology?		
	Very Accurate	70%
	Accurate	25%
	Inaccurate	5%
2. How easy is it to use the TunaTrike mobile application for booking rides and viewing fare information?		
	Very Easy	60%
	Easy	30%
	Difficult	10%
3. How effective is the feedback mechanism in resolving issues or reporting experiences?		
	Verry Effective	55%
	Effective	35%
	Ineffective	10%
4. How would you rate your overall experience using the TunaTrike application?		
	Excellent	55%
	Good	35%
	Average	10%
	Poor	0%
5. How satisfied are you with the availability of tricycles and the driver's response time to ride requests		
	Very satisfied	40%
	Satisfied	60%
	Dissatisfied	0%
6. How satisfied are you with the emergency hotline feature within the application		
	Very satisfied	55%
	Satisfied	45%
	Dissatisfied	0%
7. Do you think the system should allow drivers to accept multiple bookings at the same time?		
	Yes	90%
	No	10%
8. How satisfied are you overall with the Tunatrike app and its services?		
	Very satisfied	55%
	Satisfied	45%
	Dissatisfied	0%

Note. This table summarizes passenger survey results, showing percentages for responses to questions about the TunaTrike app's usability, satisfaction, and features

Table 3

Driver App (Drivers) Survey Results

Survey Questions	Response Options	Percentage
1. How easy was it to register and verify on the platform?	Very Easy	50%
	Easy	40%
	Difficult	10%
2. How effective is the communication between drivers and passengers for ride coordination and issue reporting?	Verry Effective	60%
	Effective	30%
	Ineffective	10%
3. How clear and transparent is the app in showing your earnings from completed rides?	Verry Clear	65%
	Clear	25%
	Unclear	10%
4. How efficient is the app in receiving and accepting ride requests?	Verry Effective	65%
	Effective	35%
	Ineffective	0%
5. How satisfied are you with the availability of tricycles and the driver's response time to ride requests	Very satisfied	40%
	Satisfied	60%
	Dissatisfied	0%
6. How would you rate your overall experience using the Tunatrike app?	Excellent	50%
	Good	45%
	Average	5%
	Poor	0%

Note. This table presents driver survey results, highlighting percentages for responses about the TunaTrike app's efficiency, communication, and overall satisfaction.

APPENDIX G. FORMAL LETTER

STI College
General Santos

January 8, 2024

Hon. Richard L. Atendido
City Councilor/ Author
Sangguniang Panlungsod
General Santos City

Dear Mr. Atendido:

Good day!

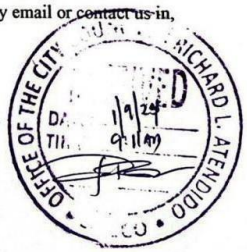
We are third-year students in the Bachelor of Science in Computer Science program at STI College, and we are writing to you with utmost respect and admiration for your work helping the tricycle drivers in General Santos City. Currently, we are working on an academic research project within the scope of the City Transportation fare meter app called "Tuna Trike ". and we are writing to seek approval for a proposed city transportation fare structure in General Santos City.

To improve accessibility, affordability, and efficiency for the locals, our thesis will analyze General Santos City's current transportation system and make recommendations for improvements. We plan to conduct surveys and compile information on the present transit fare system as part of our research. The intended modifications seek to provide a more equitable and sustainable public transport system while also attending to the requirements of the community.

For the implementation of the suggested modifications to city transport fares, I sincerely ask the Sangguniang Panlungsod to accept and approve it. The study is in line with the city's objective of ongoing advancement, and its findings may have a favourable impact on locals' quality of life.

We sincerely appreciate your consideration of this request and understand the demands placed on your esteemed organization. We are enthusiastic about the possibility of collaborating with you on this meaningful research endeavor. We are confident that our findings will not only benefit our academic pursuits but also contribute positively to the aspiring teachers who seek guidance from your esteemed institution.

Thank you for considering my request. I am looking forward to the opportunity to contribute to the betterment of General Santos City through this research initiative. You may email or contact us in, adrian.macaculop@gmail.com or 09488846085.



STI College
General Santos

Thank you for kindly considering our request.

Sincerely,

Ariel M. Guerra Jr.
Student

Raffy G. Simon
Student

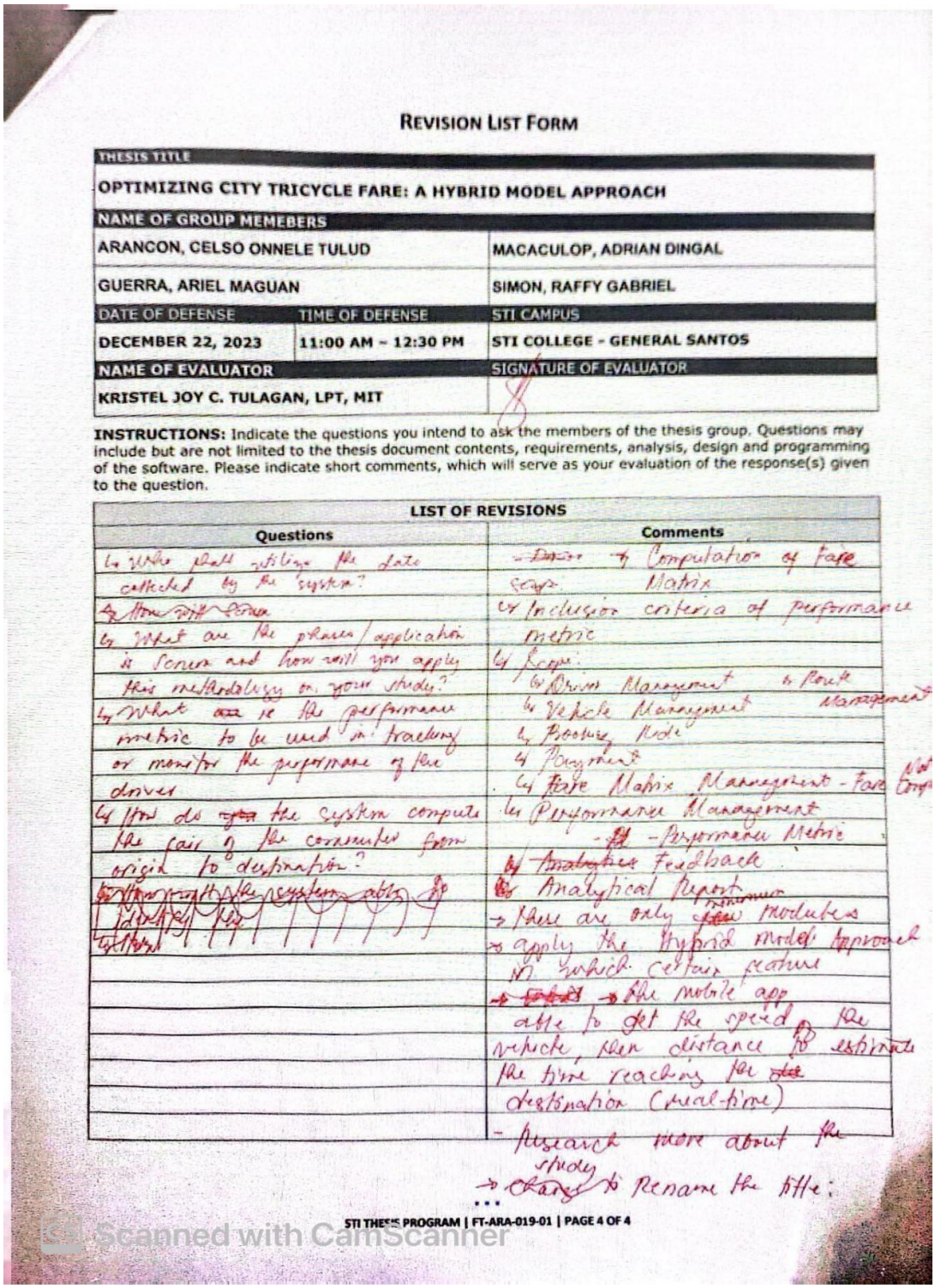
Adrian C. Macaculop
Student

Celso Omnele T. Arancon
Student

Approved By:

Darwin T. Berja
Thesis Adviser

APPENDIX H. REVISION LIST



REVISION LIST FORM

THESIS TITLE		
OPTIMIZING CITY TRICYCLE FARE: A HYBRID MODEL APPROACH		
NAME OF GROUP MEMEBERS		
ARANCON, CELSO ONNELE TULUD		MACACULOP, ADRIAN DINGAL
GUERRA, ARIEL MAGUAN		SIMON, RAFFY GABRIEL
DATE OF DEFENSE	TIME OF DEFENSE	STI CAMPUS
DECEMBER 22, 2023	11:00 AM – 12:30 PM	STI COLLEGE - GENERAL SANTOS
NAME OF EVALUATOR		SIGNATURE OF EVALUATOR
KRISTEL JOY C. TULAGAN, LPT, MIT		

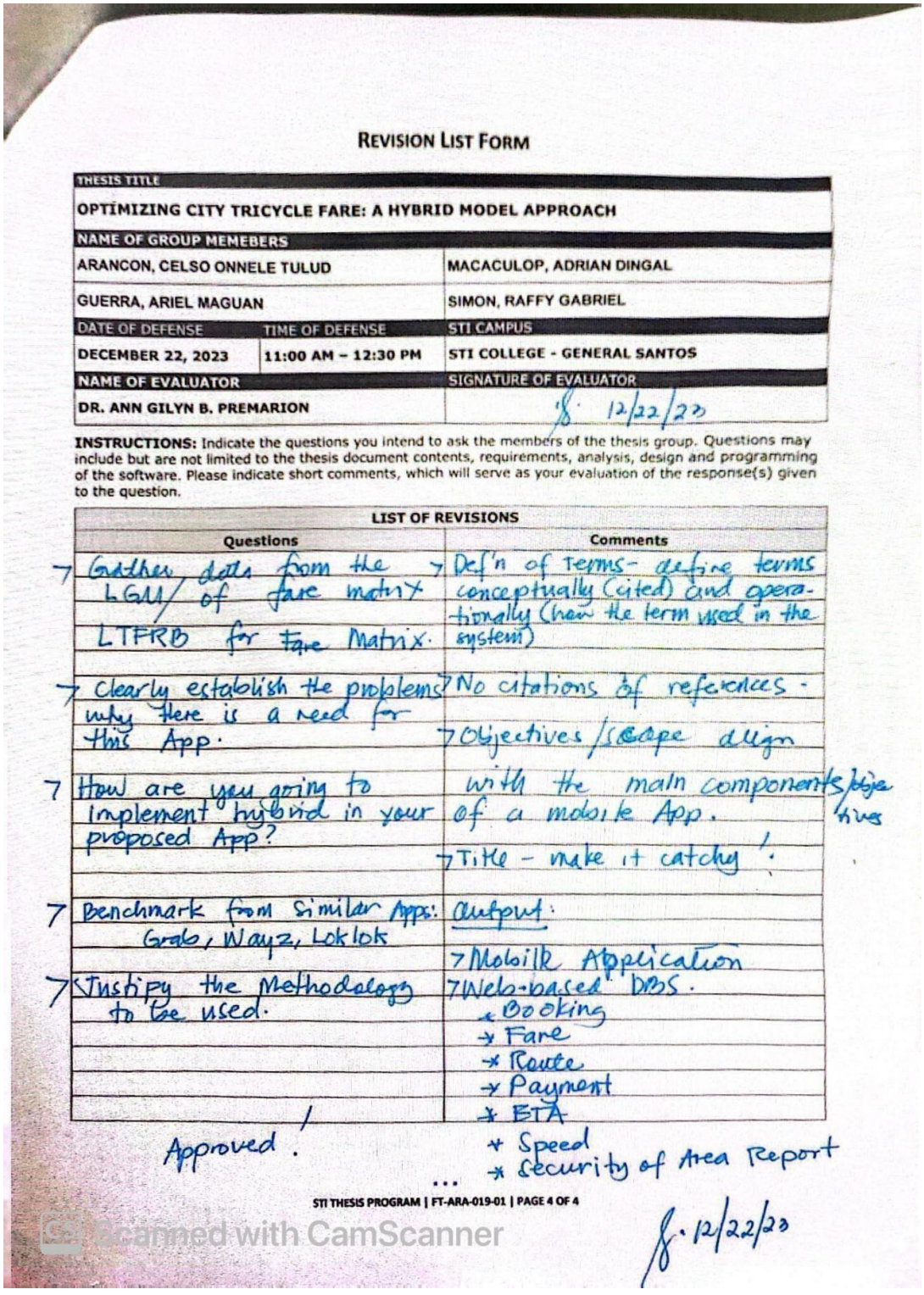
INSTRUCTIONS: Indicate the questions you intend to ask the members of the thesis group. Questions may include but are not limited to the thesis document contents, requirements, analysis, design and programming of the software. Please indicate short comments, which will serve as your evaluation of the response(s) given to the question.

LIST OF REVISIONS	
Questions	Comments
1. Where shall utilize the data collected by the system?	- Computation of fare - Matrix
2. How will Fares	2. Inclusion criteria of performance metric
3. What are the phases/application in Fares and how will you apply this methodology on your study?	3. Scope: a. Driver Management & Route Management b. Vehicle Management c. Booking Ride d. Payment
4. What are the performance metric to be used in tracking or monitor the performance of the driver	e. Fare Matrix Management - Fare Computation f. Performance Management - Performance Metric
5. How do the system compute the fair of the commuter from origin to destination?	g. Analytical Report → There are only modules to apply the Hybrid model approach in which certain feature → the mobile app able to get the speed of the vehicle, then distance to estimate the time reaching the destination (real-time)
6. How will the system ability to	- Research more about the study → Rename the title:

STI THESIS PROGRAM | FT-ARA-019-01 | PAGE 4 OF 4

Scanned with CamScanner

Title Defense



REVISION LIST FORM

THESIS TITLE		
OPTIMIZING CITY TRICYCLE FARE: A HYBRID MODEL APPROACH		
NAME OF GROUP MEMEBERS		
ARANCON, CELSO ONNELE TULUD		MACACULOP, ADRIAN DINGAL
GUERRA, ARIEL MAGUAN		SIMON, RAFFY GABRIEL
DATE OF DEFENSE	TIME OF DEFENSE	STI CAMPUS
DECEMBER 22, 2023	11:00 AM – 12:30 PM	STI COLLEGE - GENERAL SANTOS
NAME OF EVALUATOR		SIGNATURE OF EVALUATOR
DR. ANN GILYN B. PREMARION		12/22/23

INSTRUCTIONS: Indicate the questions you intend to ask the members of the thesis group. Questions may include but are not limited to the thesis document contents, requirements, analysis, design and programming of the software. Please indicate short comments, which will serve as your evaluation of the response(s) given to the question.

LIST OF REVISIONS	
Questions	Comments
7 Gather data from the LGU/ of fare matrix LTFRB for fare Matrix.	7 Def'n of terms - define terms conceptually (cited) and operationally (how the term used in the system)
7 Clearly establish the problems why there is a need for this App.	7 No citations of references.
7 How are you going to implement hybrid in your proposed App?	7 Objectives/scope align with the main components/objectives of a mobile App.
	7 Title - make it catchy!
7 Benchmark from Similar Apps: Grab, Wayz, Loklok	Output:
7 Justify the Methodology to be used.	7 Mobile Application 7 Web-based DBS. * Booking * Fare * Route * Payment * ETA * Speed * Security of Area Report

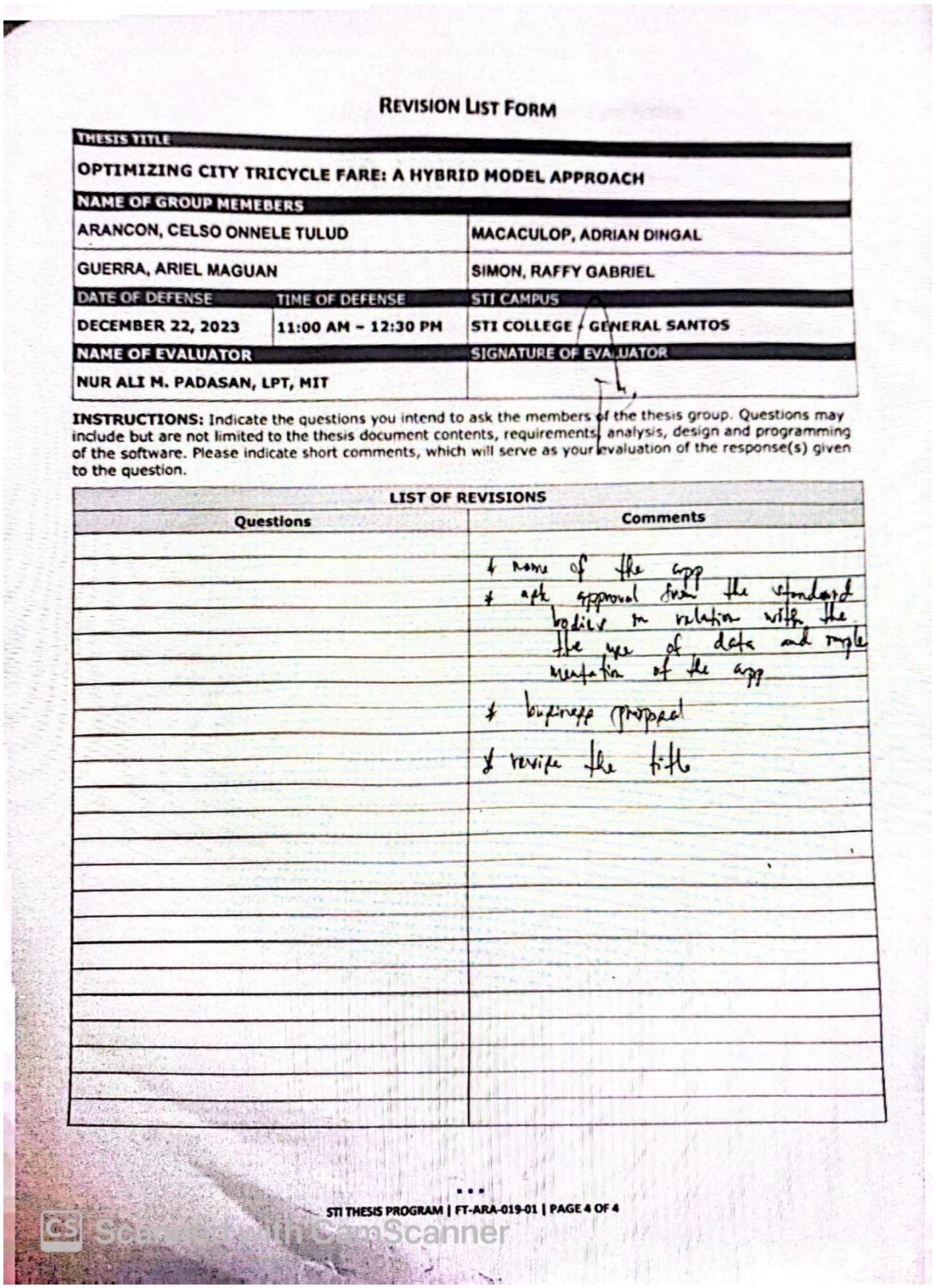
Approved.

STI THESIS PROGRAM | FT-ARA-019-01 | PAGE 4 OF 4

Scanned with CamScanner

12/22/23

Title Defense



REVISION LIST FORM

THESIS TITLE		
OPTIMIZING CITY TRICYCLE FARE: A HYBRID MODEL APPROACH		
NAME OF GROUP MEMEBERS		
ARANCON, CELSO ONNELE TULUD		MACACULOP, ADRIAN DINGAL
GUERRA, ARIEL MAGUAN		SIMON, RAFFY GABRIEL
DATE OF DEFENSE	TIME OF DEFENSE	STI CAMPUS
DECEMBER 22, 2023	11:00 AM – 12:30 PM	STI COLLEGE - GENERAL SANTOS
NAME OF EVALUATOR		SIGNATURE OF EVALUATOR
NUR ALI M. PADASAN, LPT, MIT		

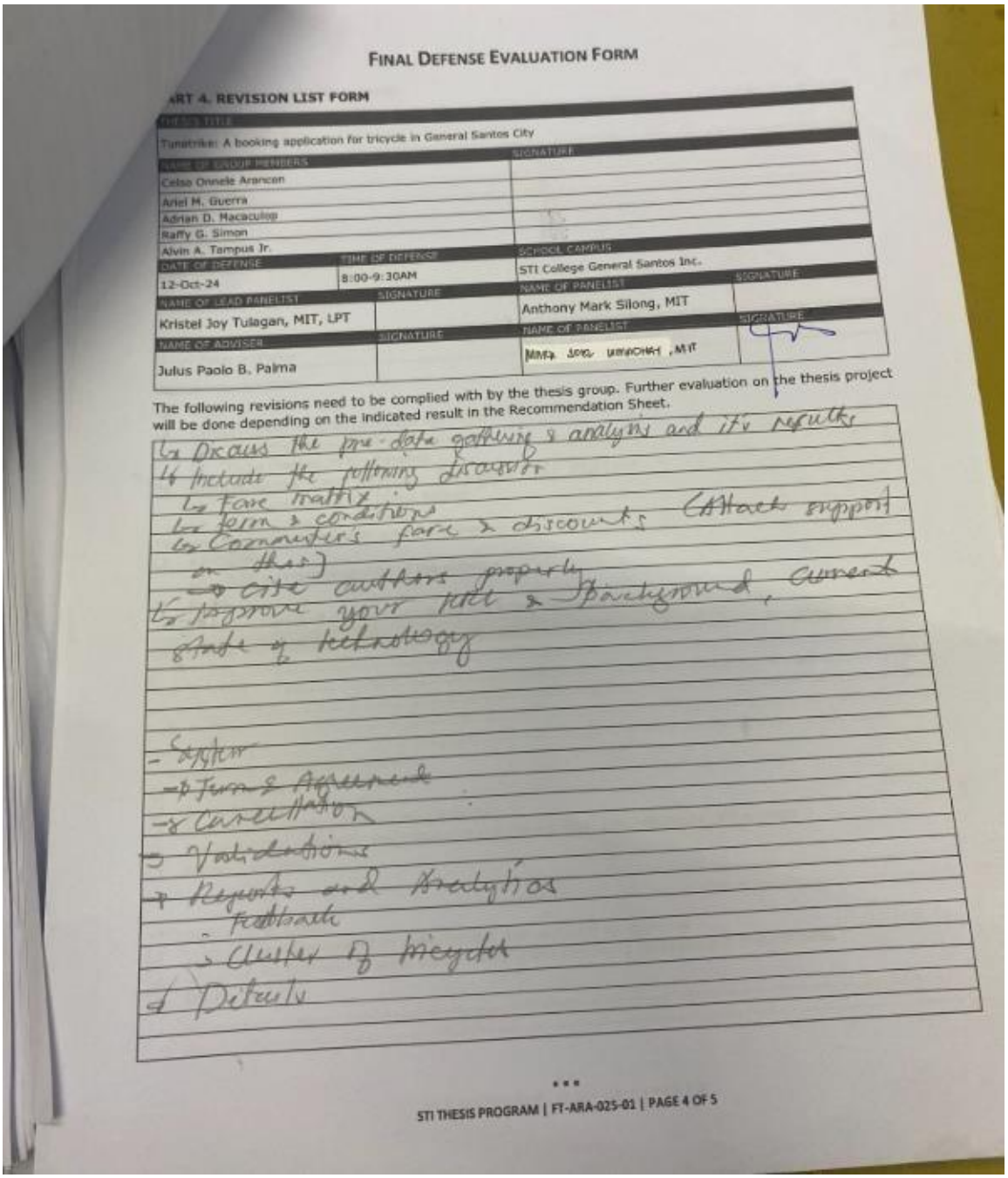
INSTRUCTIONS: Indicate the questions you intend to ask the members of the thesis group. Questions may include but are not limited to the thesis document contents, requirements, analysis, design and programming of the software. Please indicate short comments, which will serve as your evaluation of the response(s) given to the question.

LIST OF REVISIONS	
Questions	Comments
	* name of the app * apk approval from the standard bodies in relation with the the use of data and mple mentation of the app * buginess proposal * revise the title

STI THESIS PROGRAM | FT-ARA-019-01 | PAGE 4 OF 4

Scanned with CamScanner

Title Defense



FINAL DEFENSE EVALUATION FORM

ART 4. REVISION LIST FORM

THESIS TITLE
Tunatrike: A booking application for tricycle in General Santos City

NAME OF GROUP MEMBERS	SIGNATURE
Celso Onnele Arancon	
Ariel M. Guerra	
Adrian D. Macaulay	
Raffy G. Simon	
Alvin A. Tampus Jr.	

DATE OF DEFENSE	TIME OF DEFENSE	SCHOOL CAMPUS
12-Oct-24	8:00-9:30AM	STI College General Santos Inc.

NAME OF LEAD PANELIST	SIGNATURE	NAME OF PANELIST	SIGNATURE
Kristel Joy Tulagan, MIT, LPT		Anthony Mark Silong, MIT	

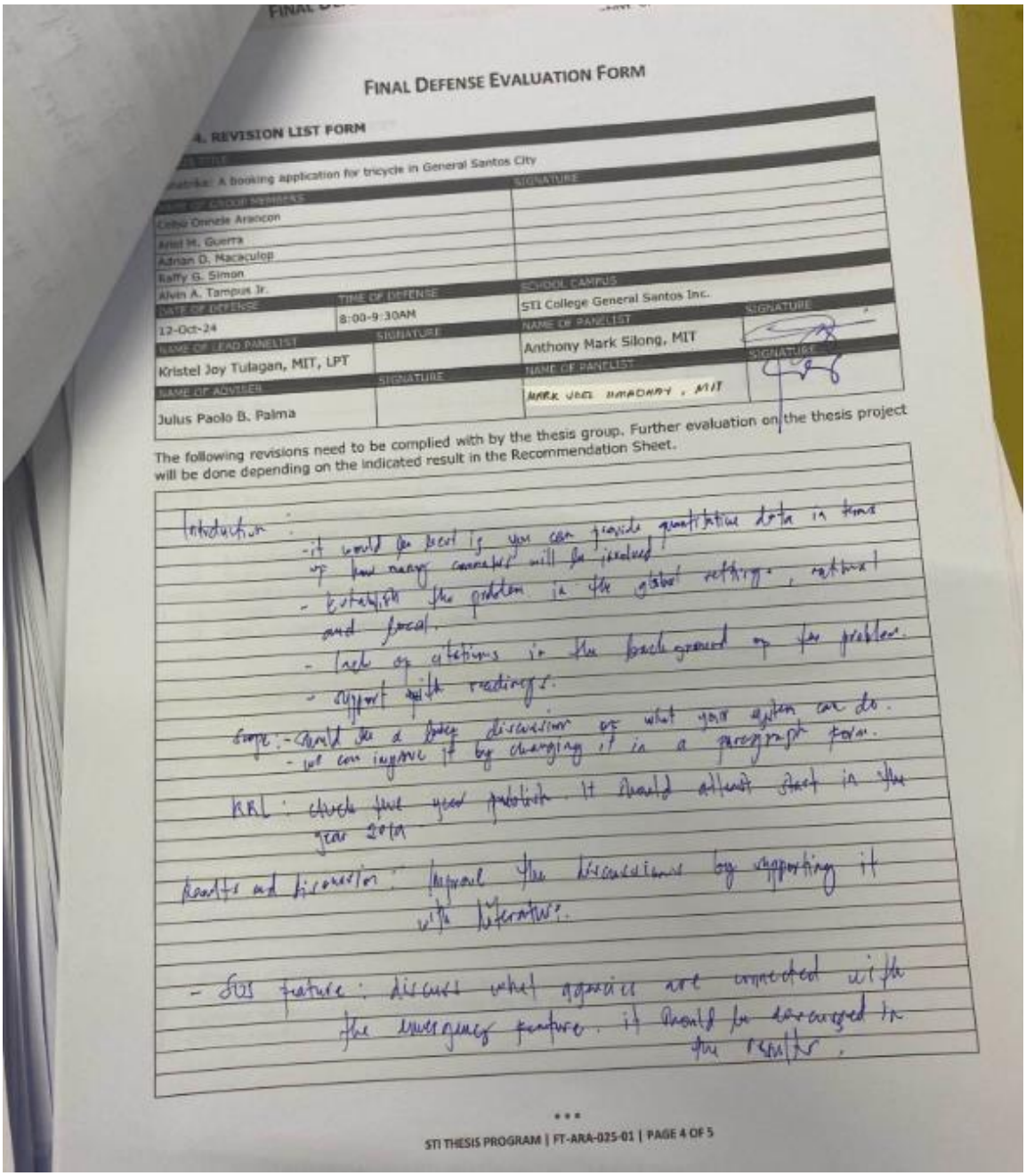
NAME OF ADVISER	SIGNATURE	NAME OF PANELIST	SIGNATURE
Julus Paolo B. Palma		[illegible]	

The following revisions need to be complied with by the thesis group. Further evaluation on the thesis project will be done depending on the indicated result in the Recommendation Sheet.

- Discuss the pre-data gathering & analysis and its results
- Include the following discussion
  - Fare matrix
  - Term & conditions
  - Commuter's fare & discounts (Attach support on this)
- cite authors properly
- Improve your title & background, current state of technology

- System
- Term & Agreement
- Cancellation
- Validations
- Reports and Analytics
  - Feedback
  - Cluster of bicycle
- Details

Thesis 2 Final Defense



FINAL DEFENSE EVALUATION FORM

4. REVISION LIST FORM

TITLE: Habitrike: A booking application for tricycle in General Santos City	
NAME OF GROUP MEMBERS	SIGNATURE
Cielbo Onnele Aracocon	
Ariel M. Guerra	
Adrian D. Macaculop	
Raffy G. Simon	
Alvin A. Tampus Jr.	
DATE OF DEFENSE: 12-Oct-24	TIME OF DEFENSE: 8:00-9:30AM
SCHOOL CAMPUS: STI College General Santos Inc.	
NAME OF LEAD PANELIST: Kristel Joy Tulagan, MIT, LPT	SIGNATURE
NAME OF PANELIST: Anthony Mark Silong, MIT	SIGNATURE
NAME OF ADVISER: Julius Paolo B. Palma	SIGNATURE
NAME OF PANELIST: MARK JOEL BARAHAN, MIT	SIGNATURE

The following revisions need to be complied with by the thesis group. Further evaluation on the thesis project will be done depending on the indicated result in the Recommendation Sheet.

Introduction:
- it would be best if you can provide quantitative data in terms of how many commuter will be involved
- Establish the problem in the global setting, national and local.
- Lack of citations in the background of the problem.
- Support with readings.

Scope: - Need to be a little discussion on what your system can do.
- we can improve it by changing it in a paragraph form.

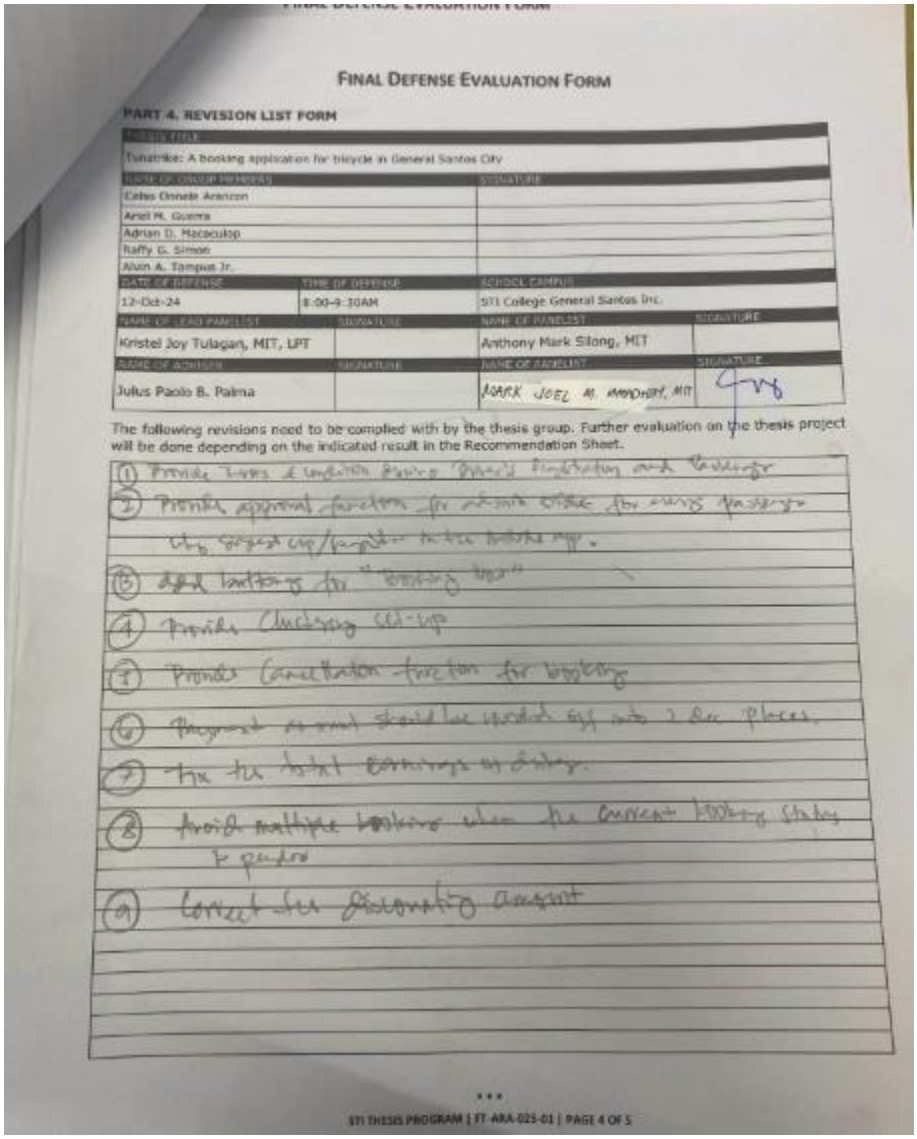
RRL: check the year published. It should atleast start in the year 2019

Results and Discussion: Improve the discussions by supporting it with literatures.

- SOS feature: discuss what agencies are connected with the emergency feature. it should be discussed in the results.

STI THESIS PROGRAM | FT-ARA-025-01 | PAGE 4 OF 5

Thesis 2 Final Defense



FINAL DEFENSE EVALUATION FORM

PART 4. REVISION LIST FORM

THESIS TITLE			
Tunatrike: A booking application for tricycle in General Santos City			
NAME OF GROUP MEMBERS		SIGNATURE	
Celso Oninde Arancon			
Ariel M. Guerra			
Adrian D. Macaculop			
Raffy G. Simon			
Alvin A. Tampus Jr.			
DATE OF DEFENSE	TIME OF DEFENSE	SCHOOL CAMPUS	
12-Oct-24	8:00-9:30AM	STI College General Santos Inc.	
NAME OF LEAD PANELIST	SIGNATURE	NAME OF PANELIST	SIGNATURE
Kristel Joy Tulagan, MIT, LPT		Anthony Mark Silong, MIT	
NAME OF ADVISER	SIGNATURE	NAME OF PANELIST	SIGNATURE
Julius Paolo B. Palma		MARK JOEL M. MADRIAN, MIT	

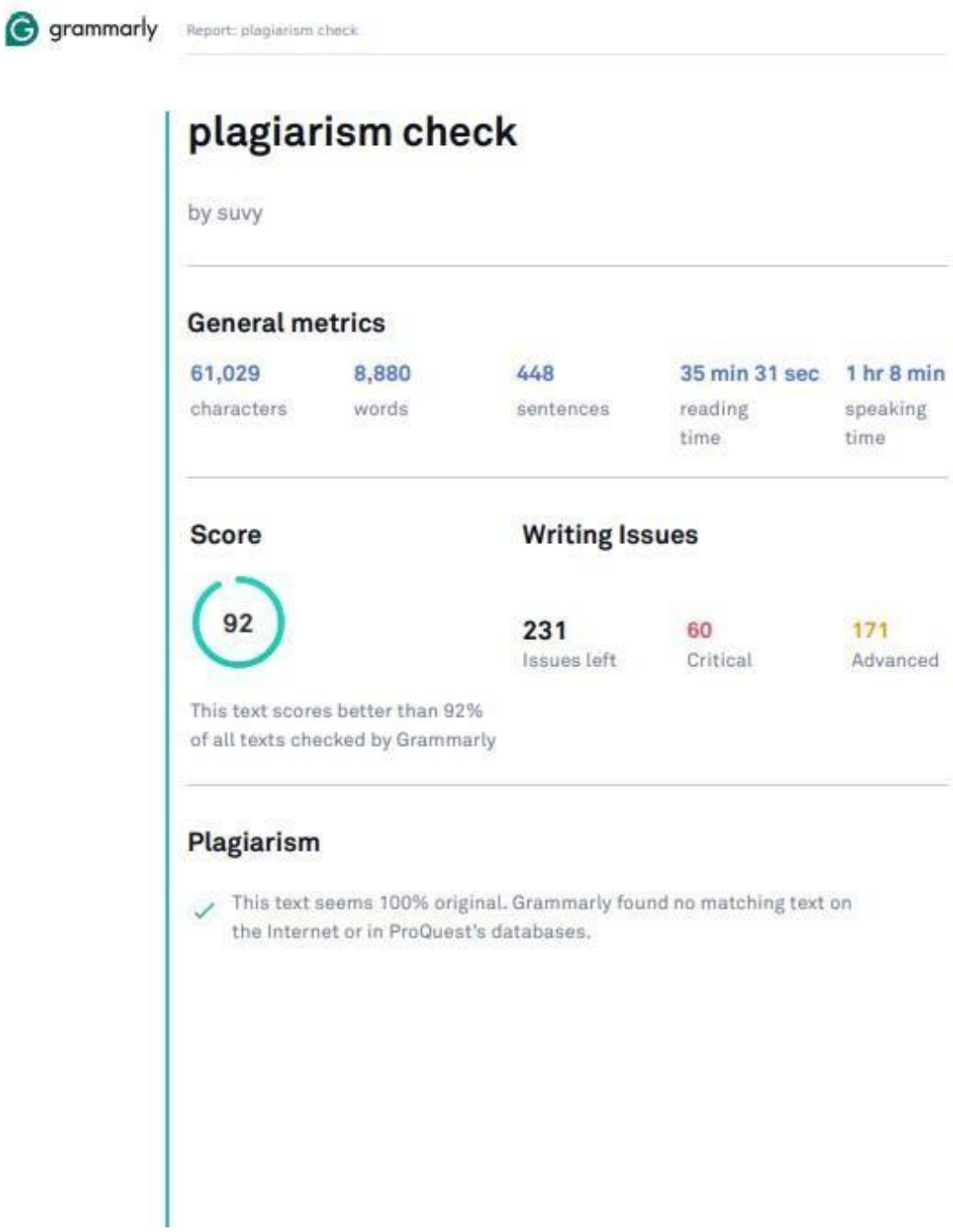
The following revisions need to be complied with by the thesis group. Further evaluation on the thesis project will be done depending on the indicated result in the Recommendation Sheet.

1. Provide terms & conditions before proceed registration and booking
2. Provide approval function for admin side for every booking
   the signed up/register in the booking app.
3. Add buttons for "booking now"
4. Provide Clustering set-up
5. Provide Cancellation function for booking
6. Payment amount should be visible before it is sent to the places.
7. Fix the total earnings display.
8. Avoid multiple booking when the current booking status is pending
9. Correct the discounting amount

STI THESIS PROGRAM | FT-ARA-025-01 | PAGE 4 OF 5

Thesis 2 Final Defense

APPENDIX I. GRAMMARIAN'S CERTIFICATION


grammarly
Report: plagiarism check
plagiarism check
by suvy
General metrics
61,029
characters
8,880
words
448
sentences
35 min 31 sec
reading time
1 hr 8 min
speaking time
Score
92
This text scores better than 92% of all texts checked by Grammarly
Writing Issues
231
Issues left
60
Critical
171
Advanced
Plagiarism
This text seems 100% original. Grammarly found no matching text on the Internet or in ProQuest's databases.

GRAMMARIAN'S CERTIFICATE

This is to certify that the undersigned has reviewed and went thoroughly all the pages of the proposal manuscript entitle "TunaTrike" by Raffy Simon; Adrian Macaculop; Ariel Guerra; and Celso Onelle Arancon, aligned with the setoff structural rules that govern the composition of sentences, phrases and words in the English Language.

Signed this day 12th day of January in the year 2024, at Glan Sarangani Province,

Signed By:

Phoebe Joy M. Laride, LPT

Grammarian

CERTIFICATION OF LANGUAGE EDITING

This is to certify that the thesis proposal titled **"TunaTrike: A Booking Application for Tricycle in General Santos City"** had undergone review and correction by the undersigned. Hence, guaranteeing the quality of this paper in terms of grammar, composition, and presentation.

This certification is issued upon the request of **Celso Onelle T. Arancon Ariel M. Guerra, Adrian D. Macaculop, Raffy G. Simon, and Alvin A. Tampus Jr.** students of Bachelor of Science in Computer Science from STI College, General Santos, Inc. for whatever purpose it may serve them.

Signed this 27th day of November in the year of our Lord, 2024 at Polomolok, South Cotabato, Philippines.

ERHARDIE PRIEL B. TEJERO, MAEd
Language Editor

APPENDIX J: STATISTICIAN'S CERTIFICATE

STI College
General Santos City

Statistician's Certificate

This is to certify that the research entitled **"Tuna trike: A Booking Application for Tricycle in General Santos City"** by Celso Onnele T. Arancon, Ariel M. Guerra, Adrian D. Macaculop, Raffy G. Simon, and Alvin A. Tampus Jr. had been check and statistically corrected by the undersigned.

This certification is issued this 12th day of October 2024 at STI College-Gen. Santos, Inc., General Santos City.

SHAMIYA GABALUNOS
Statistician

APPENDIX K: SYSTEM VALIDATION CERTIFICATE

STI College
General Santos City

System Validation Certificate

Date: December 10, 2024

This certifies that the following researchers have reviewed and validated the minimur fare of **15 pesos** in the Tunatrike: A Booking Application for Tricycle in General Santos City: Celso Onnele T. Arancon, Ariel M. Guerra, Adrian D. Macaculop, Raffy G. Simon, and Alvin A. Tampus Jr. Their analysis confirms that the fare complies with applicable regulations and standards.

Should there be any changes or updates to the fare policies, this certificate may be subject to revision to reflect the latest standards.

Certified by:

ERNIE O GALANG
VP PRESIDENT

Date of Certification: 12/10/2024

STI College
General Santos City

System Validation Certificate

Date: December 10, 2024

This certifies that the following researchers have reviewed and validated the minimum fare of **15 pesos** in the Tuna trike: A Booking Application for Tricycle in General Santos City: Celso Onnele T. Arancon, Ariel M. Guerra, Adrian D. Macaculop, Raffy G. Simon, and Alvin A. Tampus Jr. Their analysis confirms that the fare complies with applicable regulations and standards.

Should there be any changes or updates to the fare policies, this certificate may be subject to revision to reflect the latest standards.

Certified by:
RTCS. Angelito Pulmones
"North Toda"
Date of Certification: 12/10/2024

APPENDIX L: FARE METRICS OF GENERAL SANTOS CITY

Minimum tricycle fare for **first four (4) kilometers** is **eighteen pesos (₱18.00)** and an **additional two pesos (₱2.00)** for every succeeding kilometer. For **senior citizens, PWDs, and students,** fare is **fourteen pesos and forty centavoes (₱14.40) and an additional of one peso and sixty centavos (₱1.60)** for every succeeding kilometer on the following routes:

	TO	KM	FARE	SC, PWD, Student
From Central Business District / Central Public Market	Lagao Brgy. Hall	3 km	₱18	₱14.40
	City Heights Brgy. Hall	3.9 km	₱18	₱14.40
	Prk. Malakas Satellite Market	4 km	₱18	₱14.40
	San Isidro Brgy. Hall	4 km	₱18	₱14.40
	Habitat Phase B, Mabuhay	8 km	₱26	₱20.80
	VSM Heights, San Isidro	8 km	₱26	₱20.80

CPIO
GENERAL SANTOS CITY
CITY PUBLIC INFORMATION OFFICE

Tricycle Fare Matrix (WHITE CLUSTER)

Minimum tricycle fare for FIRST FOUR (4) KILOMETERS - **Fifteen Pesos (₱15.00)** and additional **One Peso (₱1.00)** for every succeeding kilometer

SM to Bulaong Terminal	2.5 km	₱15
KCC/Gaisano/SM to Bulaong Terminal	3.14 km	₱15
CBD/Public Market/SM to Leon Llido St. - NLSA Junction (Brgy. Lagao)	3.42 km	₱15
Plaza Heneral Santos to Lagao Gym	3.55 km	₱15
SM via Aparente St. to Mindanao Medical Center (Brgy. City Heights)	3.68km	₱15
SM to SM Save More (Brgy. San Isidro)	3.7km	₱15
KCC/Robinson via Digos Makar road to Greenfield Residence (Brgy. Lagao)	3.83 km	₱15
KCC/Gaisano/SM via Aparente St. to Lagao National High School (Brgy. San Isidro)	3.91 km	₱15
Plaza Heneral Santos to Dacera Ave. Yumang Junction (Brgy. San Isidro)	3.93 km	₱15
CBD/Public Market via Pioneer Ave. Aparente St. to Salvani St. (Brgy. City Heights)	3.97 km	₱15
KCC/Robinson via G.Misa St. to Sea Oil Yumang St. corner Aparente St. (Brgy. City Heights)	4 km	₱15
CBD/Public Market to Purok Malakas Satellite Market (Brgy. San Isidro)	4 km	₱15
CBD/Public Market to Brgy. Hall - San Isidro	4 km	₱15
KCC/Robinson via G.Misa St. to Lessandra (Brgy. San Isidro)	4 km	₱15
CBD/Public Market to Habitat Phase B (Brgy. Mabuhay)	8 km	₱19
CBD/Public Market to VSM Heights (Brgy. San Isidro)	8 km	₱19
CBD/Public Market to Habitat Phase A (Brgy. Mabuhay)	9 km	₱20
CBD/Public Market to Brgy. Hall - Katangawan	9 km	₱20
CBD/Public Market to Brgy. Hall - Mabuhay	13 km	₱24
CBD/Public Market to Brgy. Hall - Conel	15 km	₱26
CBD/Public Market to Brgy. Hall - Olympog	16 km	₱27
CBD/Public Market to Brgy. Hall - Upper Labay	25 km	₱36

Legal Basis: Ordinance No. 8, Series of 2023

*"**Section 120. FARES AND FARE MATRIX**. - The minimum fare of MTH in the city is at fifteen pesos (₱15.00) per passenger for the first four (4) kilometers and an additional fare of ₱ 1.00 per kilometer, if the distance traverse exceeds 4 kilometers. Provided, that the prevailing transportation fare privileges granted by law to students, senior citizens, and persons with disability (PWD) shall continue to apply. Shortchanging the passenger, as defined under "No Short Changing Act of 2016" (Republic Act No. 10909) shall constitute a violation of this section."*

"When the origin-destination is only CBD-CBD, minimum fare is still fifteen pesos (₱15.00), as the shortest distance from any point never exceeds four (4) kilometers."

Tricycle Fare Matrix (APPLE GREEN CLUSTER)

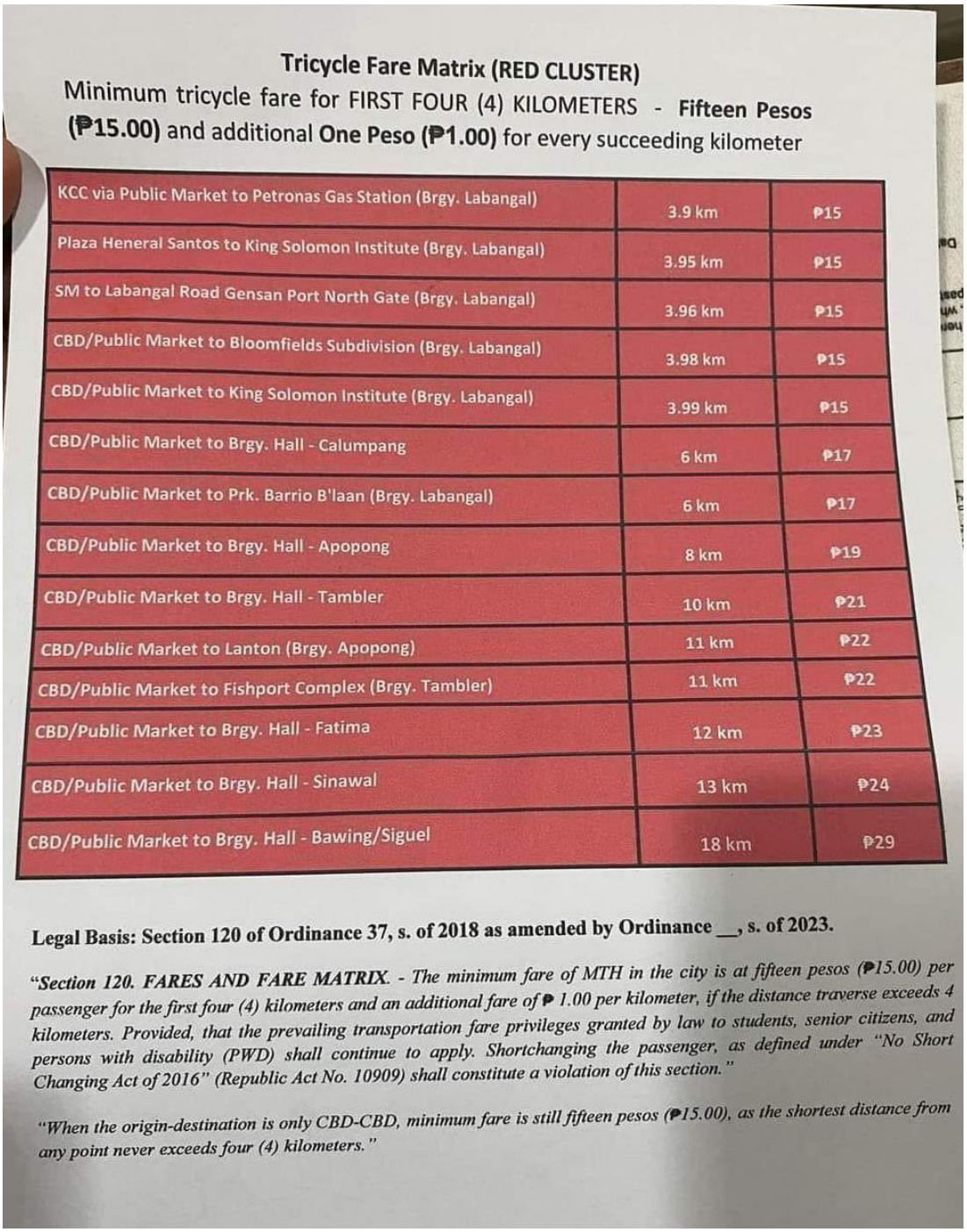
Minimum tricycle fare for FIRST FOUR (4) KILOMETERS - **Fifteen Pesos (₱15.00)** and additional **One Peso (₱1.00)** for every succeeding kilometer

CBD/Public Market to Brgy. Hall - Bula	2.1 km	₱15
KCC via Arriola St. to Gensanville (Brgy. Bula)	2.86 km	₱15
CBD/Public Market/Gaisano/KCC/Robinson to Lagao Public Market	3.1 km	₱15
SM via Public Market to Gensanville (Brgy. Bula)	3.59 km	₱15
Plaza Heneral Santos/SM/Gaisano/KCC/Robinson to Lagao Public Market	3.92 km	₱15
SM/Gaisano/KCC/Robinson to VSM Estate (Brgy. Lagao)	4 km	₱15
KCC/Robinson to Balut Park (Brgy. Baluan)	4 km	₱15
Plaza Heneral Santos to Gensanville (Brgy. Bula)	4 km	₱15
CBD/Public Market to Agan (Brgy. Ligaya)	5 km	₱16
CBD/Public Market to Brgy. Hall - Baluan	6 km	₱17
CBD/Public Market to Brgy. Hall - Buayan	7 km	₱18
CBD/Public Market to Brgy. Hall - Ligaya	8 km	₱19
CBD/Public Market to Brgy. Hall - Tinagacan	16 km	₱27

Legal Basis: **Ordinance No. 8, Series of 2023**

*"**Section 120. FARES AND FARE MATRIX**. - The minimum fare of MTH in the city is at fifteen pesos (₱15.00) per passenger for the first four (4) kilometers and an additional fare of ₱ 1.00 per kilometer, if the distance traverse exceeds 4 kilometers. Provided, that the prevailing transportation fare privileges granted by law to students, senior citizens, and persons with disability (PWD) shall continue to apply. Shortchanging the passenger, as defined under "No Short Changing Act of 2016" (Republic Act No. 10909) shall constitute a violation of this section."*

"When the origin-destination is only CBD-CBD, minimum fare is still fifteen pesos (₱15.00), as the shortest distance from any point never exceeds four (4) kilometers."



Tricycle Fare Matrix (RED CLUSTER)

Minimum tricycle fare for FIRST FOUR (4) KILOMETERS - **Fifteen Pesos (₱15.00)** and additional **One Peso (₱1.00)** for every succeeding kilometer

KCC via Public Market to Petronas Gas Station (Brgy. Labangal)	3.9 km	₱15
Plaza Heneral Santos to King Solomon Institute (Brgy. Labangal)	3.95 km	₱15
SM to Labangal Road Gensan Port North Gate (Brgy. Labangal)	3.96 km	₱15
CBD/Public Market to Bloomfields Subdivision (Brgy. Labangal)	3.98 km	₱15
CBD/Public Market to King Solomon Institute (Brgy. Labangal)	3.99 km	₱15
CBD/Public Market to Brgy. Hall - Calumpang	6 km	₱17
CBD/Public Market to Prk. Barrio B'laan (Brgy. Labangal)	6 km	₱17
CBD/Public Market to Brgy. Hall - Apopong	8 km	₱19
CBD/Public Market to Brgy. Hall - Tambler	10 km	₱21
CBD/Public Market to Lanton (Brgy. Apopong)	11 km	₱22
CBD/Public Market to Fishport Complex (Brgy. Tambler)	11 km	₱22
CBD/Public Market to Brgy. Hall - Fatima	12 km	₱23
CBD/Public Market to Brgy. Hall - Sinawal	13 km	₱24
CBD/Public Market to Brgy. Hall - Bawing/Siguel	18 km	₱29

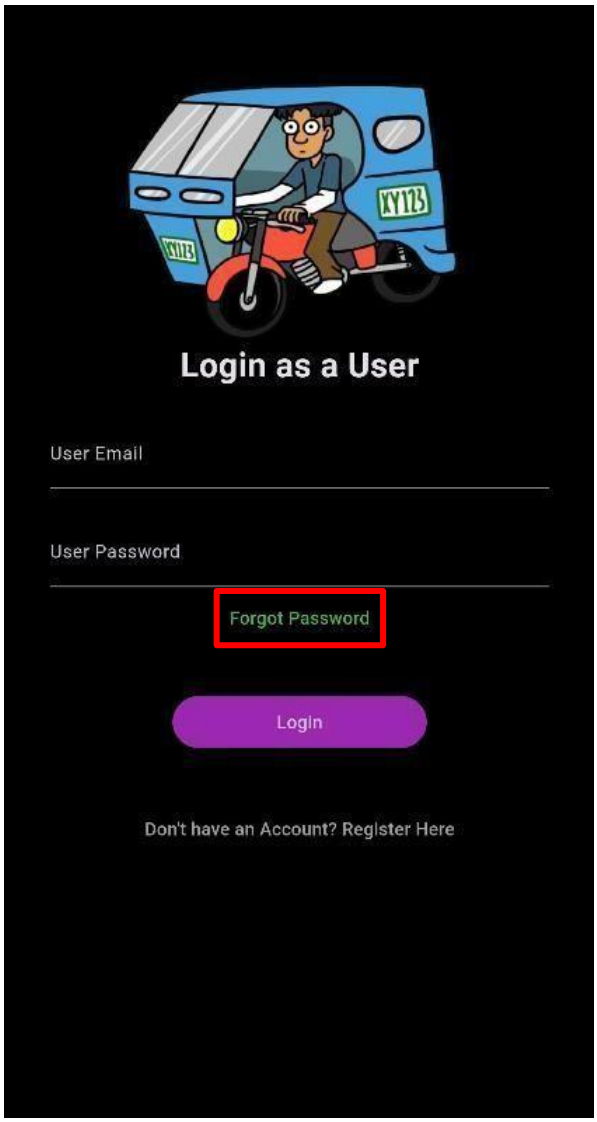
Legal Basis: Section 120 of Ordinance 37, s. of 2018 as amended by Ordinance __, s. of 2023.

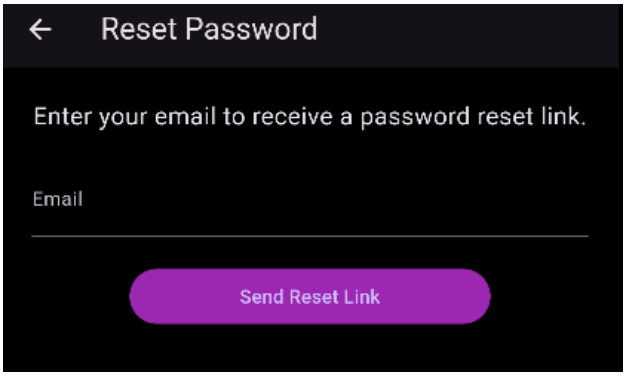
*"**Section 120. FARES AND FARE MATRIX**. - The minimum fare of MTH in the city is at fifteen pesos (₱15.00) per passenger for the first four (4) kilometers and an additional fare of ₱ 1.00 per kilometer, if the distance traverse exceeds 4 kilometers. Provided, that the prevailing transportation fare privileges granted by law to students, senior citizens, and persons with disability (PWD) shall continue to apply. Shortchanging the passenger, as defined under "No Short Changing Act of 2016" (Republic Act No. 10909) shall constitute a violation of this section."*

"When the origin-destination is only CBD-CBD, minimum fare is still fifteen pesos (₱15.00), as the shortest distance from any point never exceeds four (4) kilometers."

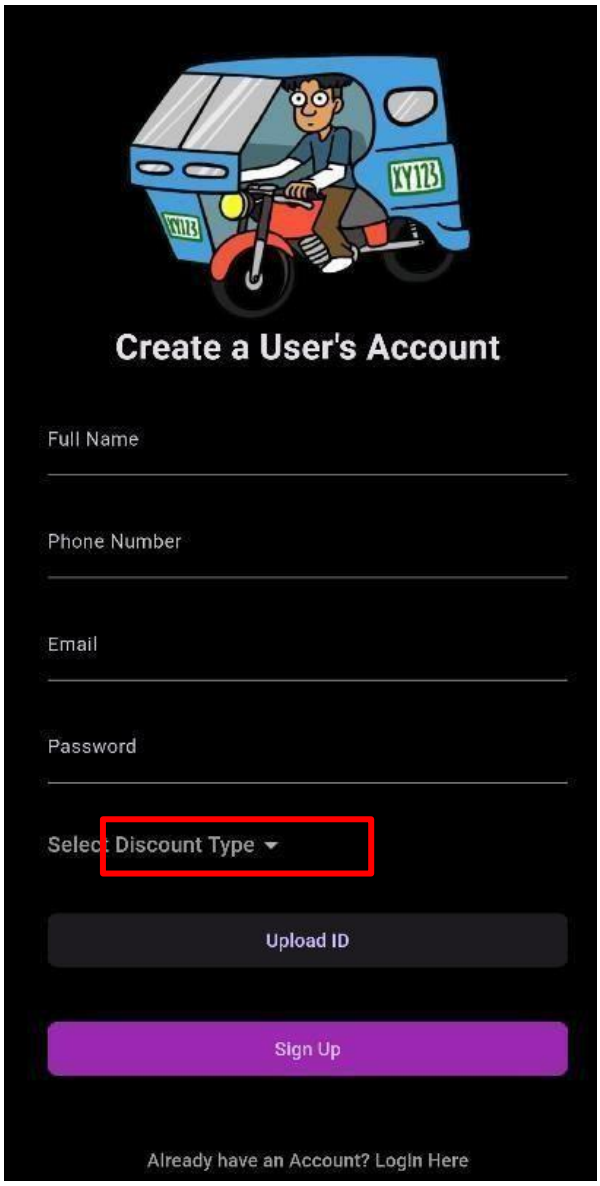
APPENDIX M. USER'S MANUAL

For passenger app



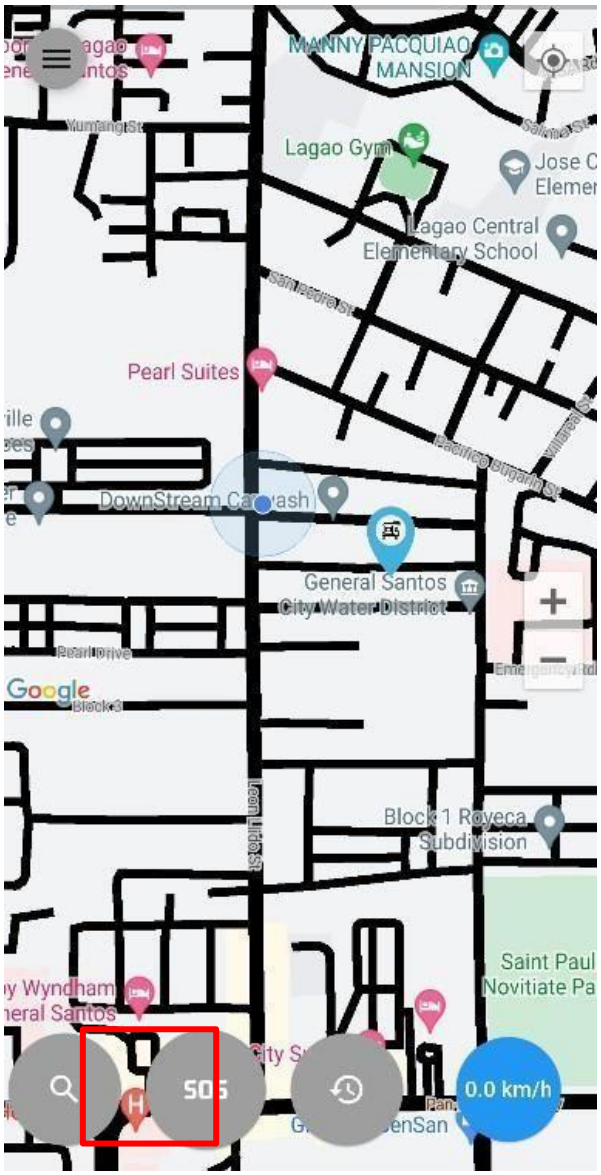


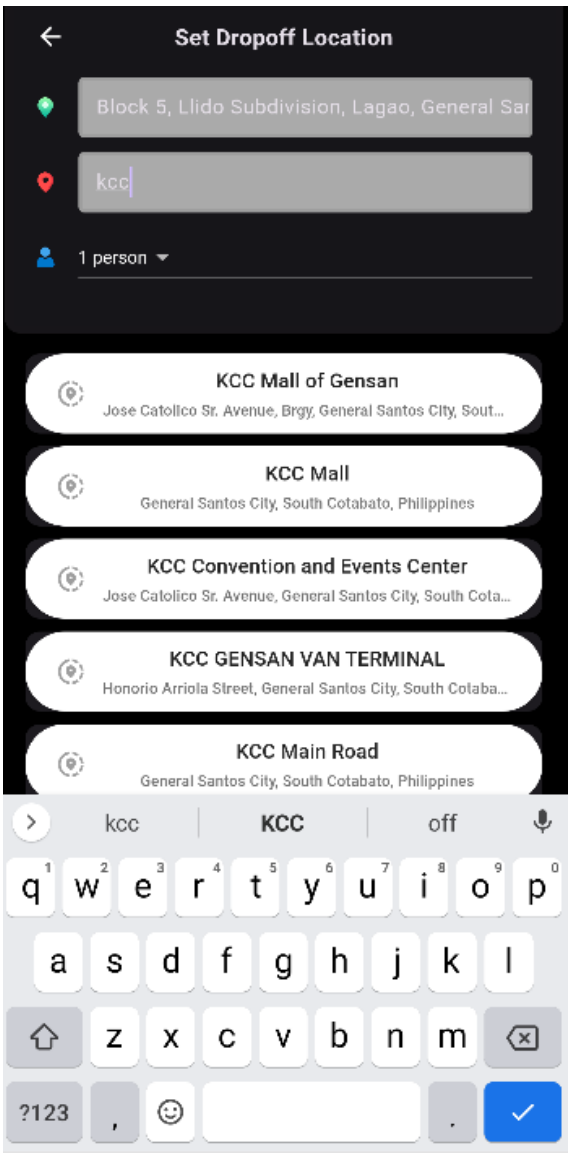
1. This is the login page where the passenger puts their email and password. You can also press forgot password if you forgot your password and enter you email to send a link for reset password.





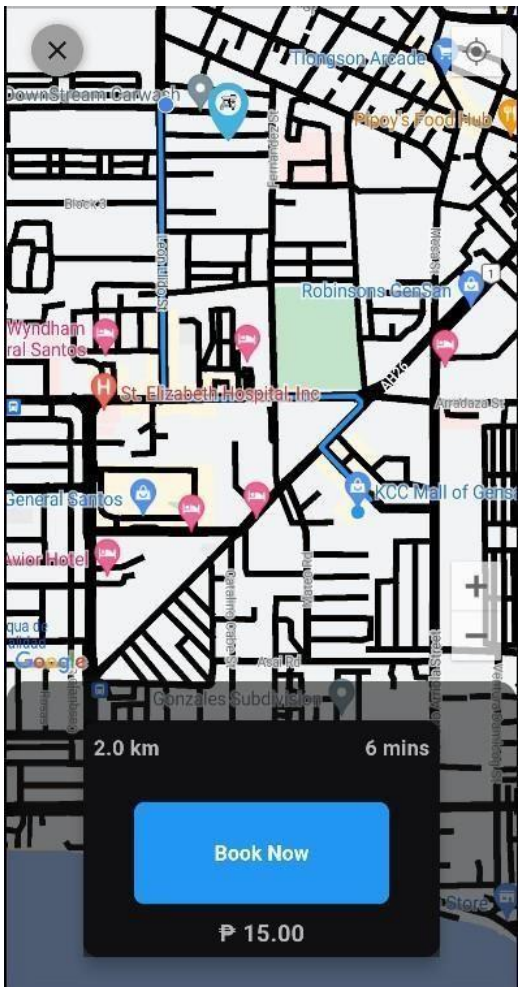
1. In this signup page, after you enter all the details needed, you can select discount type and upload a valid ID. After uploading a valid ID, it will automatically reflect on the admin side and verify whether it is still valid.

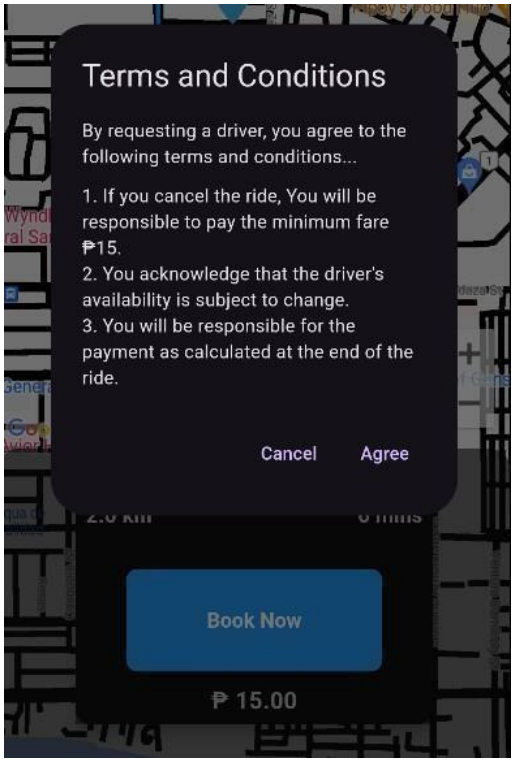


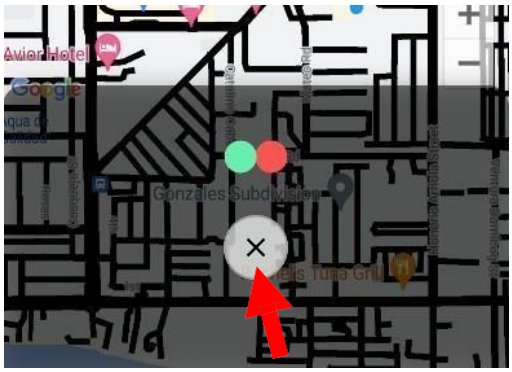


1. After logging in, you will be directed to the passenger dashboard, where you can see the nearby online drivers within 500 meters on your current location.

2. For booking, you can select the search icon, and after that, in the green icon, it will automatically show your current location, or you can select it where you want to be picked up, and in the red icon, you can input where your destination is.

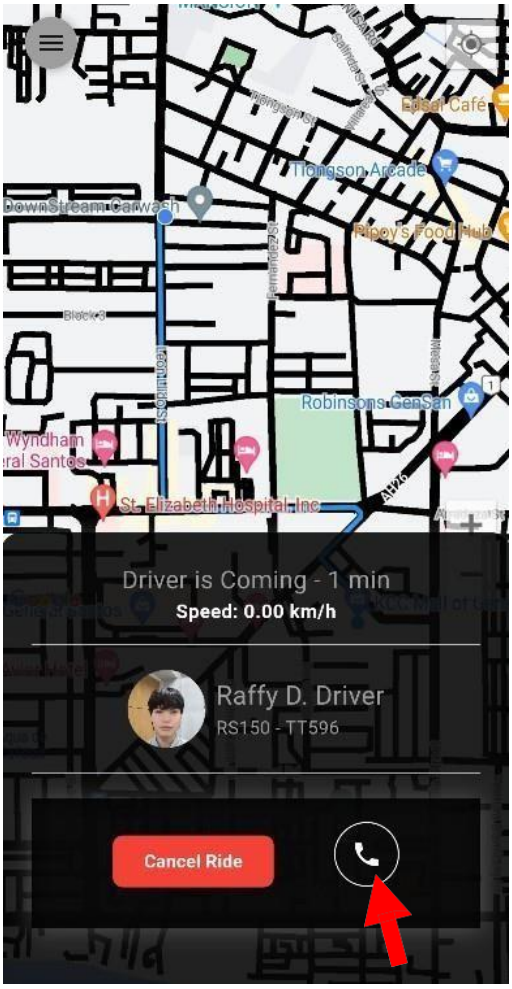




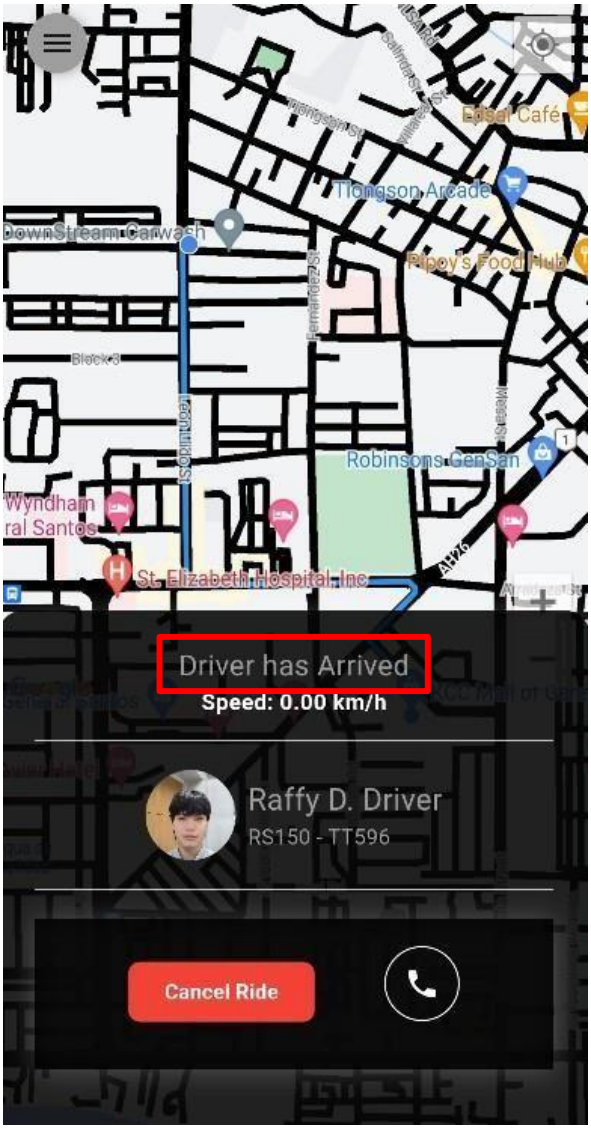


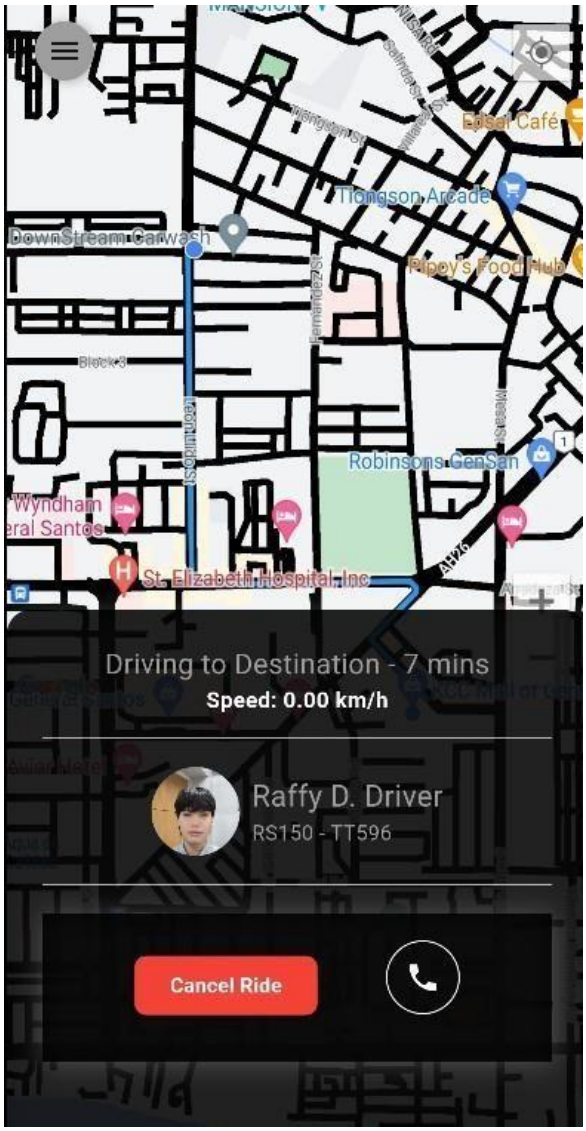
1. After you select where your destination is, it will automatically find nearby drivers and draw the fastest route.

2. After you select where your destination is, it will appear the fastest route and the estimated fare. If you click the book now there some terms and condition you need to agree to proceed the booking, after that it will automatically search for nearby online drivers, and you can also cancel the booking by clicking the “X” icon if you have some changes.

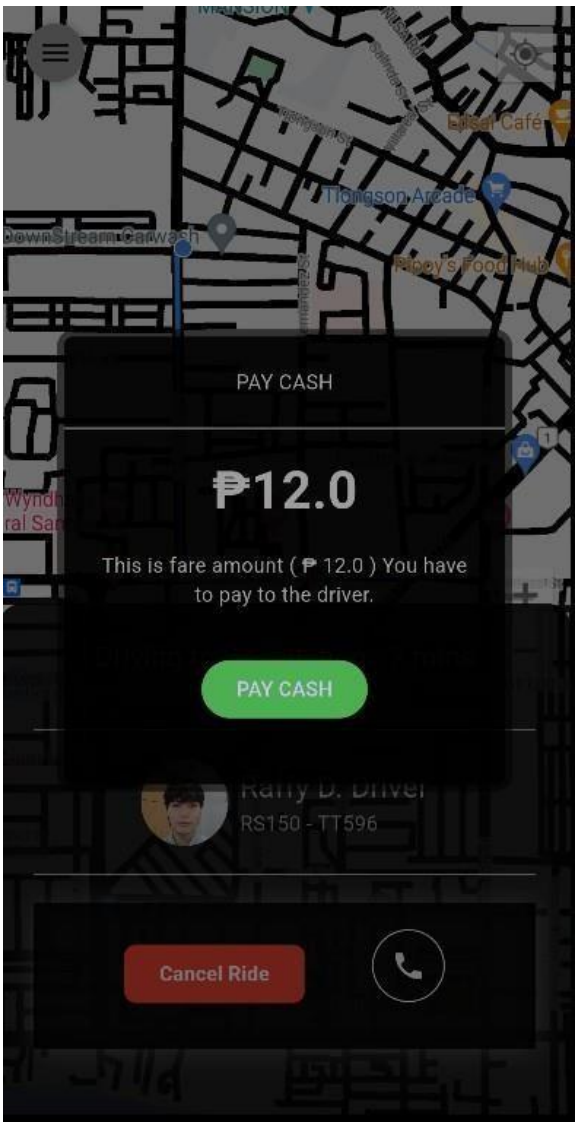


1. After the driver accepts the book, it will show the details of your driver, and you can call him by clicking the icon.

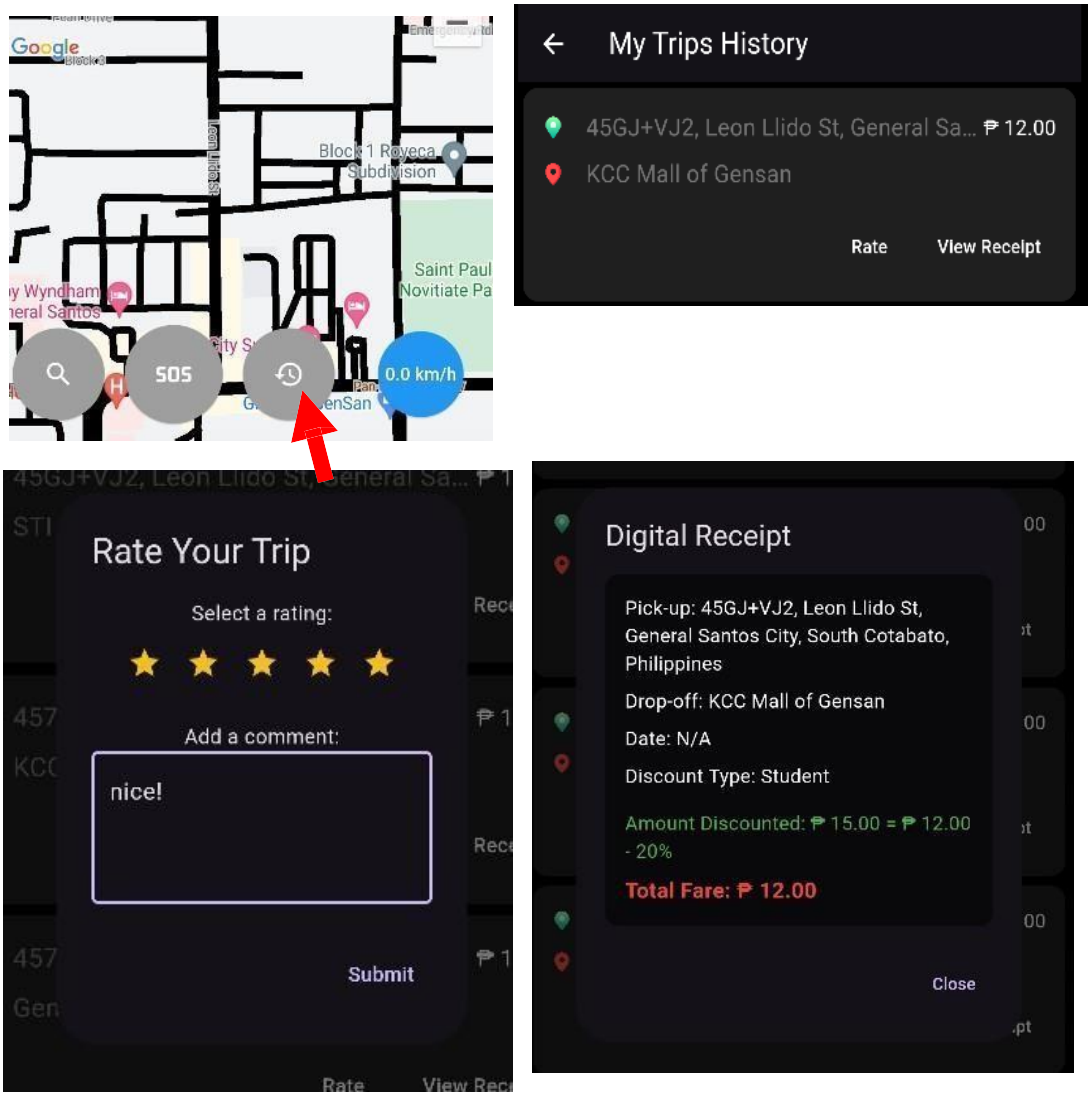




1. It will show on the box if the driver arrived and the status of the trip if it has already started.

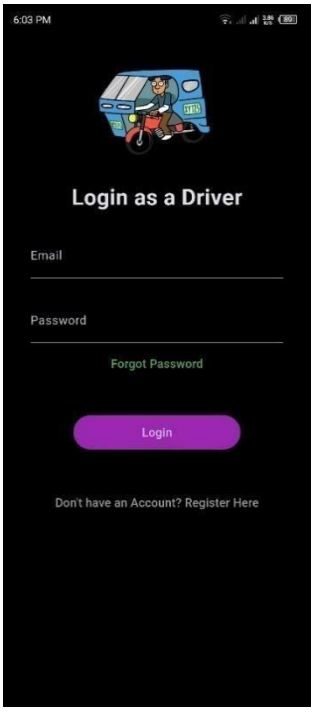


1. After the trip ends, it will show the fare you must pay; if you have availed of the discount, it will automatically be deducted.



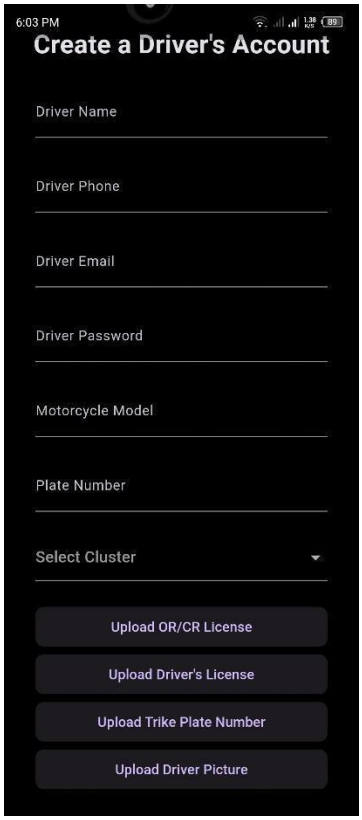
1. You can view your trip history by clicking the button on the picture shown above, and you can rate or view your digital receipt.

For driver app

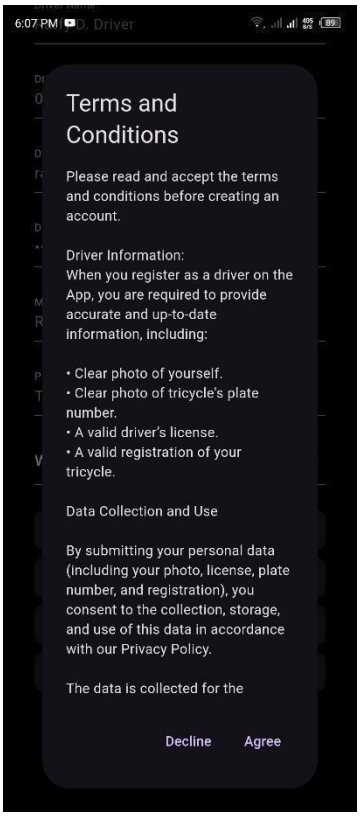


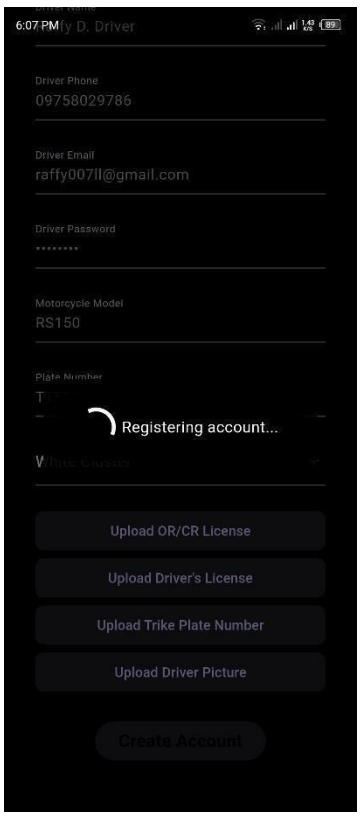
1. This is the login page where the driver put their email and password also, you can press forgot password if you forgot your password and press login button, at the very bottom the driver can press the “Register Here” to register an account.





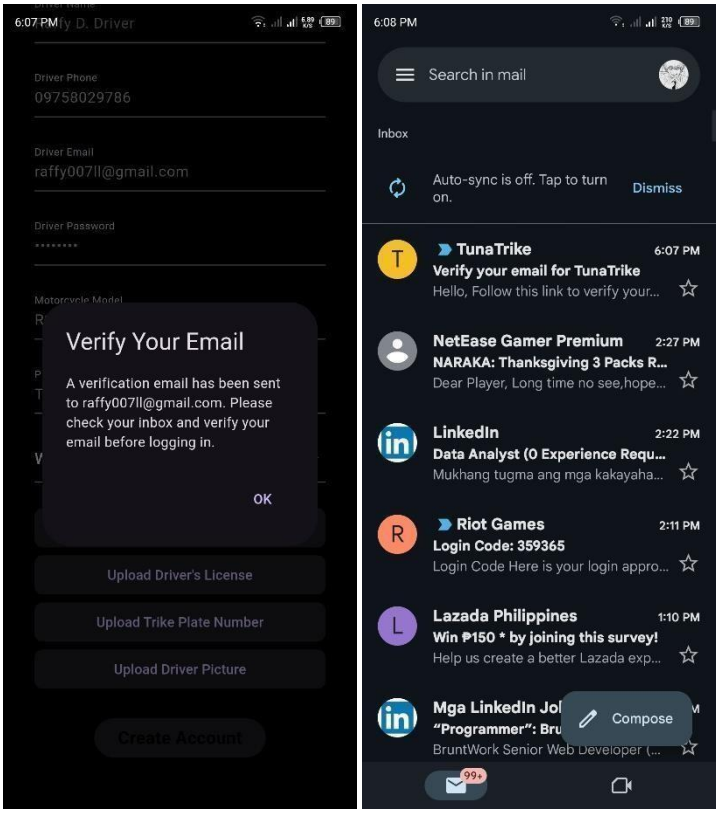
1. This is the registration page, where drivers can create their account by filling up the text boxes such as name, phone number, email, password, motorcycle model, plate number then select their cluster. Lastly upload your OR/CR, drivers license, plate number, then profile picture or a photo of your face.





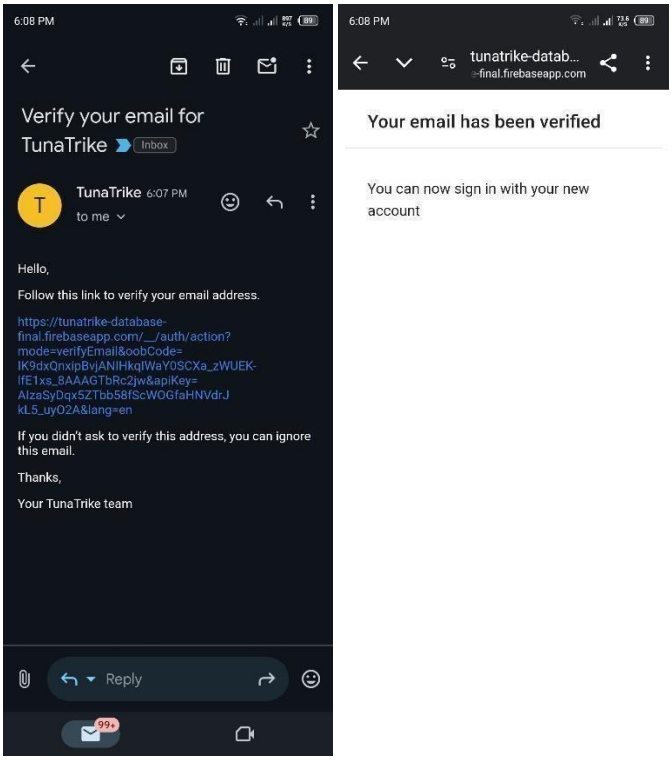
1. The terms and condition will pop-up when you press the create account button then read the terms and conditions, then you can choose to decline or agree to the terms and condition.

Then the registering account loading screen will appear upon pressing the agree button.

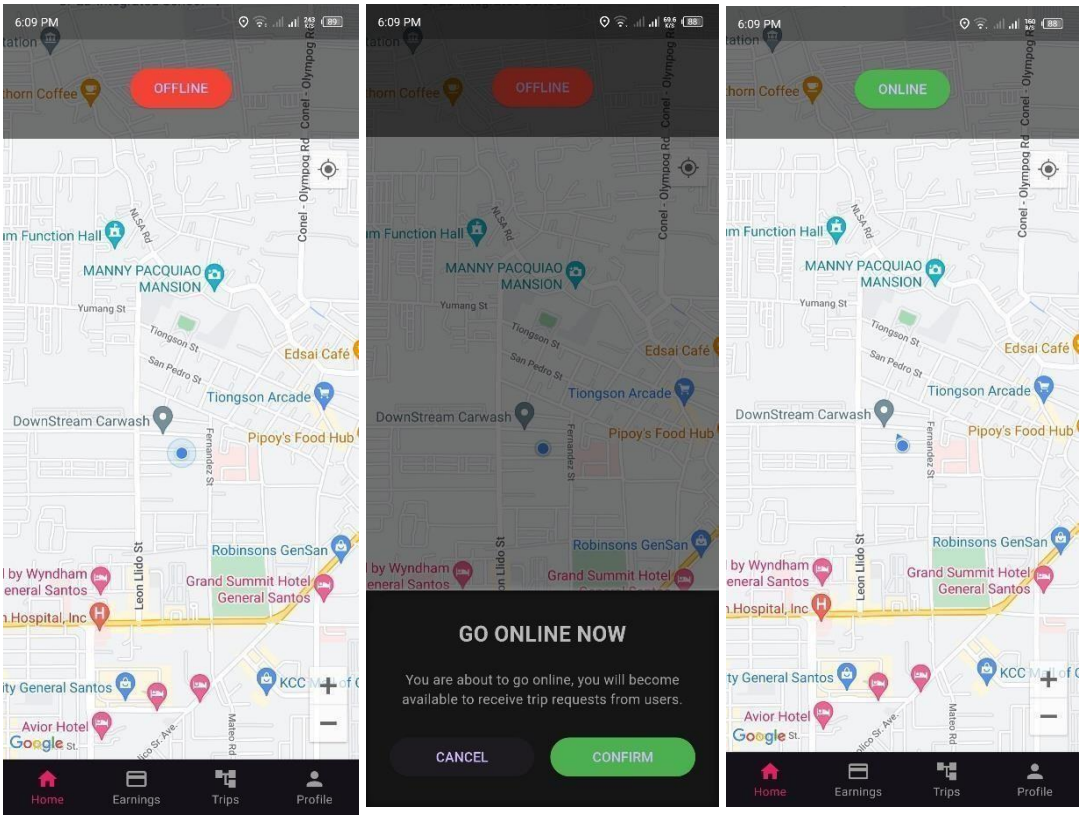


1. The email verification notification will pop-up to notify the driver that they need to go to their email to verify their account.

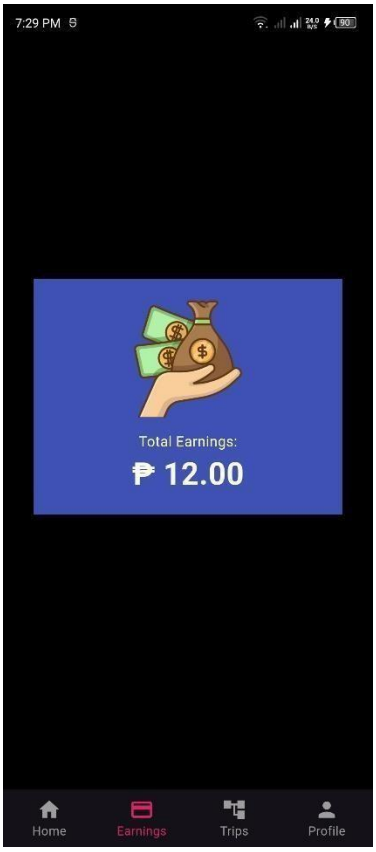
2. You will need to go to your email then press our email to go to the link for the account verification.



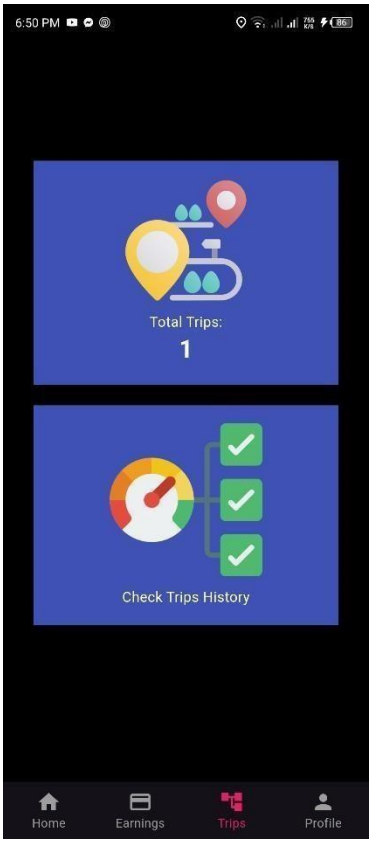
1. Now that the driver has pressed the link it will verify the drivers email that is legitimate email and now can be use to login to our app.

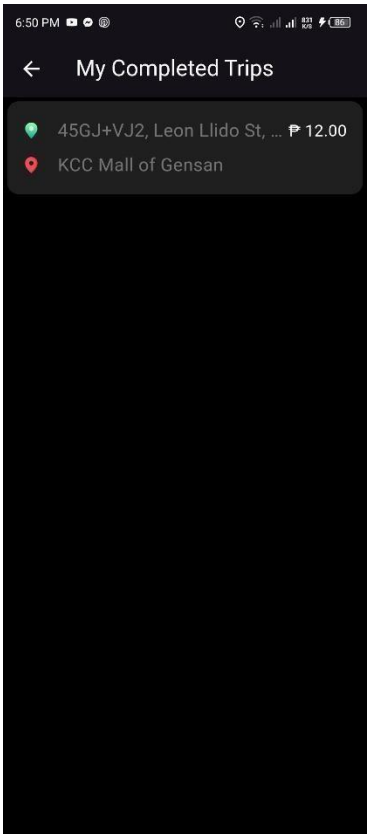


1. This is the user interface for driver app the interface shows the map and they can choose whether to go online or offline. Online if they want to find a passenger then offline if they want to rest or do not want to find a passenger.

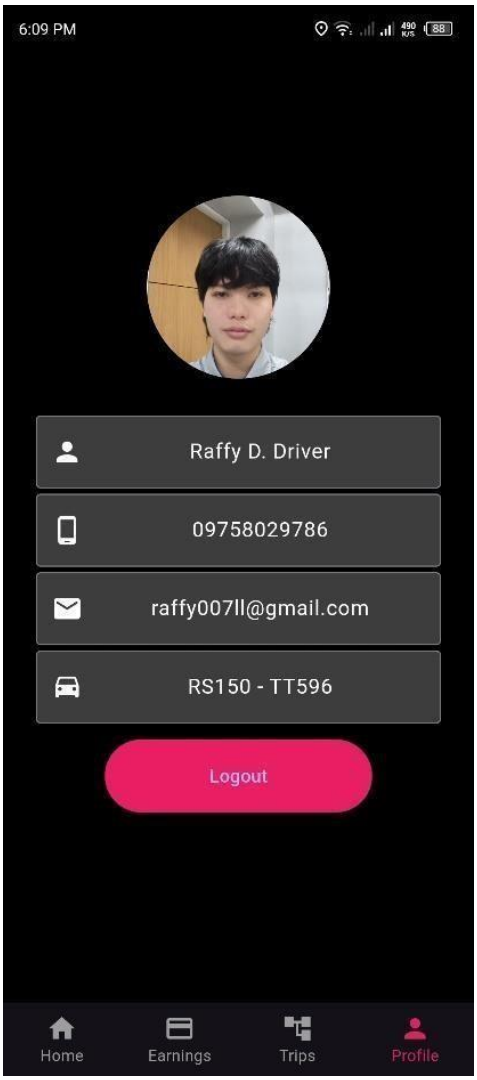


1. Total earning page where the driver can see their total earnings from the app.

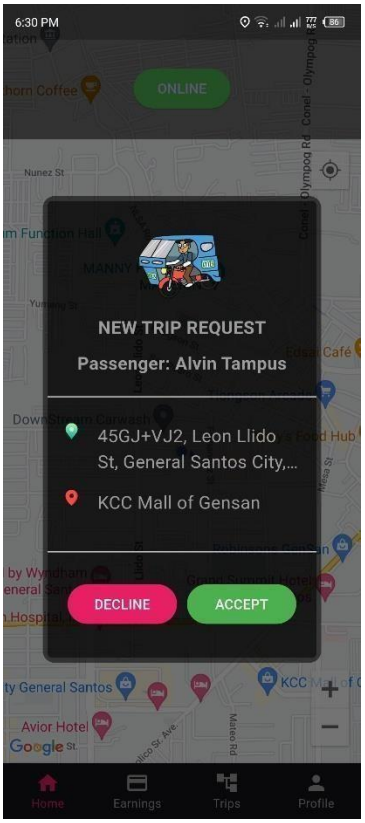


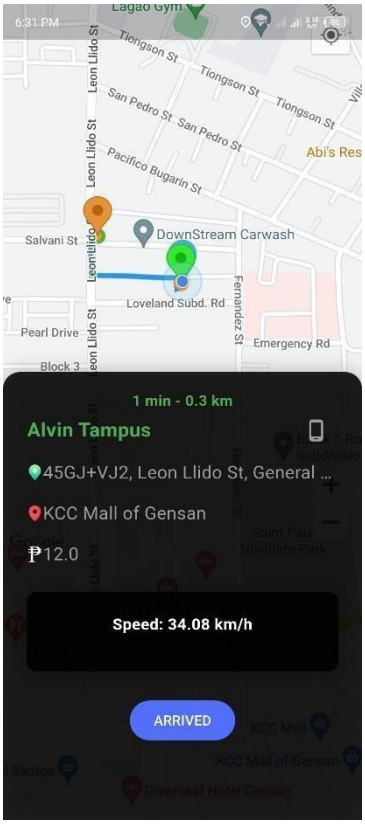


1. Trips page is where the driver can see their total trips and when they press the "check trip history" they can see their completed trips from the passenger location to the destination and the earning from that trip.



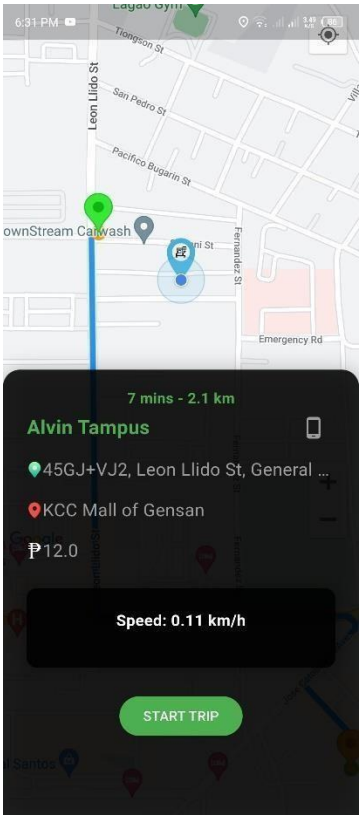
1. The profile page contains the driver's basic information like name, phone number, email, and motorcycle model and plate number.

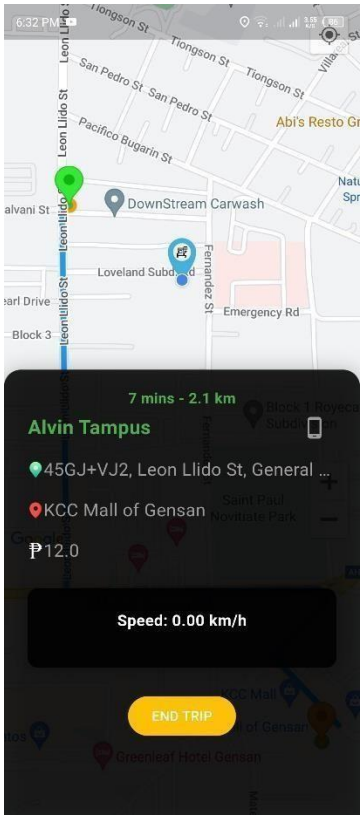




1. When the driver finds a passenger, he can see where the location of the passenger and where is the passenger's destination they have the option to decline or accept.

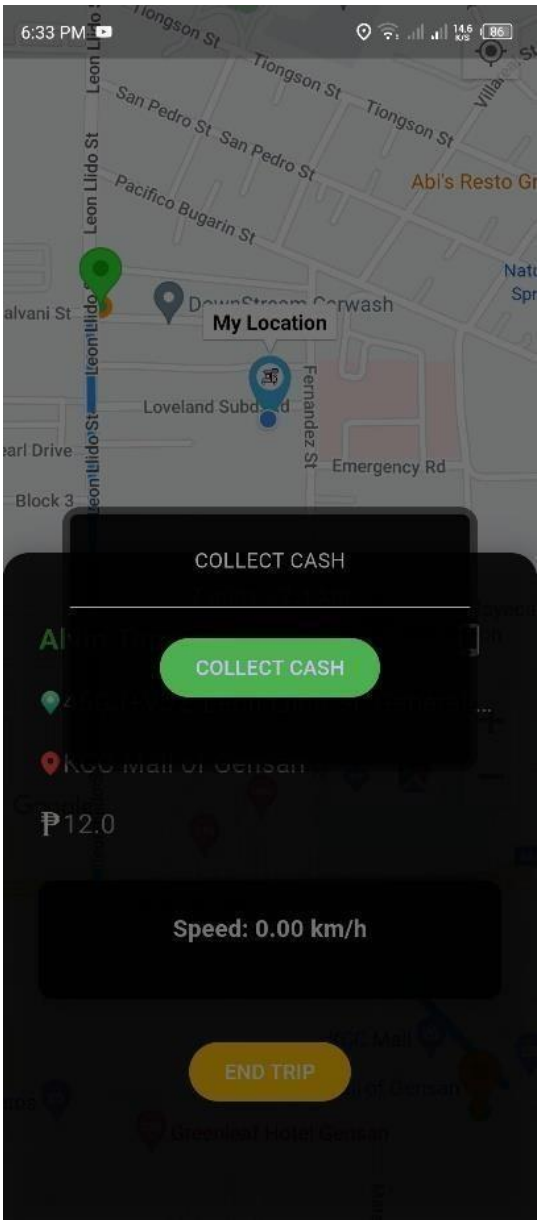
2. When the driver accepts the trip the driver can see a polyline to the passenger's destination and when he arrived at the passenger the driver can now press the arrived button.





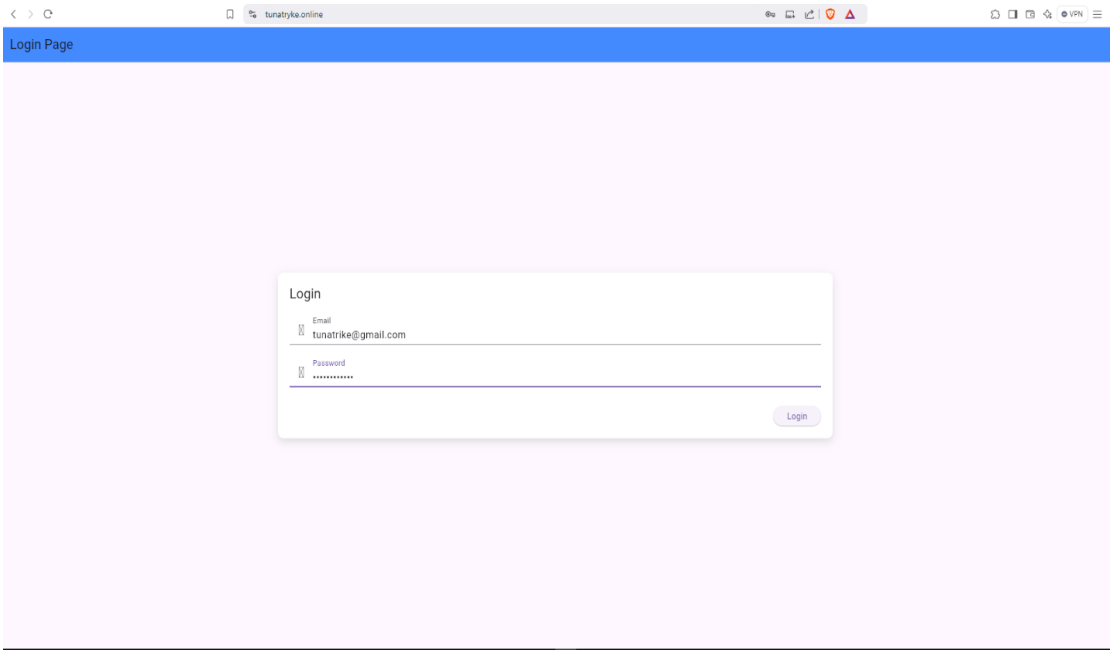
1. After the driver press arrived button, he can now press the start trip button to start the trip and can now follow the polyline to the destination. Also, the driver can see the fare price and the speed of the tricycle.

2. Now that they have arrived at the passenger's destination the driver can now press the end trip button to end the trip.

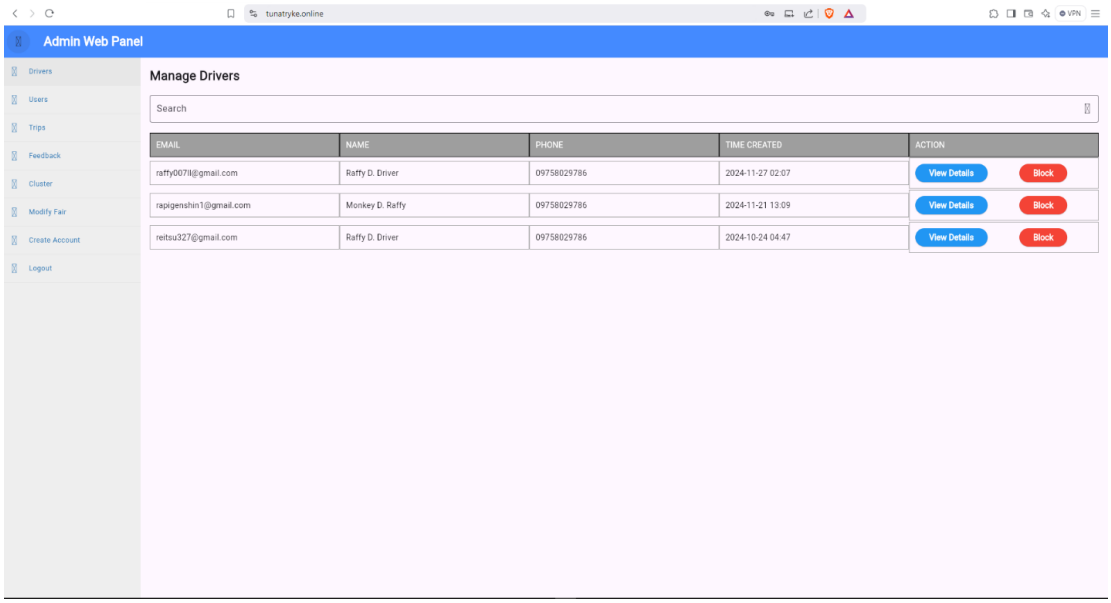


1. After pressing the end trip button, the app will remind the driver to collect the cash from the passenger.

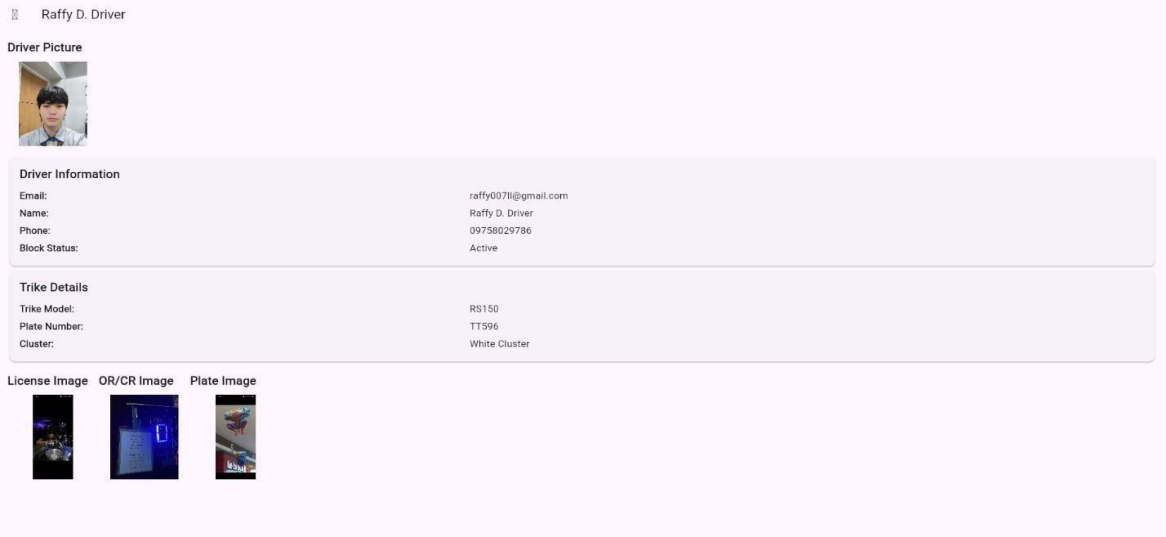
Admin User Manual



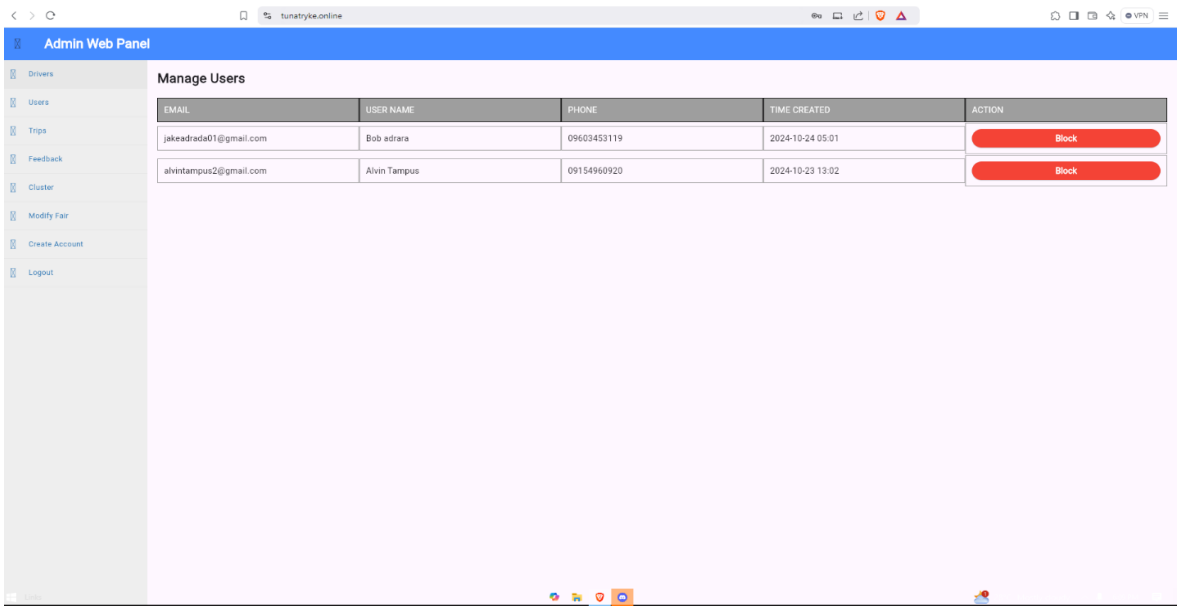
1. Search the https://tunatryke.online/ for the website URL.
2. Enter the Admin credentials using Email and the password



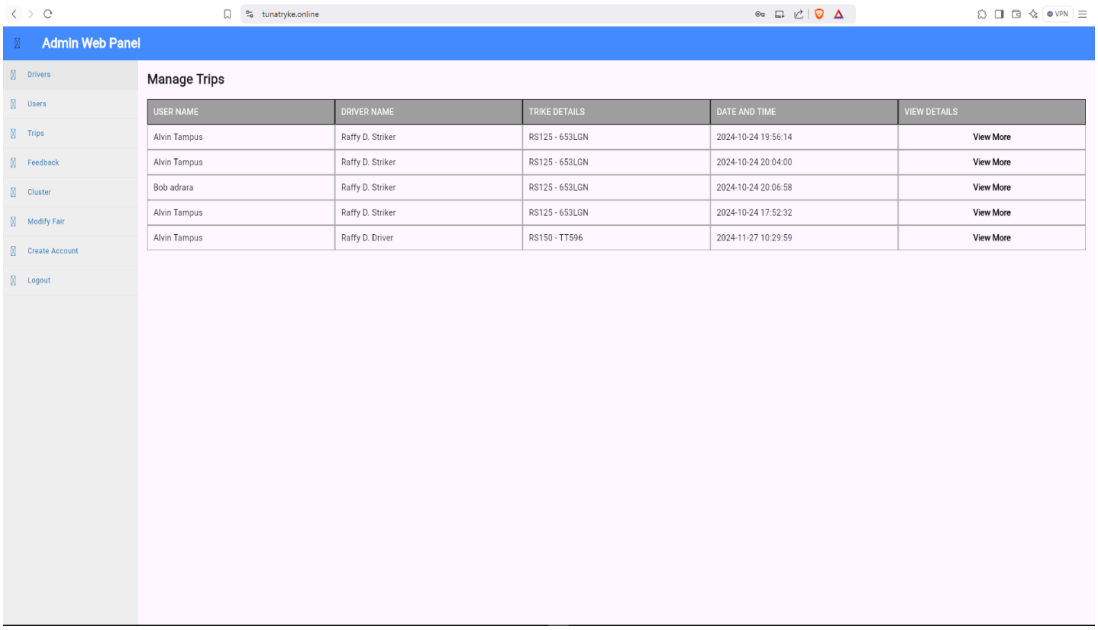
1. In the Driver's Panel, Where the Email, name, Phone number and time created can be found.

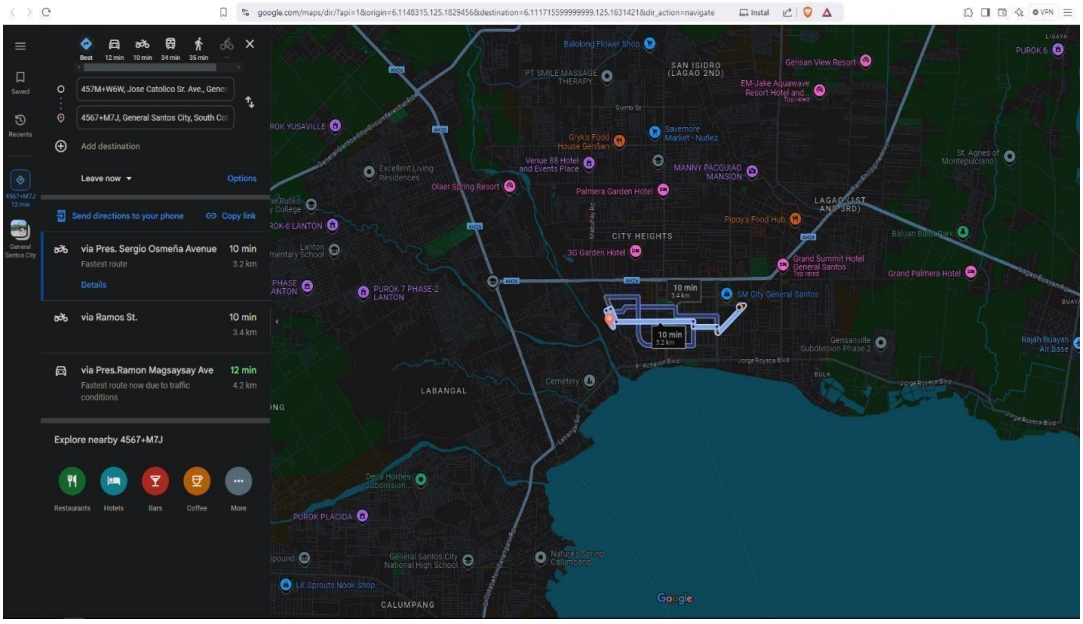


1. If you clicked the button " View Details " this page will appear and consist the additional information of the driver. In this page the admin can also check the License Image, OR/CR and the plate image of the Tricycle.

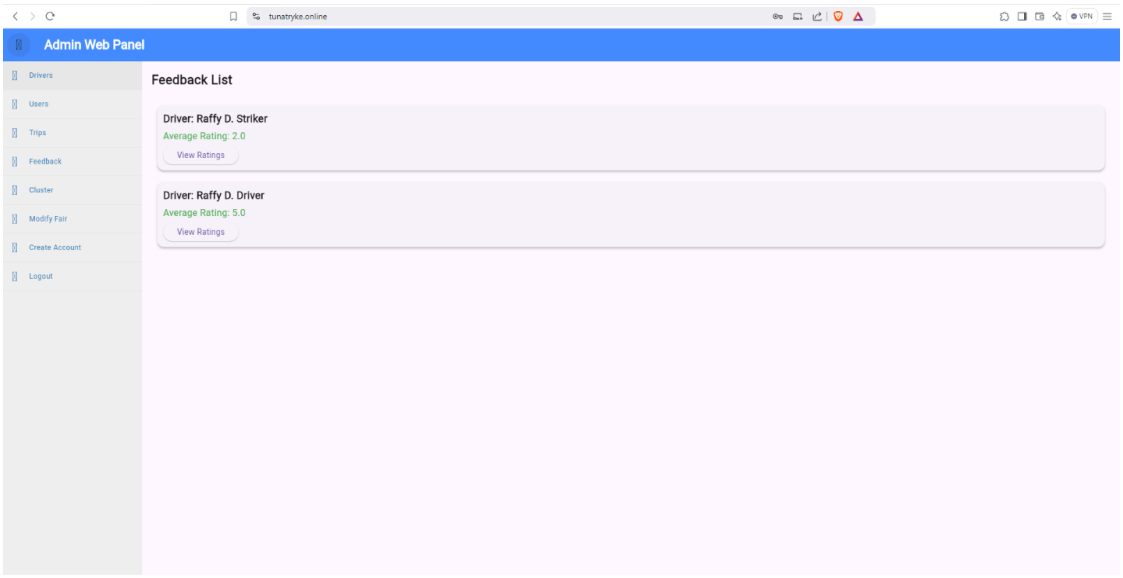


1. In the Users panel, Where the information of the Passenger can be found.

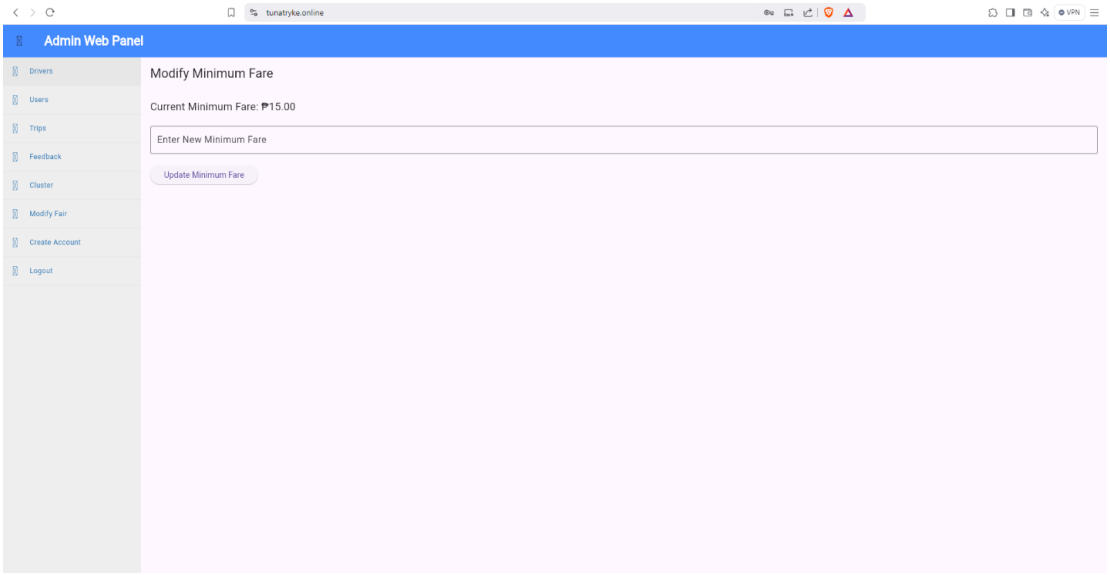




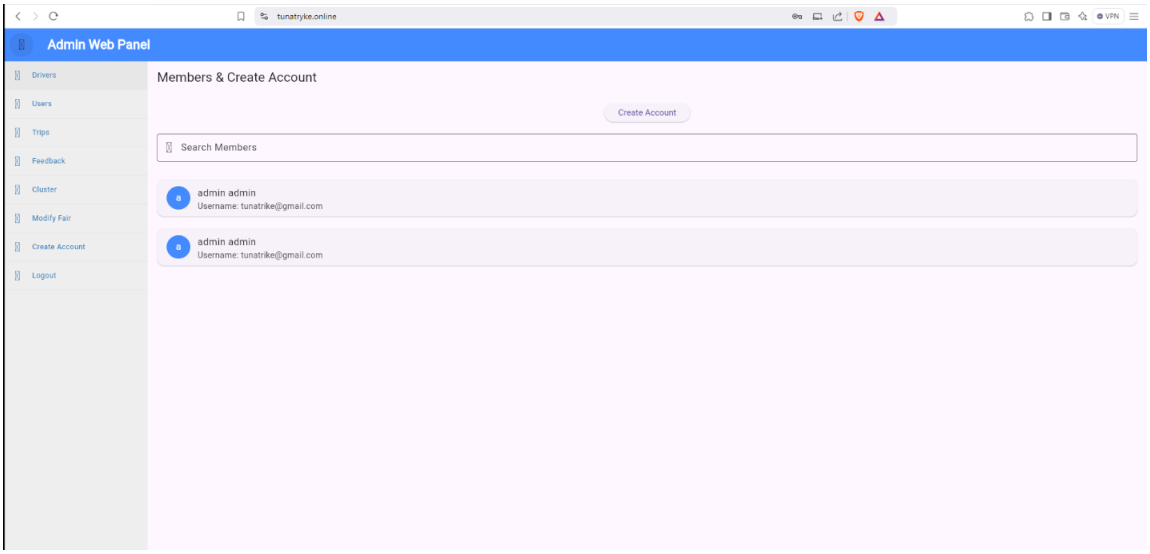
1. In the Trips panel, Where the successful transaction can be found. In the “ View more “ button you can see the past transaction of the driver and the passenger.



1. In the Feedback panel, the admin can check the rating of the passengers for the drivers.



1. In the Modify Fair panel, incase the LGU of General Santos City will implement new fair for the city transportation, the admin can change the fair using this panel.



1. In this panel the admin can create another account for logging in the admin.

APPENDIX N. CURRICULUM VITAE OF RESEARCHERS



Curriculum Vitae of

CELSO ONELLE T. ARANCON

Tual Poblacion, Kiamba, Sarangani, Province
Arancon.194150@gensan.sti.edu.ph
09260218832

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	N/A	STI College General Santos
TechVoc	2017-2018	Notre Dame Of Kiamba INC.
High School	2016-2017	Notre Dame Of Kiamba INC.
Elementary	2012-2013	Kiamaba Central Elementary School

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization
2019 - 2023	Automotive Painter	Bakal works Kiamba Sarangani
2020 - 2021	Covid 19 Isolation Volunteer	LGU Kiamba

Listed in reverse chronological order (most recent first).

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
N/A		
N/A		
N/A		
N/A		

Listed in reverse chronological order (most recent first).

SKILLS

SKILLS	Level of Competency	Date Acquired
N/A		
N/A		

TRAININGS, SEMINARS OR WORKSHOP ATTENDED

Inclusive Dates	Title of Training, Seminar or Workshop
N/A	
N/A	
N/A	
N/A	

Listed in reverse chronological order (most recent first).



Curriculum Vitae of

ARIEL M. GUERRA

Tiongson Street Lagao General Santos, City
Guerra.272228@gensan.sti.edu.ph
09268285162

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	N/A	STI College General Santos
TechVoc	April 2018	Stratford International School
High School	March 2014	General Santos City National Secondary School of Arts and Trades
Elementary	July 2020	Jose C. Catolico Elementary School

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization
N/A		
N/A		

Listed in reverse chronological order (most recent first).

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
N/A		
N/A		
N/A		
N/A		

Listed in reverse chronological order (most recent first).

SKILLS

SKILLS	Level of Competency	Date Acquired
N/A		
N/A		

TRAININGS, SEMINARS OR WORKSHOP ATTENDED

Inclusive Dates	Title of Training, Seminar or Workshop
N/A	
N/A	
N/A	
N/A	

Listed in reverse chronological order (most recent first).



Curriculum Vitae of

ADRIAN D. MACACULOP

Poblacion, Glan, Sarangani, Province
Macaculop.258269@gensan.sti.edu.ph
09488846085

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	N/A	STI College General Santos
TechVoc	N/A	
	July 2021	Glan School of Arts and Trades
High School	April 2019	Glan School of Arts and Trades
Elementary	March 2015	Glan Little Angels Academy

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization
2018 - 2023	Professional Videographer	Kirk Satura Films

Listed in reverse chronological order (most recent first).

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
N/A		

Listed in reverse chronological order (most recent first).

SKILLS

SKILLS	Level of Competency	Date Acquired
N/A		
N/A		

TRAININGS, SEMINARS OR WORKSHOP ATTENDED

Inclusive Dates	Title of Training, Seminar or Workshop
N/A	
N/A	
N/A	
N/A	

Listed in reverse chronological order (most recent first).



Curriculum Vitae of

RAFFY G. SIMON

General Santos City, Lagao, Balite
Simon.139579@gensan.sti.edu.ph
09758029786

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	2022-2023	STI College General Santos
TechVoc	N/A	
High School	April, 2018	General Santos City National Secondary School of Arts and Trades
Elementary	March, 2014	Balite, Elementary School

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization
2020 - 2021	Web Development	Custard Apple Inc.
2021 - 2020	Information Encoder	ABS Company

Listed in reverse chronological order (most recent first).

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
N/A		

Listed in reverse chronological order (most recent first).

SKILLS

SKILLS	Level of Competency	Date Acquired
N/A		
N/A		

TRAININGS, SEMINARS OR WORKSHOP ATTENDED

Inclusive Dates	Title of Training, Seminar or Workshop
N/A	

Listed in reverse chronological order (most recent first)



Curriculum Vitae of

ALVIN A. TAMPUS JR.

S WATA street, Poblacion, Glan Sarangani Province
tampus.256959@gensan.sti.edu.ph
09154960920

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	2022-2023	STI College General Santos
Vocational/Technical	N/A	
Senior High School	July 2021	Glan School of Arts and Trades
High School	April, 2019	Glan School of Arts and Trades
Elementary	March, 2015	Glan Central Elementary School SPED Center

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization
N/A		
N/A		

Listed in reverse chronological order (most recent first).

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
N/A		

Listed in reverse chronological order (most recent first).

SKILLS

SKILLS	Level of Competency	Date Acquired
N/A		
N/A		

TRAININGS, SEMINARS OR WORKSHOP ATTENDED

Inclusive Dates	Title of Training, Seminar or Workshop
N/A	

Listed in reverse chronological order (most recent first).

APPENDIX O. 30% CODE FROM THE SYSTEM

Admin driver's feedback page

```
import 'package:flutter/material.dart';
class DriverFeedbackPage extends StatelessWidget {
  final String driverName;
  final List<Map<String, dynamic>> feedbackList;

  const DriverFeedbackPage({
    Key? key,
    required this.driverName,
    required this.feedbackList,
  }) : super(key: key);

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: Text("Feedback for $driverName"),
      ),
      body: Padding(
        padding: const EdgeInsets.all(16.0),
        child: feedbackList.isNotEmpty
            ? ListView.builder(
                itemCount: feedbackList.length,
                itemBuilder: (context, index) {
                  final feedback = feedbackList[index];
                  return Card(
                    elevation: 2,
                    margin: const EdgeInsets.symmetric(vertical: 8),
```

```
                child: Padding(
                  padding: const EdgeInsets.all(10),
                  child: Column(
                    crossAxisAlignment: CrossAxisAlignment.start,
                    children: [
                      Text(
                        "Rating: ${feedback['rating'].toStringAsFixed(1)}",
                        style: const TextStyle(
                          fontWeight: FontWeight.bold,
                          fontSize: 16,
                        ),
                      ),
                      const SizedBox(height: 5),
                      Text(
                        "Comment: ${feedback['comment']}",
                        style: const TextStyle(fontSize: 14),
                      ),
                    ],
                  ),
                ),
              );
            },
          )
        : const Center(child: Text("No feedback available")),
    ),
  );
 }
}
```

Passenger app home page

```
import 'dart:async';

import 'dart:convert';

import 'dart:typed_data';

import 'package:firebase_auth/firebase_auth.dart';

import 'package:firebase_database/firebase_database.dart';

import 'package:flutter/cupertino.dart';

import 'package:flutter/material.dart';

import 'package:flutter/services.dart';

import 'package:flutter_geofire/flutter_geofire.dart';

import 'package:flutter_polyline_points/flutter_polyline_points.dart';

import 'package:geolocator/geolocator.dart';

import 'package:google_maps_flutter/google_maps_flutter.dart';

import 'package:image_picker/image_picker.dart';

import 'package:loading_animation_widget/loading_animation_widget.dart';

import 'package:provider/provider.dart';

import 'package:restart_app/restart_app.dart';

import 'package:url_launcher/url_launcher.dart';

import 'package:users_app/authentication/login_screen.dart';

import 'package:users_app/global/global_var.dart';
```

```
import 'package:users_app/global/trip_var.dart';

import 'package:users_app/methods/common_methods.dart';

import 'package:users_app/methods/manage_drivers_methods.dart';

import 'package:users_app/models/direction_details.dart';

import 'package:users_app/models/online_nearby_drivers.dart';

import 'package:users_app/pages/about_page.dart';

import 'package:users_app/pages/feedback_page.dart';

import 'package:users_app/pages/search_destination_page.dart';

import 'package:users_app/pages/trips_history_page.dart';

import 'package:users_app/utils.dart';

import '../appInfo/app_info.dart';

import  '../methods/push_notification_service.dart';

import '../widgets/info_dialog.dart';

import '../widgets/loading_dialog.dart';

import  '../widgets/payment_dialog.dart';

import 'package:intl/intl.dart';

class HomePage extends StatefulWidget {

  const HomePage({super.key});
```

```
  @override
  State<HomePage> createState() => _HomePageState();
}

class _HomePageState extends State<HomePage> {
  final Completer<GoogleMapController> googleMapCompleterController =
      Completer<GoogleMapController>();
  GoogleMapController? controllerGoogleMap;
  bool hasAgreedToTerms = false;
  Position? currentPositionOfUser;
  GlobalKey<ScaffoldState> sKey = GlobalKey<ScaffoldState>();
  CommonMethods cMethods = CommonMethods();
  double searchContainerHeight = 276;
  double bottomMapPadding = 0;
  double currentSpeed = 0.0;
  double rideDetailsContainerHeight = 0;
  double requestContainerHeight = 0;
  double tripContainerHeight = 0;
  DirectionDetails? tripDirectionDetailsInfo;
```

```
Future<String?>? fareAmountFuture;

List<LatLng> polylineCoOrdinates = [];

Set<Polyline> polylineSet = {};

Set<Marker> markerSet = {};

Set<Circle> circleSet = {};

bool isDrawerOpened = true;

String stateOfApp = "normal";

bool nearbyOnlineDriversKeysLoaded = false;

BitmapDescriptor? carIconNearbyDriver;

DatabaseReference? tripRequestRef;

List<OnlineNearbyDrivers>? availableNearbyOnlineDriversList;

StreamSubscription<DatabaseEvent>? tripStreamSubscription;

bool requestingDirectionDetailsInfo = false;

Uint8List? image;


void initState() {
  super.initState();
  if (tripDirectionDetailsInfo != null) {
    fareAmountFuture =
        cMethods.calculateFareAmount(tripDirectionDetailsInfo!);
```

```
  }

  getCurrentSpeed();

}

void getCurrentSpeed() async {

  LocationSettings locationSettings = LocationSettings(

    accuracy: LocationAccuracy.bestForNavigation,

    distanceFilter: 1,

  );


  Geolocator.getPositionStream(locationSettings: locationSettings)

      .listen((Position position) {

    if (position != null) {

    setState(() {

        currentSpeed = position.speed * 3.6;

      });

    }

  });

}
```

```
makeDriverNearbyCarIcon() {
  if (carIconNearbyDriver == null) {
    ImageConfiguration configuration =
        createLocalImageConfiguration(context, size: Size(0.5, 0.5));
    BitmapDescriptor.fromAssetImage(
            configuration, "assets/images/trisikol_marker.png")
        .then((iconImage) {
      carIconNearbyDriver = iconImage;
    });
  }
}

void updateMapTheme(GoogleMapController controller) {
  getJsonFileFromThemes("themes/dark_style.json")
      .then((value) => setGoogleMapStyle(value, controller));
}

Future<String> getJsonFileFromThemes(String mapStylePath) async {
  ByteData byteData = await rootBundle.load(mapStylePath);
  var list = byteData.buffer
```

```
      .asUint8List(byteData.offsetInBytes, byteData.lengthInBytes);

  return utf8.decode(list);

}

setGoogleMapStyle(String googleMapStyle, GoogleMapController controller) {

  controller.setMapStyle(googleMapStyle);

}

getCurrentLiveLocationOfUser() async {

  Position positionOfUser = await Geolocator.getCurrentPosition(

      desiredAccuracy: LocationAccuracy.bestForNavigation);

  currentPositionOfUser = positionOfUser;
  LatLng positionOfUserInLatLng = LatLng(

      currentPositionOfUser!.latitude, currentPositionOfUser!.longitude);

  CameraPosition cameraPosition =

      CameraPosition(target: positionOfUserInLatLng, zoom: 18);

  controllerGoogleMap!

      .animateCamera(CameraUpdate.newCameraPosition(cameraPosition));
```

```
  await CommonMethods.convertGeoGraphicCoOrdinatesIntoHumanReadableAddress(
      currentPositionOfUser!, context);
  await getUserInfoAndCheckBlockStatus();
  await initializeGeoFireListener();
}

getUserInfoAndCheckBlockStatus() async {
  DatabaseReference usersRef = FirebaseDatabase.instance
      .ref()
      .child("users")
      .child(FirebaseAuth.instance.currentUser!.uid);
  await usersRef.once().then((snap) {
    if (snap.snapshot.value != null) {
      if ((snap.snapshot.value as Map)["blockStatus"] == "no") {
        setState(() {
          userName = (snap.snapshot.value as Map)["name"];
```

```
          userPhone = (snap.snapshot.value as Map)["phone"];
          discountType = (snap.snapshot.value as Map)["discountType"];
        });
      } else {
        FirebaseAuth.instance.signOut();

        Navigator.push(
            context, MaterialPageRoute(builder: (c) => LoginScreen()));

        cMethods.displaySnackBar(
            "You are blocked. Contact admin: tunatrike@gmail.com", context);
      }
    } else {
      FirebaseAuth.instance.signOut();
      Navigator.push(
          context, MaterialPageRoute(builder: (c) => LoginScreen()));
    }
  });
}
```

```
displayUserRideDetailsContainer() async {
  ///Directions API
  await retrieveDirectionDetails();

  setState(() {
    searchContainerHeight = 0;
    bottomMapPadding = 240;
    rideDetailsContainerHeight = 242;
    isDrawerOpened = false;
  });
}

retrieveDirectionDetails() async {
  var pickUpLocation =
      Provider.of<AppInfo>(context, listen: false).pickUpLocation;
  var dropOffDestinationLocation =
      Provider.of<AppInfo>(context, listen: false).dropOffLocation;


  var pickupGeoGraphicCoOrdinates = LatLng(
      pickUpLocation!.latitudePosition!, pickUpLocation.longitudePosition!);
```

```
var dropOffDestinationGeoGraphicCoOrdinates = LatLng(
    dropOffDestinationLocation!.latitudePosition!,
    dropOffDestinationLocation.longitudePosition!);

showDialog(
  barrierDismissible: false,
  context: context,
  builder: (BuildContext context) =>
      LoadingDialog(messageText: "Please Wait..."),
);

///Directions API
var detailsFromDirectionAPI =
    await CommonMethods.getDirectionDetailsFromAPI(
        pickupGeoGraphicCoOrdinates,
        dropOffDestinationGeoGraphicCoOrdinates);
setState(() {
  tripDirectionDetailsInfo = detailsFromDirectionAPI;
});
setState(() {
```

```
  tripDirectionDetailsInfo = detailsFromDirectionAPI;


  // Trigger fare calculation here if direction details are available
  if (tripDirectionDetailsInfo != null) {
    fareAmountFuture =
        cMethods.calculateFareAmount(tripDirectionDetailsInfo!);
     debugPrint("Fare calculation triggered after direction details set.");
  } else {
    debugPrint(
         "Error: tripDirectionDetailsInfo is null after direction API call.");
  }
});
Navigator.pop(context);
///draw route from pickup to dropOffDestination
PolylinePoints pointsPolyline = PolylinePoints();
List<PointLatLng> latLngPointsFromPickUpToDestination =
    pointsPolyline.decodePolyline(tripDirectionDetailsInfo!.encodedPoints!);


polylineCoOrdinates.clear();
```

```
if (latLngPointsFromPickUpToDestination.isNotEmpty) {
  latLngPointsFromPickUpToDestination.forEach((PointLatLng latLngPoint) {
  polylineCoOrdinates
      .add(LatLng(latLngPoint.latitude, latLngPoint.longitude));
  });
}

polylineSet.clear();
setState(() {
  Polyline polyline = Polyline(
    polylineId: const PolylineId("polylineID"),
    color: Colors.blue,
    points:  polylineCoOrdinates,
    jointType:  JointType.round,
    width: 4,
    startCap: Cap.roundCap,
    endCap:  Cap.roundCap,
    geodesic: true,
  );
```

```
  polylineSet.add(polyline);
});


//fit the polyline into the map
LatLngBounds boundsLatLng;
if (pickupGeoGraphicCoOrdinates.latitude >
        dropOffDestinationGeoGraphicCoOrdinates.latitude &&
    pickupGeoGraphicCoOrdinates.longitude >
        dropOffDestinationGeoGraphicCoOrdinates.longitude) {
  boundsLatLng = LatLngBounds(
    southwest: dropOffDestinationGeoGraphicCoOrdinates,
    northeast: pickupGeoGraphicCoOrdinates,
  );
} else if (pickupGeoGraphicCoOrdinates.longitude >
  dropOffDestinationGeoGraphicCoOrdinates.longitude) {
  boundsLatLng = LatLngBounds(
    southwest: LatLng(pickupGeoGraphicCoOrdinates.latitude,
        dropOffDestinationGeoGraphicCoOrdinates.longitude),
    northeast: LatLng(dropOffDestinationGeoGraphicCoOrdinates.latitude,
        pickupGeoGraphicCoOrdinates.longitude),
```

```
  );
} else if (pickupGeoGraphicCoOrdinates.latitude >
  dropOffDestinationGeoGraphicCoOrdinates.latitude) {
  boundsLatLng = LatLngBounds(
    southwest: LatLng(dropOffDestinationGeoGraphicCoOrdinates.latitude,
        pickupGeoGraphicCoOrdinates.longitude),
    northeast: LatLng(pickupGeoGraphicCoOrdinates.latitude,
        dropOffDestinationGeoGraphicCoOrdinates.longitude),
  );
} else {
  boundsLatLng = LatLngBounds(
    southwest: pickupGeoGraphicCoOrdinates,
    northeast: dropOffDestinationGeoGraphicCoOrdinates,
  );
}

controllerGoogleMap!
    .animateCamera(CameraUpdate.newLatLngBounds(boundsLatLng,  72));


//add markers to pickup and dropOffDestination points
```

```
Marker pickUpPointMarker = Marker(
  markerId: const MarkerId("pickUpPointMarkerID"),
  position: pickupGeoGraphicCoOrdinates,
  icon: BitmapDescriptor.defaultMarkerWithHue(BitmapDescriptor.hueRed),
  infoWindow: InfoWindow(
     title: pickUpLocation.placeName, snippet: "Pickup Location"),
);

Marker dropOffDestinationPointMarker = Marker(
  markerId: const MarkerId("dropOffDestinationPointMarkerID"),
  position: dropOffDestinationGeoGraphicCoOrdinates,
  icon: BitmapDescriptor.defaultMarkerWithHue(BitmapDescriptor.hueGreen),
  infoWindow: InfoWindow(
     title: dropOffDestinationLocation.placeName,
     snippet: "Destination Location"),
);

setState(() {
  markerSet.add(pickUpPointMarker);
  markerSet.add(dropOffDestinationPointMarker);
```

```
});

//add circles to pickup and dropOffDestination points
Circle pickUpPointCircle = Circle(
  circleId: const CircleId('pickupCircleID'),
  strokeColor: Colors.blue,
  strokeWidth: 4,
  radius: 14,
  center: pickupGeoGraphicCoOrdinates,
  fillColor: Colors.blue,
);
Circle dropOffDestinationPointCircle = Circle(
  circleId: const CircleId('dropOffDestinationCircleID'),
  strokeColor: Colors.blue,
  strokeWidth: 4,
  radius: 14,
  center: dropOffDestinationGeoGraphicCoOrdinates,
  fillColor: Colors.blue,
);
```

```
  setState(() {
    circleSet.add(pickUpPointCircle);
    circleSet.add(dropOffDestinationPointCircle);
  });
}

resetAppNow() {
  setState(() {
    polylineCoOrdinates.clear();
    polylineSet.clear();
    markerSet.clear();
    circleSet.clear();
    rideDetailsContainerHeight = 0;
    requestContainerHeight = 0;
    tripContainerHeight = 0;
    searchContainerHeight = 276;
    bottomMapPadding = 300;
    isDrawerOpened = true;
```

```
    status = "";
    nameDriver = "";
    photoDriver = "";
    phoneNumberDriver = "";
    trikeDetailsDriver = "";
    tripStatusDisplay = 'Driver is Arriving';
  });
}

cancelRideRequest() {
  if (tripRequestRef != null) {
    tripRequestRef!.remove();
    if (tripStreamSubscription != null) {
      tripStreamSubscription!.cancel();
      tripStreamSubscription = null;
    }
    setState(() {
      stateOfApp = "normal";
      tripContainerHeight = 0;
      searchContainerHeight = 276;
```

```
      bottomMapPadding = 300;
    });
    cMethods.displaySnackBar("Ride has been canceled", context);
  }
}

displayRequestContainer() {
  setState(() {
  rideDetailsContainerHeight = 0;
    requestContainerHeight = 220;
    bottomMapPadding = 200;
    isDrawerOpened = true;
  });

  //send ride request
  makeTripRequest();
}

updateAvailableNearbyOnlineDriversOnMap() {
  setState(() {
```

```
  markerSet.clear();
});

Set<Marker> markersTempSet = Set<Marker>();
for (OnlineNearbyDrivers eachOnlineNearbyDriver
    in ManageDriversMethods.nearbyOnlineDriversList) {
  LatLng driverCurrentPosition = LatLng(
      eachOnlineNearbyDriver.latDriver!, eachOnlineNearbyDriver.lngDriver!);

  Marker driverMarker = Marker(
    markerId: MarkerId(
        "driver ID = " + eachOnlineNearbyDriver.uidDriver.toString()),
    position: driverCurrentPosition,
    icon: carIconNearbyDriver!,
  );
  markersTempSet.add(driverMarker);
}
```

```
  setState(() {
    markerSet = markersTempSet;
  });
}

initializeGeoFireListener() {
  Geofire.initialize("onlineDrivers");
  Geofire.queryAtLocation(currentPositionOfUser!.latitude,
        currentPositionOfUser!.longitude, 0.5)!
      .listen((driverEvent) {
    if (driverEvent != null) {
      var onlineDriverChild = driverEvent["callBack"];
      switch (onlineDriverChild) {
        case Geofire.onKeyEntered:
          OnlineNearbyDrivers onlineNearbyDrivers = OnlineNearbyDrivers();
          onlineNearbyDrivers.uidDriver = driverEvent["key"];
          onlineNearbyDrivers.latDriver = driverEvent["latitude"];
          onlineNearbyDrivers.lngDriver = driverEvent["longitude"];
          ManageDriversMethods.nearbyOnlineDriversList
```

```
        .add(onlineNearbyDrivers);

    if (nearbyOnlineDriversKeysLoaded == true) {
      //update drivers on google map
      updateAvailableNearbyOnlineDriversOnMap();
    }

    break;

  case Geofire.onKeyExited:
    ManageDriversMethods.removeDriverFromList(driverEvent["key"]);

    updateAvailableNearbyOnlineDriversOnMap();
    break;

  case Geofire.onKeyMoved:
    OnlineNearbyDrivers onlineNearbyDrivers = OnlineNearbyDrivers();
    onlineNearbyDrivers.uidDriver = driverEvent["key"];
    onlineNearbyDrivers.latDriver = driverEvent["latitude"];
    onlineNearbyDrivers.lngDriver = driverEvent["longitude"];
```

```
            ManageDriversMethods.updateOnlineNearbyDriversLocation(
                onlineNearbyDrivers);

            //update drivers on google map
            updateAvailableNearbyOnlineDriversOnMap();

            break;

          case Geofire.onGeoQueryReady:
            nearbyOnlineDriversKeysLoaded = true;
            //display nearest online drivers
            updateAvailableNearbyOnlineDriversOnMap();
            break;
        }
      }
    });
  }

  void makeTripRequest() async {
    tripRequestRef =
```

```
    FirebaseDatabase.instance.ref().child("tripRequests").push();


var pickUpLocation =
    Provider.of<AppInfo>(context, listen: false).pickUpLocation;
var dropOffDestinationLocation =
    Provider.of<AppInfo>(context, listen: false).dropOffLocation;


Map pickUpCoOrdinatesMap = {
  "latitude": pickUpLocation!.latitudePosition.toString(),
  "longitude": pickUpLocation.longitudePosition.toString(),
};


Map dropOffDestinationCoOrdinatesMap = {
  "latitude": dropOffDestinationLocation!.latitudePosition.toString(),
  "longitude": dropOffDestinationLocation.longitudePosition.toString(),
};


Map driverCoOrdinates = {
  "latitude": "",
  "longitude": "",
```

```
};
String? fareAmountStr = await fareAmountFuture ?? "Not available";
double fareAmount = double.tryParse(fareAmountStr ?? "0") ?? 0;
if (discountType == "Student" ||
  discountType == "PWD" ||
  discountType == "Senior Citizen") {
  fareAmount = fareAmount * 0.8;
}

Map dataMap = {
  "tripID": tripRequestRef!.key,
  "publishDateTime":
      DateFormat('yyyy-MM-dd HH:mm:ss').format(DateTime.now()),
  "userName": userName,
  "discountType": discountType,
  "userPhone": userPhone,
  "userID": userID,
  "pickUpLatLng": pickUpCoOrdinatesMap,
  "dropOffLatLng": dropOffDestinationCoOrdinatesMap,
  "pickUpAddress": pickUpLocation.placeName,
```

```
  "dropOffAddress": dropOffDestinationLocation.placeName,
  "driverID": "waiting",
  "trike_details": "",
  "driverLocation": driverCoOrdinates,
  "driverName": "",
  "driverPhone": "",
  "driverPictureUrl": "",
  "fareAmount": fareAmount.toStringAsFixed(2),
  "status": "new",
};

tripRequestRef!.set(dataMap);

tripStreamSubscription =
  tripRequestRef!.onValue.listen((eventSnapshot) async {
  if (eventSnapshot.snapshot.value == null) {
    return;
  }

  if ((eventSnapshot.snapshot.value as Map)["driverName"] != null) {
```

```
  nameDriver = (eventSnapshot.snapshot.value as Map)["driverName"];
}

if ((eventSnapshot.snapshot.value as Map)["driverPhone"] != null) {
  phoneNumberDriver =
      (eventSnapshot.snapshot.value as Map)["driverPhone"];
}

if ((eventSnapshot.snapshot.value as Map)["driverPictureUrl"] != null) {
  photoDriver = (eventSnapshot.snapshot.value as Map)["driverPictureUrl"];
}

if ((eventSnapshot.snapshot.value as Map)["trike_details"] != null) {
  trikeDetailsDriver =
      (eventSnapshot.snapshot.value as Map)["trike_details"];
}

if ((eventSnapshot.snapshot.value as Map)["status"] != null) {
  status = (eventSnapshot.snapshot.value as Map)["status"];
}
```

```
if ((eventSnapshot.snapshot.value as Map)["driverLocation"] != null) {
  double driverLatitude = double.parse(
      (eventSnapshot.snapshot.value as Map)["driverLocation"]["latitude"]
          .toString());
  double driverLongitude = double.parse(
      (eventSnapshot.snapshot.value as Map)["driverLocation"]["longitude"]
          .toString());
  LatLng driverCurrentLocationLatLng =
      LatLng(driverLatitude, driverLongitude);


  if (status == "accepted") {
    updateFromDriverCurrentLocationToPickUp(driverCurrentLocationLatLng);
  } else if (status == "arrived") {
    setState(() {
      tripStatusDisplay = 'Driver has Arrived';
    });
  } else if (status == "ontrip") {
    updateFromDriverCurrentLocationToDropOffDestination(
        driverCurrentLocationLatLng);
```

```
  }
}

if (status == "accepted") {
  displayTripDetailsContainer();

  Geofire.stopListener();
  setState(() {
  markerSet.removeWhere(
      (element) => element.markerId.value.contains("driver"));
  });
}

if (status == "ended") {
  if ((eventSnapshot.snapshot.value as Map)["fareAmount"] != null) {
    double fareAmount = double.parse(
        (eventSnapshot.snapshot.value as Map)["fareAmount"].toString());

    var responseFromPaymentDialog = await showDialog(
      context: context,
```

```
        builder: (BuildContext context) =>
            PaymentDialog(overAllTotalFareAmount: fareAmount.toString()),
      );

      if (responseFromPaymentDialog == "paid") {
        tripRequestRef!.onDisconnect();
        tripRequestRef = null;

        tripStreamSubscription!.cancel();
        tripStreamSubscription = null;

        resetAppNow();
        Restart.restartApp();
      }
    }
  }
});
}

displayTripDetailsContainer() {
```

```
  setState(() {
    requestContainerHeight = 0;
    tripContainerHeight = 350;
    bottomMapPadding = 281;
  });
}

updateFromDriverCurrentLocationToPickUp(driverCurrentLocationLatLng) async {
  if (!requestingDirectionDetailsInfo) {
    requestingDirectionDetailsInfo = true;

    var userPickUpLocationLatLng = LatLng(
        currentPositionOfUser!.latitude, currentPositionOfUser!.longitude);

    var directionDetailsPickup =
        await CommonMethods.getDirectionDetailsFromAPI(
            driverCurrentLocationLatLng, userPickUpLocationLatLng);

    if (directionDetailsPickup == null) {
      return;
```

```
    }

    setState(() {
      tripStatusDisplay =
          "Driver is Coming - ${directionDetailsPickup.durationTextString}";
    });

    requestingDirectionDetailsInfo = false;
  }
}

updateFromDriverCurrentLocationToDropOffDestination(
    driverCurrentLocationLatLng) async {
  if (!requestingDirectionDetailsInfo) {
    requestingDirectionDetailsInfo = true;

    var dropOffLocation =
        Provider.of<AppInfo>(context, listen: false).dropOffLocation;
    var userDropOffLocationLatLng = LatLng(dropOffLocation!.latitudePosition!,
        dropOffLocation.longitudePosition!);
```

```
    var directionDetailsPickup =
        await CommonMethods.getDirectionDetailsFromAPI(
            driverCurrentLocationLatLng, userDropOffLocationLatLng);

    if (directionDetailsPickup == null) {
      return;
    }

    setState(() {
      tripStatusDisplay =
          "Driving to Destination - ${directionDetailsPickup.durationTextString}";
    });

    requestingDirectionDetailsInfo = false;
  }
}

noDriverAvailable() {
  showDialog(
```

```
      context: context,
      barrierDismissible: false,
      builder: (BuildContext context) => InfoDialog(
          title: "No Driver Available",
          description:
              "No driver found in the nearby location. Please try again shortly.",
          ));
}

void showAgreementDialog() {
  showDialog(
    context: context,
    builder: (BuildContext context) {
      return AlertDialog(
        title: const Text('Terms and Conditions'),
        content: SingleChildScrollView(
          child: Column(
            children: const [
            Text(
                'By requesting a driver, you agree to the following terms and conditions...',
```

```
            style: TextStyle(fontSize: 14),
          ),
          SizedBox(height: 10),
          Text(
            '1. If you cancel the ride, You will be responsible to pay the minimum fare ₱15.\n'
            '2. You acknowledge that the driver\'s availability is subject to change.\n'
            '3. You will be responsible for the payment as calculated at the end of the ride.',
            style: TextStyle(fontSize: 14),
          ),
        ],
      ),
    ),
    actions: <Widget>[
      TextButton(
        child: const Text('Cancel'),
        onPressed: () {
          Navigator.of(context).pop(); // Close the dialog
        },
      ),
```

```
      TextButton(
        child: const Text('Agree'),
        onPressed: () {
          Navigator.of(context).pop(); // Close the dialog
          // Proceed to find the driver
          setState(() {
            stateOfApp = "requesting";
          });
          displayRequestContainer();

          //get nearest available online drivers
          availableNearbyOnlineDriversList =
              ManageDriversMethods.nearbyOnlineDriversList;

          //search driver
          searchDriver();
        },
      ),
    ],
  );
```

```
    },
  );
}

searchDriver() {
  if (hasAgreedToTerms) {
    if (availableNearbyOnlineDriversList!.length == 0) {
      cancelRideRequest();
      resetAppNow();
      noDriverAvailable();
      return;
    }
  } else {
    print("User did not agree to terms");
  }

  var currentDriver = availableNearbyOnlineDriversList![0];

  //send notification to this currentDriver
  sendNotificationToDriver(currentDriver);
```

```
  availableNearbyOnlineDriversList!.removeAt(0);
}

sendNotificationToDriver(OnlineNearbyDrivers currentDriver) {
  //update driver's newTripStatus - assign tripID to current driver
  DatabaseReference currentDriverRef = FirebaseDatabase.instance
      .ref()
      .child("drivers")
      .child(currentDriver.uidDriver.toString())
      .child("newTripStatus");

  currentDriverRef.set(tripRequestRef!.key);

  //get current driver device recognition token
  DatabaseReference tokenOfCurrentDriverRef = FirebaseDatabase.instance
      .ref()
      .child("drivers")
      .child(currentDriver.uidDriver.toString())
      .child("deviceToken");
```

```
tokenOfCurrentDriverRef.once().then((dataSnapshot) {
  if (dataSnapshot.snapshot.value != null) {
    String deviceToken = dataSnapshot.snapshot.value.toString();

    //send notification
    PushNotificationService.sendNotificationToSelectedDriver(
        deviceToken, context, tripRequestRef!.key.toString());
  } else {
    return;
  }

  const oneTickPerSec = Duration(seconds: 1);

  var timerCountDown = Timer.periodic(oneTickPerSec, (timer) {
    requestTimeoutDriver = requestTimeoutDriver - 1;

    //when trip request is not requesting means trip request cancelled - stop timer
    if (stateOfApp != "requesting") {
      timer.cancel();
```

```
  currentDriverRef.set("cancelled");
  currentDriverRef.onDisconnect();
  requestTimeoutDriver = 20;
}

//when trip request is accepted by online nearest available driver
currentDriverRef.onValue.listen((dataSnapshot) {
  if (dataSnapshot.snapshot.value.toString() == "accepted") {
    timer.cancel();
    currentDriverRef.onDisconnect();
    requestTimeoutDriver = 20;
  }
});

//if 20 seconds passed - send notification to next nearest online available driver
if (requestTimeoutDriver == 0) {
  currentDriverRef.set("timeout");
  timer.cancel();
  currentDriverRef.onDisconnect();
  requestTimeoutDriver = 20;
```

```
        //send notification to next nearest online available driver
        searchDriver();
      }
    });
  });
}

void selectImage() async {
  Uint8List img = await pickImage(ImageSource.gallery);
  setState(() {
    image = img;
  });
}

@override
Widget build(BuildContext context) {
  makeDriverNearbyCarIcon();
  return Scaffold(
    key: sKey,
```

```
drawer: Container(
  width: 285,
  color: Colors.black87,
  child: Drawer(
    backgroundColor: Colors.white10,
    child: ListView(
      children: [
        const Divider(
        height: 1,
          color: Colors.grey,
          thickness: 1,
        ),

        //header
        Container(
          color: Colors.black54,
          height: 160,
          child: DrawerHeader(
            decoration: const BoxDecoration(
              color: Colors.white10,
```

```
),
child: Row(
  children: [
  Stack(
     children:       [
      image != null
          ? CircleAvatar(
              radius: 30,
              backgroundImage: MemoryImage(image!),
            )
          : const CircleAvatar(
              radius: 30,
              backgroundImage:
                  AssetImage("assets/images/user.png"),
            ),
       Positioned(
         bottom: -15,
         left: 23,
         child: IconButton(
           onPressed: () {
```

```
            selectImage();
          },
          icon: const Icon(Icons.add_a_photo),
          iconSize: 15,
          color: Colors.black,
        ),
      ),
    ],
  ),
  const SizedBox(
    width: 16,
  ),
  Column(
    mainAxisAlignment: MainAxisAlignment.center,
    children: [
      Text(
        userName,
        style: const TextStyle(
          fontSize: 16,
          color: Colors.grey,
```

```
            fontWeight: FontWeight.bold,
          ),
        ),
        const SizedBox(
          height: 4,
        ),
        const Text(
          "Profile",
          style: TextStyle(
            color: Colors.white38,
          ),
        ),
      ],
    ),
  ],
 ),
),

const Divider(
```

```
  height: 1,
  color: Colors.grey,
  thickness: 1,
),

const SizedBox(
  height: 10,
),

//body
GestureDetector(
onTap: () {
    Navigator.push(context,
        MaterialPageRoute(builder: (c) => TripsHistoryPage()));
  },
  child: ListTile(
    leading: IconButton(
    onPressed: () {},
    icon: const Icon(
        Icons.history,
```

```
        color: Colors.grey,
      ),
    ),
    title: const Text(
      "History",
      style: TextStyle(color: Colors.grey),
    ),
  ),
),

GestureDetector(
  onTap: () {
  Navigator.push(
      context, MaterialPageRoute(builder: (c) => AboutPage()));
  },
  child: ListTile(
    leading: IconButton(
    onPressed: () {},
    icon: const Icon(
      Icons.info,
```

```
        color: Colors.grey,
      ),
    ),
    title: const Text(
      "About",
      style: TextStyle(color: Colors.grey),
    ),
  ),
),
GestureDetector(
  onTap: () {
    Navigator.push(context,
        MaterialPageRoute(builder: (c) => FeedbackPage()));
  },
  child: ListTile(
    leading: IconButton(
    onPressed: () {},
    icon: const Icon(
        Icons.chat_bubble_outline,
        color: Colors.grey,
```

```
      ),
    ),
    title: const Text(
      "Feedback",
      style: TextStyle(color: Colors.grey),
    ),
  ),
),

GestureDetector(
  onTap: () {
    FirebaseAuth.instance.signOut();

    Navigator.push(context,
        MaterialPageRoute(builder: (c) => LoginScreen()));
  },
  child: ListTile(
    leading: IconButton(
    onPressed: () {},
    icon: const Icon(
```

```
                Icons.logout,
                color: Colors.grey,
              ),
            ),
            title: const Text(
              "Logout",
              style: TextStyle(color: Colors.grey),
            ),
          ),
        ),


      ],
    ),
  ),
),
body: Stack(
  children: [
    ///google map
    GoogleMap(
      padding: EdgeInsets.only(top: 26, bottom: bottomMapPadding),
```

```
    mapType: MapType.normal,
    myLocationEnabled: true,
    polylines: polylineSet,
    markers: markerSet,
    circles: circleSet,
    initialCameraPosition: googlePlexInitialPosition,
    onMapCreated: (GoogleMapController mapController) {
    controllerGoogleMap = mapController;
    updateMapTheme(controllerGoogleMap!);

      googleMapCompleterController.complete(controllerGoogleMap);

      setState(() {
        bottomMapPadding = 300;
      });

      getCurrentLiveLocationOfUser();
    },
  ),
```

```
///drawer button
Positioned(
top: 36,
  left: 19,
  child: GestureDetector(
    onTap: () {
      if (isDrawerOpened == true) {
        sKey.currentState!.openDrawer();
      } else {
        resetAppNow();
      }
    },
    child: Container(
      decoration: BoxDecoration(
        color: Colors.white,
        borderRadius: BorderRadius.circular(20),
        boxShadow: const [
          BoxShadow(
            color: Colors.black26,
            blurRadius: 5,
```

```
            spreadRadius: 0.5,
            offset: Offset(0.7, 0.7),
          ),
        ],
      ),
      child: CircleAvatar(
        backgroundColor: Colors.grey,
        radius: 20,
        child: Icon(
          isDrawerOpened == true ? Icons.menu : Icons.close,
          color: Colors.black87,
        ),
      ),
    ),
  ),
),
```

Driver app home page

```
import 'dart:async';

import 'dart:convert';

import 'dart:typed_data';

import 'package:drivers_app/methods/map_theme_methods.dart';

import 'package:drivers_app/pushNotification/push_notification_system.dart';

import 'package:firebase_auth/firebase_auth.dart';

import 'package:firebase_database/firebase_database.dart';

import 'package:flutter/material.dart';

import 'package:flutter/services.dart';

import 'package:flutter_geofire/flutter_geofire.dart';

import 'package:geolocator/geolocator.dart';

import 'package:google_maps_flutter/google_maps_flutter.dart';


import '../global/global_var.dart';


class HomePage extends StatefulWidget {

  const HomePage({super.key});


  @override
```

```
  State<HomePage> createState() => _HomePageState();
}

class _HomePageState extends State<HomePage> {
  final Completer<GoogleMapController> googleMapCompleterController =
      Completer<GoogleMapController>();
  GoogleMapController? controllerGoogleMap;
  Position? currentPositionOfDriver;
  Color colorToShow = Colors.red;
  String titleToShow = "OFFLINE";
  bool isDriverAvailable = false;
  DatabaseReference?                newTripRequestReference;
  MapThemeMethods themeMethods = MapThemeMethods();

  getCurrentLiveLocationOfDriver() async {
    Position positionOfUser = await Geolocator.getCurrentPosition(
        desiredAccuracy: LocationAccuracy.bestForNavigation);
    currentPositionOfDriver = positionOfUser;
    driverCurrentPosition = currentPositionOfDriver;
```

```
  LatLng positionOfUserInLatLng = LatLng(
      currentPositionOfDriver!.latitude, currentPositionOfDriver!.longitude);

  CameraPosition cameraPosition =
      CameraPosition(target: positionOfUserInLatLng, zoom: 15);
  controllerGoogleMap!
      .animateCamera(CameraUpdate.newCameraPosition(cameraPosition));
}

goOnlineNow() {
  //all drivers who are Available for new trip requests
  Geofire.initialize("onlineDrivers");

  Geofire.setLocation(
    FirebaseAuth.instance.currentUser!.uid,
    currentPositionOfDriver!.latitude,
    currentPositionOfDriver!.longitude,
  );

  newTripRequestReference = FirebaseDatabase.instance
```

```
      .ref()
      .child("drivers")
      .child(FirebaseAuth.instance.currentUser!.uid)
      .child("newTripStatus");
  newTripRequestReference!.set("waiting");

  newTripRequestReference!.onValue.listen((event) {});
}

setAndGetLocationUpdates() {
  positionStreamHomePage =
      Geolocator.getPositionStream().listen((Position position) {
    currentPositionOfDriver = position;

    if (isDriverAvailable == true) {
      Geofire.setLocation(
      FirebaseAuth.instance.currentUser!.uid,
      currentPositionOfDriver!.latitude,
      currentPositionOfDriver!.longitude,
      );
```

```
    }


    LatLng positionLatLng = LatLng(position.latitude, position.longitude);
    controllerGoogleMap!
        .animateCamera(CameraUpdate.newLatLng(positionLatLng));
  });
}

goOfflineNow() {
  //stop sharing driver live location updates
  Geofire.removeLocation(FirebaseAuth.instance.currentUser!.uid);

  //stop listening to the newTripStatus
  newTripRequestReference!.onDisconnect();
  newTripRequestReference!.remove();
  newTripRequestReference = null;
}

initializePushNotificationSystem() {
  PushNotificationSystem notificationSystem = PushNotificationSystem();
```

```
  notificationSystem.generateDeviceRegistrationToken();
  notificationSystem.startListeningForNewNotification(context);
}

retrieveCurrentDriverInfo() async {
  await FirebaseDatabase.instance
      .ref()
      .child("drivers")
      .child(FirebaseAuth.instance.currentUser!.uid)
      .once()
      .then((snap) {
    driverName = (snap.snapshot.value as Map)["name"];
    driverPhone = (snap.snapshot.value as Map)["phone"];
    driverPictureUrl = (snap.snapshot.value as Map)["driverPictureUrl"];
    trikeModel = (snap.snapshot.value as Map)["trike_details"]["trikeModel"];
    plateNumber =
        (snap.snapshot.value as Map)["trike_details"]["plateNumber"];
  });

  initializePushNotificationSystem();
```

```
}

@override
void initState() {
  // TODO: implement initState
  super.initState();

  retrieveCurrentDriverInfo();
}

@override
Widget build(BuildContext context) {
  return Scaffold(
    body: Stack(
      children: [
        ///google map
        GoogleMap(
          padding: const EdgeInsets.only(top: 136),
          mapType: MapType.normal,
          myLocationEnabled: true,
```

```
  initialCameraPosition: googlePlexInitialPosition,
  onMapCreated: (GoogleMapController mapController) {
  controllerGoogleMap = mapController;
  themeMethods.updateMapTheme(controllerGoogleMap!);

    googleMapCompleterController.complete(controllerGoogleMap);

    getCurrentLiveLocationOfDriver();
  },
),

Container(
  height: 136,
  width: double.infinity,
  color: Colors.black54,
),

///go online offline button
Positioned(
  top: 61,
```

```
left: 0,
right: 0,
child: Row(
 mainAxisAlignment: MainAxisAlignment.center,
 children: [
  ElevatedButton(
   onPressed: () {
    showModalBottomSheet(
     context: context,
     isDismissible: false,
     builder: (BuildContext context) {
      return Container(
       decoration: const BoxDecoration(
        color: Colors.black87,
        boxShadow: [
         BoxShadow(
          color: Colors.grey,
          blurRadius: 5.0,
          spreadRadius: 0.5,
          offset: Offset(
```

```
          0.7,
          0.7,
        ),
      ),
    ],
  ),
  height: 221,
  child: Padding(
    padding: const EdgeInsets.symmetric(
        horizontal: 24, vertical: 18),
    child: Column(
      children: [
        const SizedBox(
          height: 11,
        ),
        Text(
          (!isDriverAvailable)
              ? "GO ONLINE NOW"
              : "GO OFFLINE NOW",
          textAlign: TextAlign.center,
```

```
                style: const TextStyle(
                  fontSize: 22,
                  color: Colors.white70,
                  fontWeight: FontWeight.bold,
                ),
              ),
              const SizedBox(
                height: 21,
              ),
              Text(
                (!isDriverAvailable)
                    ? "You are about to go online, you will become available to receive trip requests from users."
                    : "You are about to go offline, you will stop receiving new trip requests from users.",
                textAlign: TextAlign.center,
                style: const TextStyle(
                color: Colors.white60,
                ),
              ),
              const SizedBox(
```

```
  height: 25,
),
Row(
  children: [
  Expanded(
     child: ElevatedButton(
       onPressed: () {
       Navigator.pop(context);
       },
       child: const Text("CANCEL"),
     ),
   ),
   const SizedBox(
     width: 16,
   ),
   Expanded(
     child: ElevatedButton(
       onPressed: () {
         if (!isDriverAvailable) {
           goOnlineNow();
```

```
      setAndGetLocationUpdates();

      Navigator.pop(context);

      setState(() {
        colorToShow = Colors.green;
        titleToShow = "ONLINE";
        isDriverAvailable = true;
      });
    } else {
      goOfflineNow();

      Navigator.pop(context);

      setState(() {
        colorToShow = Colors.red;
        titleToShow = "OFFLINE";
        isDriverAvailable = false;
      });
    }
```

```
                    },
                    style: ElevatedButton.styleFrom(
                      backgroundColor:
                          (titleToShow == "GO ONLINE NOW")
                              ? Colors.red
                              : Colors.green,
                    ),
                    child: const Text("CONFIRM"),
                  ),
                ),
              ],
            ),
          ],
        ),
      ),
    );
  });
},
style: ElevatedButton.styleFrom(
  backgroundColor: colorToShow,
```

```
                ),
                child: Text(
                  titleToShow,
                ),
              ),
            ),
          ],
        ),
      ),
    ],
  ),
);
  }
}
```